



NE450

Nuclear Weapons

Effects



NE450 NUCLEAR WEAPONS EFFECTS

ABET eCOURSEBOOK

AY2025-2026

TABLE OF CONTENTS

AY2026 Course Syllabus

SECTION I. COURSE ASSESSMENTS

TAB A	AY2026 Course Assessment
TAB B	AY2025 Course Assessment
TAB C	AY2024 Course Assessment
TAB D	AY2023 Course Assessment
TAB E	AY2022 Course Assessment
TAB F	AY2021 Course Assessment

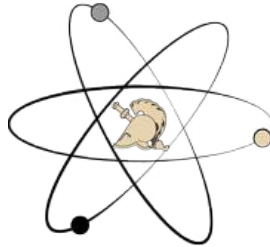
SECTION II. EXAMINATIONS

TAB G	WPR #1
TAB H	WPR #2
TAB I	Term End Examination
TAB J	Final Course Grades

SECTION III. ASSIGNMENTS

TAB K	Problem Sets
TAB L	Extension Problems

UNITED STATES MILITARY ACADEMY



Nuclear Engineering Program Course Syllabus

NE450 Nuclear Weapons Effects

AY 26-2

January 4, 2026

Nuclear Engineering Program Director:

Professor Kenneth Allen, PhD, PE

Course Director:

Major Dave Pert

Contents

1	Course Introduction	1
2	Conduct and Expectations	2
2.1	What can I expect from the Course Director?	2
2.2	What does the Course Director expect from me?	2
2.3	Sexual Harassment Policy	3
2.4	Chain of Command	4
3	Course Objectives	5
3.1	Nuclear Engineering Program Educational Objectives (NE PEOs)	5
3.2	Course Objectives Mapped to ABET Student Outcomes	6
3.3	Course Objectives Mapped to Academic Program Goals	8
4	Course Requirements and Grading	10
4.1	Course Point Allocation	10
4.2	Grade Cutoffs	12
4.3	Late Assignment Policy	13
5	Course Resources	14
5.1	Course Text	14
5.2	Course Outline	14
5.3	Calculators	15
5.4	Canvas	16
5.5	Examination Materials	16
5.6	Additional Instruction (AI)	16
6	Documentation of Academic Work (DAW)	18
6.1	Documentation of Previous Work	19
6.2	Course Common Knowledge	19
7	Works Cited	21
	Enclosures	22

1 Course Introduction

Nuclear Weapons Effects (NE450) provides a broad introduction to the design, prompt effects, residual effects, and strategic policy of nuclear weapons. Weapon design will examine the theory of fission and fusion with broad application to the assembly and disassembly of a critical mass of fissile nuclear material (core) and non-nuclear materials (case) surrounding the core. Prompt effects include X-rays, thermal radiation, air blast, cratering and ground shock, neutron and gamma radiation, and electromagnetic pulse. Residual effects include the fractionation of radioactive fission products throughout the volume and on the surface of condensed particles, distribution of radioactive particles (fallout), as well as dose and associated biological effects of exposure to radiation from fallout. Quantitative estimates of many effects will be obtained from empirical relationships, graphs, and tables that were developed from data measured during atmospheric, under-ground, and under-water detonations of nuclear weapons that occurred in the United States from 1945 - 1992.

2 Conduct and Expectations

2.1 What can I expect from the Course Director?

I will provide engaging and relevant lecture material, resources, and evaluation tools to build and assess cadets' mastery of nuclear weapons concepts and skills as outlined in this syllabus. While the basics of nuclear weapon physics and effects are unclassified, conversation about US weapon engineering can quickly require clearance above what the classroom is able to provide. Accordingly, I will table questions that fall into classified space for the CNWDI brief lesson, which will be held in an appropriately cleared room.

I will select points of emphasis to evaluate throughout the course using board problems, writs, WPRs, and the TEE. You should expect timely and objective assessment of your performance on graded events with the complementary goals of teaching you how analyze and synthesize solutions to complex problems while continually building your knowledge of nuclear weapons topics. These points of emphasis support areas in which graduates have shown need for improvement in the past and serve to underpin the accomplishment of Course Objectives.

2.2 What does the Course Director expect from me?

Cadets are expected to execute their duties, both implicit and explicit, with the intellect and deportment of the commissioned officers they aspire to soon become. Always live the Cadet Honor Code and conduct yourself in a professional manner.

You are expected to arrive at each class meeting wholly prepared to discuss the day's material, including any pertinent questions you may have over the assigned reading or study problems. During class, you should remain engaged in the discussion at hand by taking meaningful notes, constructively contributing to lecture through positing or answering

questions as applicable, putting forth genuine thought and effort at boards, and confidently delivering briefings or presentations as required. As professionals and future commissioned leaders of character, you are expected to respect the instructor and the time allotted for each class meeting. As such, refrain from using laptops, iPads, etc. to view unrelated material during class. Leave cell phones, materials unrelated to NE450, and any other potential distractions outside the classroom. Stand and move to the rear of the classroom if you become drowsy. Quietly excuse yourself and return quickly from any necessary trip to the latrine.

2.3 Sexual Harassment Policy

Prevention and elimination of sexual harassment and sexual assault is the number one priority at the United States Military Academy. The Army "I. A. M. STRONG" campaign combats sexual harassment and sexual assault by engaging all Soldiers to "Intervene," "Act," and "Motivate" others in preventing sexual assaults before they occur. Sexual harassment and sexual assault are reprehensible, hurt other team members, and wound our Army. It is a betrayal of the trust inherent in serving in the Profession of Arms. Your instructor is dedicated to fostering a positive SHARP environment.

If you need to speak to anyone about SHARP, contact your instructor, the unit representative for Cadets Against Sexual Harassment and Assault (CASHA), or one of the following resources:

- West Point Sexual Assault 24/7 Helpline: (845) 659-7467
- DoD Safe Helpline: (877) 995-5247
- SHARP Victim Advocate: (845) 938-3532 or (845) 938-1938.

2.4 Chain of Command

Below is the chain of command for NE450. The Department of Physics and Nuclear Engineering has an open-door policy. Feel free to address questions, problems, or concerns through the chain of command.

- **Course Director, NE450**

MAJ Dave Pert

BH472A, x4297

- **Program Director, Nuclear Engineering**

Prof. Kenneth Allen

BH472B, x3548

- **Department Head, Physics and Nuclear Engineering**

COL Peter Chapman

BH480B, x3901

3 Course Objectives

Upon completion of this course, students should be able to:

- a. **Apply** physical concepts and principles to solve relevant problems from radiation interactions with matter related to the design, prompt, and residual effects of nuclear weapons.
- b. **Comprehend** empirical relationships and engineering tools used by the Department of Defense nuclear weapon effects communities.
- c. **Communicate** effectively, both verbally and in writing, the critical examination and analysis of engineering problems.
- d. **Demonstrate** knowledge of contemporary nuclear issues.

The NE450 course objectives are apart of a broader set of nuclear engineering program education objectives that satisfy both Accreditation Board for Engineering and Technology (ABET) requirements and USMA Academic Program Goals (APG).

3.1 Nuclear Engineering Program Educational Objectives (NE PEOs)

The nuclear engineering program educational objectives describe the career and professional accomplishments that the NE program prepares Cadets to achieve in a few years after graduation. The NE PEOs are designed to meet the needs of our primary constituency: the United States Army. The NE student outcomes are what Cadets are expected to know and be able to do upon graduation. They are given below.

1. As Army leaders of character, graduates solve complex, multi-disciplinary problems

for the Army and the Nation.

2. Graduates demonstrate the necessary leadership and teamwork skills to lead technical combined-arms Army teams solving problems in support of the Army mission.
3. Graduates are prepared to provide appropriate nuclear and radiological engineering expertise to the Army.
4. Graduates communicate effectively, orally and in writing; providing clear instructions to subordinates and clear feedback to supervisors.
5. Graduates and continue to develop and grow their intellect, professionalism, and character – as Army officers and as engineers.

3.2 Course Objectives Mapped to ABET Student Outcomes

The Nuclear Engineering Program Student Outcomes describe what Cadets are expected to know and be able to do at the time of graduation. The student outcomes relate to the skills, knowledge, and behaviors that a Cadet acquires during progression through the Nuclear Engineering program, as defined by ABET. Below are the stated NE program outcomes (1 - 7) which align directly with ABET accreditation student outcomes.

Cadets must attain:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. an ability to communicate effectively with a range of audiences;

4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgements, which must consider the impact of engineering solutions in global, economic, environmental , and social contexts;
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and,
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

The nuclear engineering program has determined NE450 best supports student outcomes 4 and 7. These outcomes are shown in the matrix below in Table 1 along with the embedded indicators that will be used to assess the outcomes against established rubrics.

Table 1: NE450 Course Objectives mapped to ABET Student Outcomes

Embedded Indicators	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Homework Sets							
Response Paper				X			
Extension Problems				X			X
Instructor Writs							
LO Poster							X
WPRs							X
TEE							

3.3 Course Objectives Mapped to Academic Program Goals

NE450 will be used to assess cadet attainment of the Academic Program Goals (APG) and What Graduates Can Do (WGCD) statements for the Core Engineering Sequence (CES). The APGs include the following selected WGCD Statements:

2.1 (Critical Thinking and Creativity) Identify the essential aspects of a situation and ask relevant questions.

2.4 (Critical Thinking and Creativity) Reason both quantitatively and qualitatively.

2.5 (Critical Thinking and Creativity) Think innovatively and accept risk to pursue solutions in the face of ambiguity.

5.1 (Science, Technology, Engineering, and Mathematics) Apply mathematics, science, and computing to model devices, systems, processes or behaviors.

5.4 (Science, Technology, Engineering, and Mathematics) Apply an engineering design process to create effective and adaptable solutions.

7.4 (Disciplinary Depth) Synthesize knowledge and concepts from across their chosen disciplines.

7.5 (Disciplinary Depth) Contribute disciplinary knowledge and skills as a part of a collaborative effort engaging challenges that span multiple disciplines.

Table 2 below depicts how NE450 introduces and/or reinforces the APG/WGCD statements listed above.

Table 2: NE450 Course Objectives mapped to Academic Program Goals

	APG/WGCD						
Embedded Indicators	2.1	2.4	2.5	5.1	5.4	7.4	7.5
Homework Sets		X		X	X		
Policy Paper	X	X				X	
Extension Problems	X	X	X	X	X	X	
Instructor Writs		X		X			
LO Poster	X		X		X		X
WPRs		X		X			
TEE		X		X			

4 Course Requirements and Grading

Assessment of course block objectives, NE student outcomes, and APG/WGCD statements will guide evaluations of reinforcement and extension of concepts, principles, and problem-solving.

1. **Reinforcement.** Cadets are required to know the underlying concepts and principles along with how to solve basic problems using those concepts and principles. Cadets demonstrate reinforcement knowledge by communicating baseline knowledge outlined in the lesson learning objectives, lesson vocabulary, and lesson notes. Evaluation mechanisms are primarily in the form of timed, written events such as instructor writs, WPRs, and the TEE. Problem sets may also be used to reinforce certain topics.
2. **Extension.** Synthesizing multiple concepts and principles beyond reinforcement is extension. Additionally, successful extension demonstrates a higher degree of understanding of the course material which requires creative and critical thinking to solve more-complex problems. Evaluation mechanisms will generally take the form of untimed events such as extension problems, laboratory reports, individual research/presentations, and bonus opportunities.

4.1 Course Point Allocation

Table 3 shows the allocation of course marks and the percentage of total for all graded events in NE450.

Term End Exam (TEE): The term end exam is a comprehensive reinforcement event which evaluates key learning objectives. Problems posed on TEE are both qualitative (explain, sketch, define) and quantitative (calculate, derive, solve) in nature. Authorized

Table 3: NE450 Course Marks Allocation

Basis of Grade	Possible Marks	Percentage of Total
Homework Sets (4 @ 125)	500	25%
Extension Problems (2 @ 150)	300	15%
Instructor Points	100	5%
Fundamental Knowledge WRITs (2 @ 50)	100	5%
WPRs (2 @ 250 each)	500	25%
TEE	500	25%
Total Marks Available	2000	100%

references are the NERD-P, NE450 Reference Card, and an approved calculator.

Problem Sets: The four assigned problem sets may serve as reinforcement and/or extension events. These problems may be more complex than text problems and may require synthesis of multiple concepts, principles, and ideas. Each set is worth 100 points. Due dates and times will be identified with each assignment.

Collaboration on problem sets is anticipated and should be used to refine one's understanding of the learning objectives encompassed by the problems as noted in each assignment. Accordingly, documentation is required IAW the Documentation of Academic Work guidebook and the memorandum "You and the Department of Physics and Nuclear Engineering." Both references are available on Canvas.

Written Partial Review (WPR): The two WPRs are reinforcement events which assess key learning objectives from corresponding blocks of instruction. Problems posed on the WPRs will be both qualitative (explain, sketch, define) and quantitative (calculate, derive, solve) in nature. Authorized references are the NERD-P, NE450 Reference Card, and an approved calculator.

Fundamental Nuclear Engineering Knowledge (FNEK) Writ: During the course you will be administered two 50-point fundamental knowledge writs. **The first will be given during Lesson 1 and the second will be given during Lesson 30.** The fun-

fundamental knowledge writs are designed to assess your retention of key nuclear engineering facts and equations you should know without the use of reference materials. As such, the fundamental knowledge writs will be reference-free with no calculator use permitted.

Extension Problems: During the course you will be assigned two extension level problems to be completed in formal groups established by your instructor. The extension problems are designed to assess your ability to synthesize multiple concepts and principles beyond material reinforcement. The extension problems are due according to the “Remarks” column in the course schedule. Both extension problems must be satisfactorily completed and turned in to receive course credit.

Bonus Points: Bonus points may be offered at the instructor’s discretion. The total bonus marks awarded to any one cadet may equal no more than 20 (1% of course grade). If assistance is received in completion of a bonus opportunity, documentation must conform to that in the DAW. No marks are awarded for copied bonus problems.

4.2 Grade Cutoffs

Each cadet will earn numerical marks (points) in accordance with his or her performance on each graded event throughout the term. Based on the total percentage of earned marks in the course, Cadets will be awarded a letter grade. The maximum cutoff percentage for each letter grade is shown in the Table 4.

Grades are usually available in CIS/Canvas as soon as they are posted by the instructor. In the case of major graded events (WPRs and TEE), grades will be visible to cadets after approval by the head of the Department of Physics and Nuclear Engineering.

A+	94.0%	C+	73.0%
A	90.5%	C	69.5%
A-	87.0%	C-	66.0%
B+	83.5%	D	60.0%
B	80.0%	F	< 60.0%
B-	76.5%		

Table 4: Letter grade cutoffs by percentage

4.3 Late Assignment Policy

A Cadet who believes he/she will fail to meet a suspense for submission of a graded assignment will inform the instructor prior to the missed deadline. Failure of a Cadet to notify the instructor of such an event may result in a negative Cadet Observation Report (COR). **Additionally, turning in homework late without prior notification to the instructor will result in a 25% reduction per calendar day.**

5 Course Resources

5.1 Course Text

The textbook resources for this course are listed below. All of the course material has been combined into a single pdf located on our Canvas page. Please download this pdf prior to the start of the semester.

1. *The Effects of Nuclear Weapons*, (3rd Edition, 1977) by Samuel Glasstone and Philip J. Dolan.
2. Various readings from authors, all of which are available in the course resources.

5.2 Course Outline

All lesson assignments are found in Lesson Summaries (published separately). For each assignment, Cadets may be tasked to **Scan**, **Read**, or **Study** certain material(s). Cadets are required to understand the learning objectives, study and know the lesson notes and vocabulary, and are provided a list of additional problem(s) that they can choose to do for practice and reinforcement of concepts. Specific definitions of these tasks and items are given below.

1. **Scan** tasks are those which require Cadets to become familiar with a given set of material prior to a lecture. Often this material will be review from NE300 or fundamental knowledge. The cadet should be able to understand how it supports concepts in follow on NE courses.
2. **Read** tasks are those which require Cadets to read a given set of material prior to a lecture in order to gain insights which will facilitate better understanding of the

material as well as any “study” requirements. Reading assignments are typically given to provide depth, clarity, and examples beyond those provided during lecture and support specific learning objectives.

3. **Study** tasks are those which require Cadets to read and recall certain material in order to demonstrate the ability to comprehend, analyze, and synthesize material by solving assigned problems and/or discussing and evaluating common applications.
4. **Lesson Learning Objectives (LOs)** are key principles which guide Cadets’ study and preparation for each lesson. Studying nuclear engineering requires much more than merely reading text(s). Contained within each text reading assignment are derivations and example problems. After understanding the LOs from the text, Cadets review model problems and attempt to solve study problems. The LOs use two types of highlighted verbs which are defined in the following paragraphs.
 - (a) **List, explain, sketch, draw,** and/or **describe** are used to indicate an objective which is evaluated by short-answer questions. In response to such questions, Cadets must correctly state pertinent facts about a given situation and then provide a conclusion based on those facts. Cadets may be required to sketch a graph or process to help illustrate the conclusion.
 - (b) **Calculate, construct, solve,** and/or **derive** are used to indicate an objective which is evaluated by methods used above or by performing algebraic, diagrammatic, and/or numerical calculations. Cadets may be required to synthesize multiple concepts in order to solve a problem.

5.3 Calculators

The only calculators authorized for use during in-class, graded events are those approved for use on the Fundamentals of Engineering Examination. A current list is available at [NCEES Website](#). If in doubt, use the issued TI-30XIIS calculator.

5.4 Canvas

Course material for NE450 can be accessed online using the Canvas interface. Canvas will provide you information about assignments, grades, etc. and is updated frequently. You will find all course administrative publications (i.e., syllabus, course outline, course schedule, reference cards, NERD-P, and other supplementary materials), lesson information, links to other related materials, and more on the Canvas page. Instructions for accessing Canvas will be provided by your instructor.

5.5 Examination Materials

A sterile NE450 Reference Data Card and NERD-P will be provided to you for each examination. The reference card and NERD-P contain the equations, data, and charts that you will need as reference for all examinations.

It is recommended you study, do your homework, and do your in-class board work with your copy of the reference cards and the NERD-P. We encourage you to make notes (identify parameters, write conditions, write units, etc.) on your reference cards and NERD-P as you study. This will provide you a greater degree of familiarity with these materials when you receive a clean copy for the examinations.

5.6 Additional Instruction (AI)

While not required, Cadets are encouraged to schedule additional instruction (AI) with their instructor. The primary purpose of AI is to provide answers to specific questions a Cadet may have after attempting to work a problem set on their own. Additional instruction may also be scheduled to assist in clarifying concepts from the reading or answers from returned, graded assignments.

Once an AI appointment is requested and confirmed by the instructor, it becomes that Cadet's place of duty unless rescheduled in advance. The Cadet should come to AI prepared with detailed questions in order to facilitate efficient use of AI time.

6 Documentation of Academic Work (DAW)

The purpose of documentation is to give credit where credit is due by delineating original work from portions which originated elsewhere. Proper documentation identifies the source(s) of borrowed ideas, quotations, and assistance in the preparation of assignments (DAW, paragraph II.A.1). Proper documentation reflects good scholarship and personal integrity.

Any homework assignment prepared outside of class that Cadets turn in for grade must be documented in accordance with the DAW and must include a cover sheet. Documents submitted with improper documentation and/or missing a signed cover sheet will be returned back to the cadet ungraded for clarification. Furthermore, improperly documented assignment submissions will be marked “late” in accordance with the policy for late assignments until the assignment is resubmitted.

Collaboration is highly encouraged; however, each Cadet must document all substantive assistance received. Endnotes should clearly distinguish the author’s personal effort from that of his/her collaborators, such as in the following example:

Jerrad P. Auxier, D-1, '20; Matthew T. Cook, D-2, '19; George K. Schweitzer, H-3, '21; collaboration with the author, verbal and written discussion, West Point, NY, 22 September 2018. We worked through the derivation of the flux for the diffusion equation as an informal group. CDT Cook set up the general equation. CDT Schweitzer added the proper form of the Laplacian. CDT Auxier determined the correct source boundary condition.

Properly documented copying is subject to an academic cut of no less than 50% of the achieved score for that portion of the work that is copied. For bonus problems, copying will result in no credit for the problem(s) copied.

If you are ever in doubt about how to correctly document your work, refer to the most

current edition of the *Documentation of Academic Work* (DAW) guidebook published by the Dean's office. The DAW is also posted on the course Canvas site. If still uncertain after consulting the DAW, ask your instructor for further guidance.

6.1 Documentation of Previous Work

If a cadet reuses or resubmits work in accordance with course policies, he or she must still take efforts to avoid what is known as “self-plagiarism”: the reuse of your own previous work in another venue while presenting it as new work. Cadets are required to cite and document that work like any other source cited or assistance received IAW the *Documentation of Academic Work* [1].

6.2 Course Common Knowledge

Documentation is not required for ideas and materials regarded as common knowledge. A list of such material follows.

1. Any equation, numerical value, or diagram listed on the NE450 Reference Data Card or contained within the Nuclear Engineering Reference Data Pamphlet (NERD-P).
2. Any equation or numerical value found in the required course textbook. The figures and text contained within the textbooks are not considered common knowledge and must be appropriately documented.
3. Material provided by the instructor, to include: board notes taken by the Cadet during a formal class meeting; example problem solutions worked out in a formal class meeting; transcribed discussions, explanations, or definitions given during a formal class.

4. Any discussions with your instructor are considered common knowledge; discussions with other faculty NOT identified as an NE450 instructor must be documented.

Materials and ideas outside of those directly listed above are not common knowledge and must be documented IAW the DAW. Furthermore, the use of any diagram outside of the NERD-P is not common knowledge and must be documented. Finally, any notes copied from another Cadet (e.g., CDT Smith is absent during lesson 3 and copies CDT Jones' notes) are not considered common knowledge and must be cited appropriately.

7 Works Cited

- [1] Office of the Dean Academic Affairs and Registrar Services. *Documentation of Academic Work*. United States Military Academy, 2017.

ENCLOSURES

Enclosure 1: NE450 AY26-2 Schedule and Calendar

Enclosure 2: NE450 AY26-2 Course Outline

Enclosure 3: NE450 AY26-2 RDC



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2026 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
AY 26-2 Course Assessment



ARMY
WEST POINT™
NUCLEAR ENGINEERING

MAJ Dave Pert
NE450 Course Director

Agenda

AY 26-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE450 Assessment Process
- AY26-2 Course Assessment
- Recommended Changes for AY27-1

Program Director's Areas of Focus

Academic Year 2025

NE450

- ✓ NE Outcomes assessment are: 4, 7
- ✓ Implement CAT recommendations (complete)
- ✓ Develop sustained solution for text requirements for course (on-going)
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up (complete)

Sustain & Improve from AY26-1 (for 26-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Guest speakers when available.
- Improve:
 - Find a replacement for MINES or redesign EP2. DTRA has made some changes to MINES and it is only available on NIPR. Using the NIPR remote desktop may solve this. This is left as an exercise from the next CD. (Replaced with NukeMap critical analysis)
 - Combine reactors and enrichment into a single lesson focused on the pathways to weapons grade material. (Complete)
 - Consider removal of radius of sustained burned calculation (Considered, now in agreement)

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

AYT	#SECT/SIZE	CPBLTY	ENRLD	WAIT	SEATS	CLOSED	DETAILS
2026 - 2	3	20	60	59	1	1	N Hours
2027 - 1	3	18	54	40	1	14	N Hours
2027 - 2	4	18	72	57	0	15	N Hours
2028 - 1	3	18	54	33	0	21	N Hours
2028 - 2	5	18	90	79	0	11	N Hours
2029 - 1	3	18	54	2	0	52	Y Hours
2029 - 2	3	18	54	41	0	13	Y Hours

Recommend name change to: Nuclear Weapons

Recommend scope change to: This course focuses on the operation, effects, and policy of nuclear weapons. Specific topics include special nuclear material, fission mechanics, fusion mechanics, thermal radiation, blast effects, initial and delayed radiation, biological effects, conventional nuclear integration, and nuclear policy. Extension problems with design components apply the concepts presented to the solution of practical problems.

Recommend capping AY 27-2 sections to 3.



NE450—CAT Briefing AY26-2

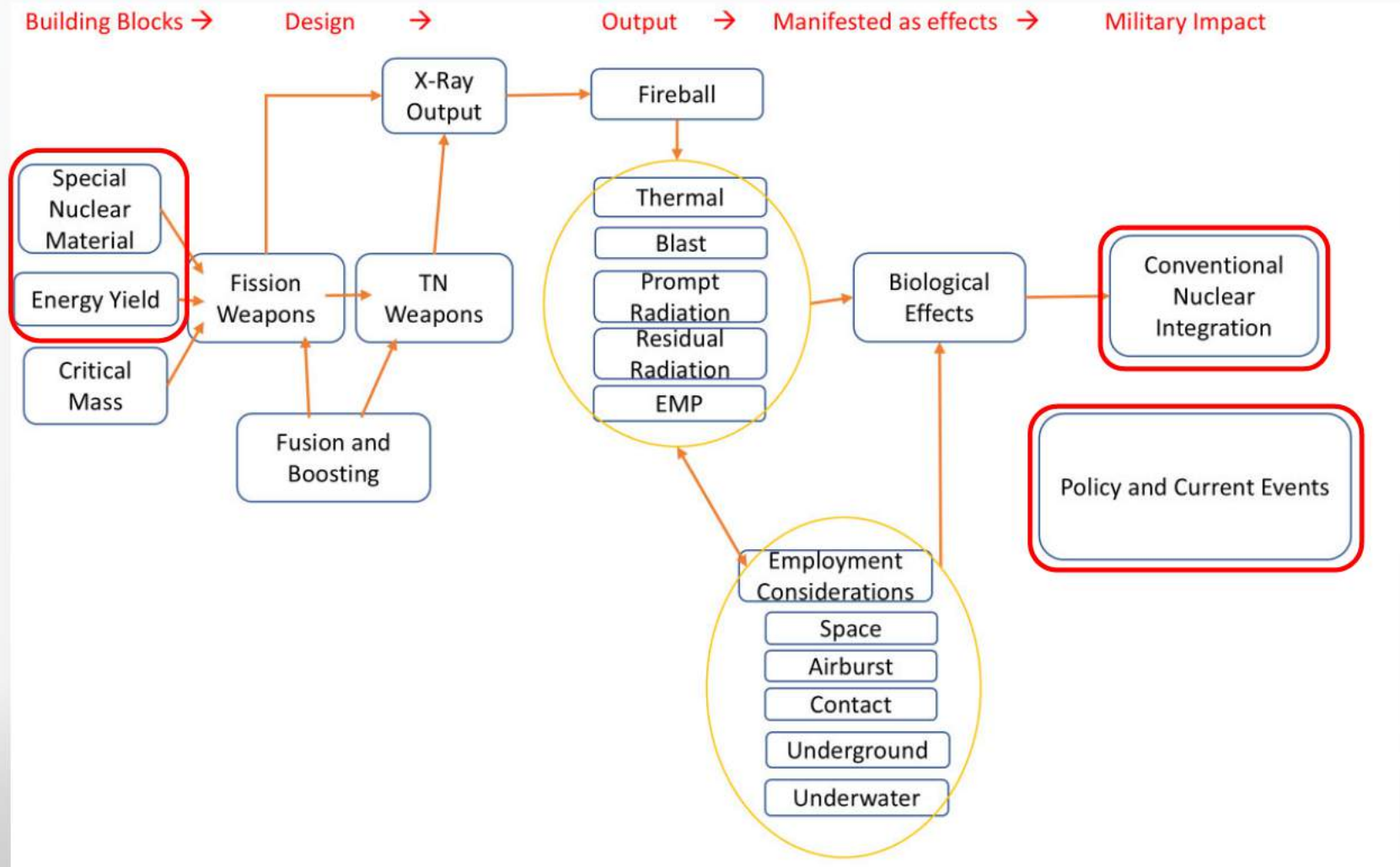
Department of Physics & Nuclear Engineering, USMA

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Map/Topic Coverage



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Indicator	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Problem Sets							
Extension Problems				X			X
FNEK Writs							
WPRs							X
TEE							

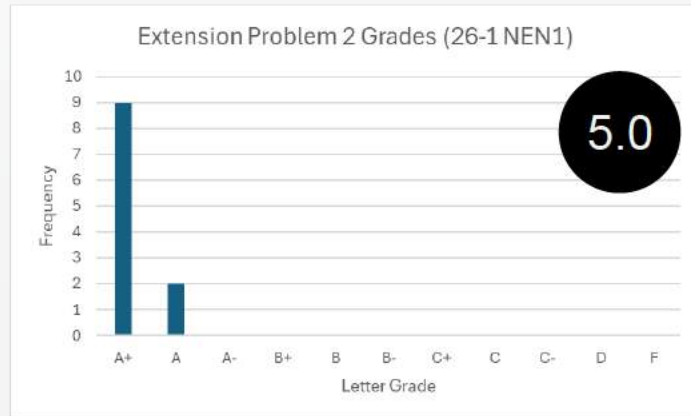
Best Assessment Instrument

- SO 4: EP 2 (150 Points)
- SO 7:
 - EP 1 (150 Points)
 - WPR 1 Thermonuclear Calculation (60 Points)
 - WPR 2 Initial Rad Dose Calculation (30 Points)

Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

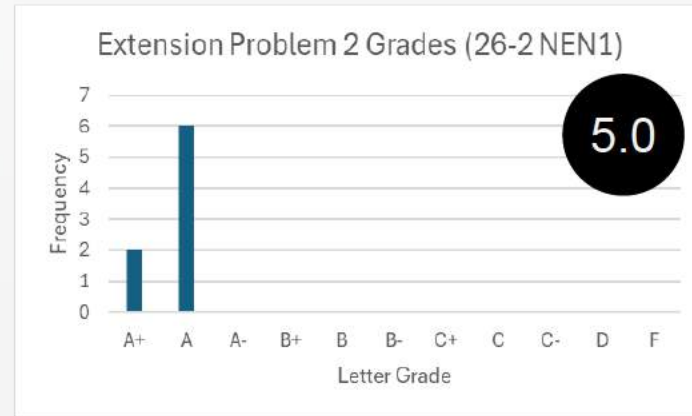
Indicator: Extension Problem 2

AY 26-1



Excellent. NEN1 cadets continue to impress, though they struggled a bit when the MINES tool was unavailable this semester.

AY 26-2

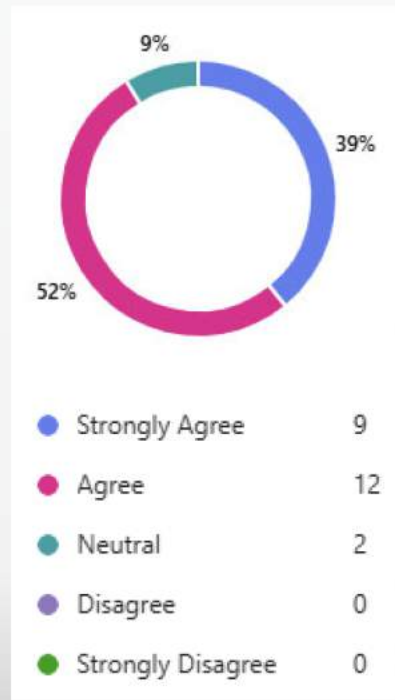


Excellent. EP2 was rewritten for 26-2 due to lack of MINES access. Cadets were tasked with validating NukeMap against NE450 effects calculations and planning maneuver through fallout.

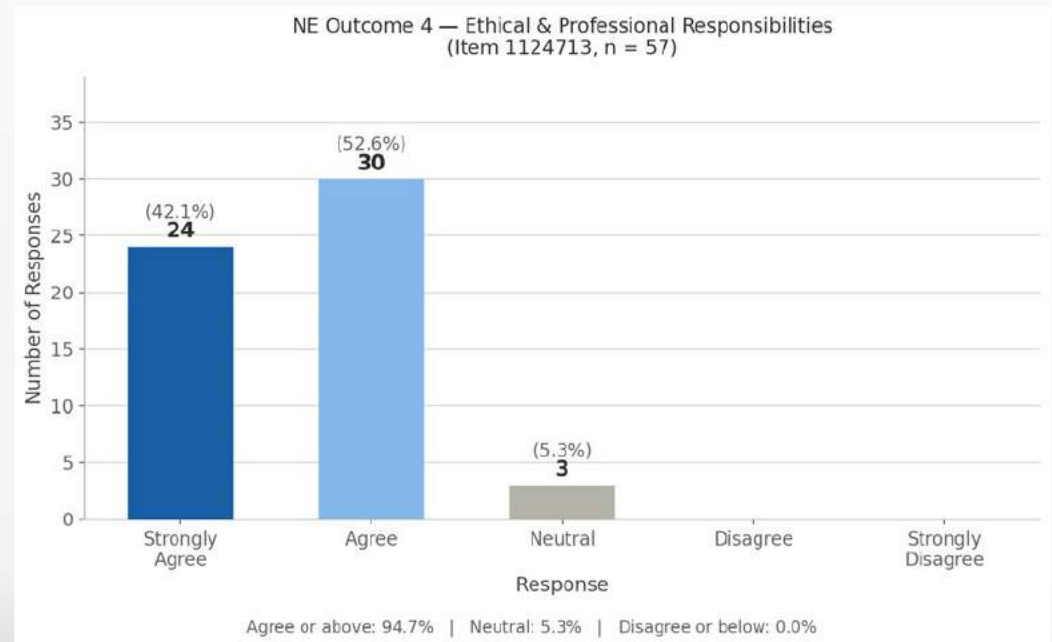
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Course-Level Feedback

AY 26-1



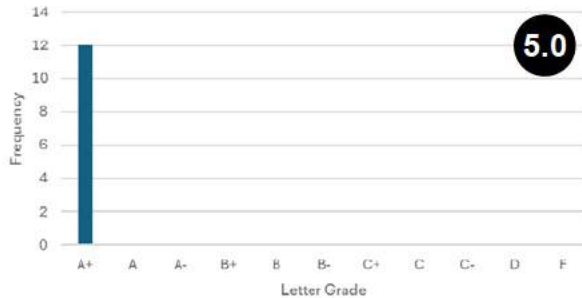
AY 26-2



Cadets must have the ability to acquire and apply new knowledge as needed, using appropriate learning strategies (SO7)

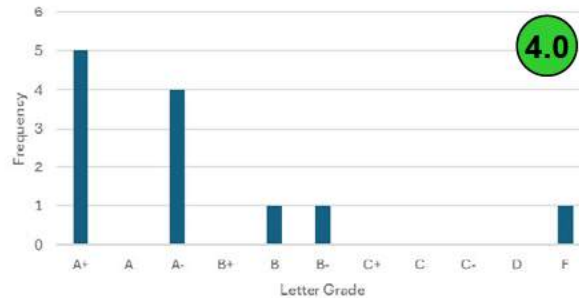
AY 26-1

SO 7 (Extension Problem 1)



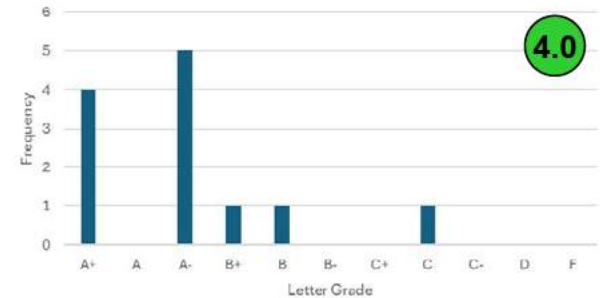
5.0

SO 7 (WPR 1 Fusion)



4.0

SO 7 (WPR 2 Initial Rad Dose)



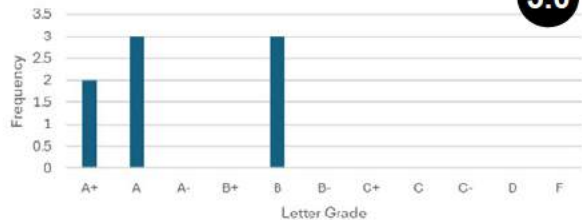
4.0

Block
4.0

Good. The NEN1 cadets excelled at the first extension problem. As in past years, the longer calculations challenge some of them. They struggle with the thermonuclear calculations because it is not an equation directly from the reference card. Initial rad dose has several steps, and many of them will miss one or two steps along the way.

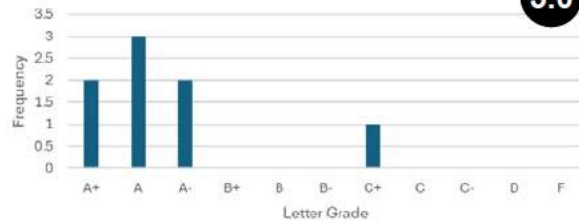
AY 26-2

SO 7 (Extension Problem 1)



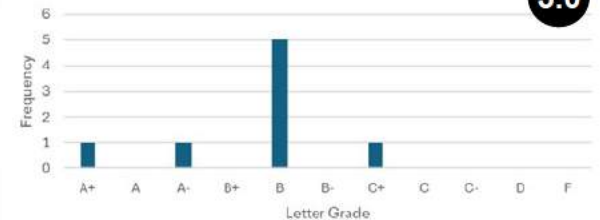
5.0

SO 7 (WPR 1 Fusion)



5.0

SO 7 (WPR 2 Initial Rad Dose)



5.0

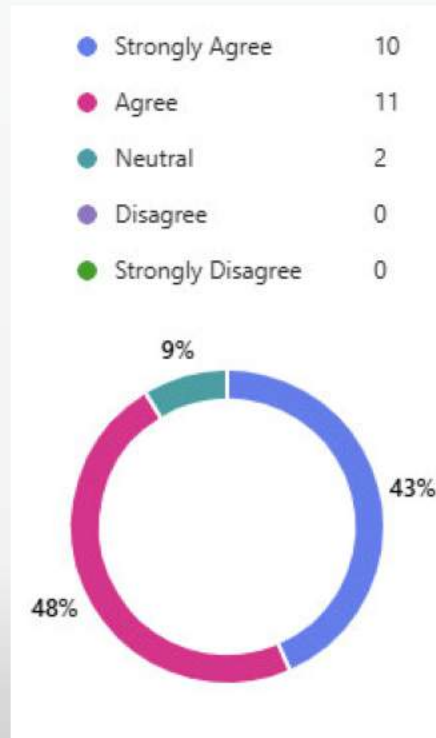
Block
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Excellent. The NEN1 cadets were challenged with EP1 and some were intermixed in groups with non-majors. Thermonuclear calculations improved overall. Initial rad dose was worth less points overall, so mistakes may have had a larger impact on letter grade calculations.

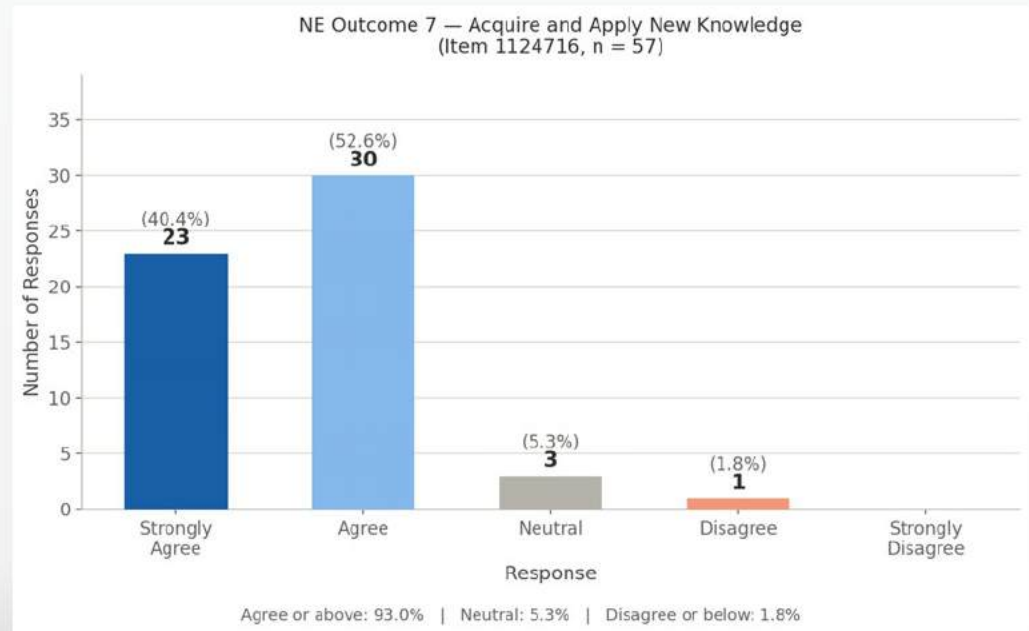
Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

Course-Level Feedback

AY 26-1



AY 26-2



NE Outcomes Assessment

AY	NE Outcome	Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
26-1	4	4.3	5.0	4.5	EP2 was difficult to assess during AY26-1, as the MINES tool was unavailable which required some paring down of the assignment and limited time to discuss ethical considerations around nuclear use.
26-2		4.4	5.0	4.5	The EP2 rewrite is a good start to bringing in a multi-dimensional approach to weapon effects understanding and situational planning from an engineering and (partially) an ethical perspective. Further refinement will improve this indicator.
26-1	7	4.3	4.0	4.5	The NEN majors performed well on the more complex problems during timed, graded events. There was one F on the WPR1 indicator, but the most frequent grade was A or A+.
26-2		4.3	5.0	4.5	Complex problems were generally well handled. There was a growing apathy as the end of their curriculum approached that will be addressed in following semesters.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5 ● -- Excellent
- 4 ● -- Good
- 3 ● -- Acceptable
- 2 ● -- Marginally Acceptable
- 1 ● -- Unacceptable

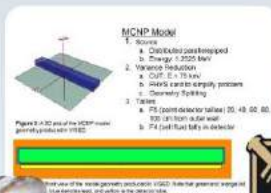
Sustain & Improve from AY26-2 (for 27-1)

Sustain Theme	Sustain Summary
Extension Problems (EP1 & EP2)	Mentioned by 15+ cadets as top sustain. EP2 received especially strong praise as practical and engaging.
Classified/CNWDI Briefing	Repeatedly cited as a highlight and motivator. Many requested additional classified sessions throughout the semester.
Pre-Class Surveys	Appreciated as a primer for lessons; most cadets found the format useful for engagement.
Problem Sets & Board Problems	Problem sets rated as fair difficulty and well-aligned to WPRs. Board problems cited as essential for learning.
Current Events & Policy Discussion	Tying nuclear policy to current events was consistently praised for adding real-world context.
Interactive Equation Sheet (IRDC)	Specifically called out as an innovative resource that should be shared across physics/NE courses.
Points Earn-Back AI Policy	Strongly appreciated; cadets felt it encouraged learning from mistakes. Several said all courses should adopt it.
Instructor Energy & Enthusiasm	Near-universal praise. Cadets noted enthusiasm was critical for keeping second-semester firsties engaged.
Pace and Lesson Sequencing	The arc from heavy math in Block 1 to "beer math" at end of semester was viewed as appropriate and well-designed.
Cat Pictures & Humor	Mentioned multiple times as a small but meaningful touch that improved classroom atmosphere.
Improve Theme	Improve Summary
More Board Problems	~10 cadets requested more in-class practice and specifically asked to interleave board problems with content
Earlier Problem Set Feedback	~7 cadets noted that receiving PS grades before the WPR would allow them to correct errors before the exam.
EP1 Coding Guidance	~5 cadets found EP1 coding requirements unclear or inaccessible, especially for those without prior coding experience.
Laptop Policy	~5 cadets advocated for restricting laptop use, arguing it would improve focus, class culture, and responsiveness.
Solution Accessibility	~5 cadets wanted hand-written, step-by-step solutions for suggested problems.
More Classified Content	~5 cadets wanted more SCIF/classified lessons throughout the semester, not just one.
AI / Gradescope Grading	~4 cadets preferred human grading feedback; found AI grading insufficient for understanding where they went wrong.
Pre-Class Survey Format	~4 cadets noted that pre-class surveys took up too much class time, or that answers were covered in slides anyway.
WPR 2 Timing	1 cadet specifically suggested moving WPR 2 before spring break to avoid the long gap.
Side Tracks / Focus	~3 cadets mentioned frequent tangents to niche topics as diluting study time and making it hard to separate testable content from enrichment.

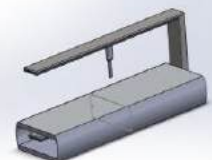
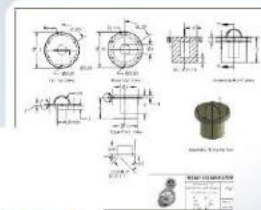
Resource Requirements

- Hoberman Sphere (acquired)
- 1000 mouse traps
- 1000 ping pong ball
- Large open room

Questions?



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UNITED STATES MILITARY ACADEMY
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TAB B

AY2025 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
AY 26-2 Course Assessment



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NUCLEAR ENGINEERING

MAJ Dave Pert
NE450 Course Director

Agenda

AY 26-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE450 Assessment Process
- AY26-2 Course Assessment
- Recommended Changes for AY27-1

Program Director's Areas of Focus

Academic Year 2025

NE450

- ✓ NE Outcomes assessment are: 4, 7
- ✓ Implement CAT recommendations (complete)
- ✓ Develop sustained solution for text requirements for course (on-going)
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up (complete)

Sustain & Improve from AY26-1 (for 26-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Guest speakers when available.
- Improve:
 - Find a replacement for MINES or redesign EP2. DTRA has made some changes to MINES and it is only available on NIPR. Using the NIPR remote desktop may solve this. This is left as an exercise from the next CD. (Replaced with NukeMap critical analysis)
 - Combine reactors and enrichment into a single lesson focused on the pathways to weapons grade material. (Complete)
 - Consider removal of radius of sustained burned calculation (Considered, now in agreement)

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

AYT	#SECT/SIZE	CPBLTY	ENRLD	WAIT	SEATS	CLOSED	DETAILS
2026 - 2	3	20	60	59	1	1	N Hours
2027 - 1	3	18	54	40	1	14	N Hours
2027 - 2	4	18	72	57	0	15	N Hours
2028 - 1	3	18	54	33	0	21	N Hours
2028 - 2	5	18	90	79	0	11	N Hours
2029 - 1	3	18	54	2	0	52	Y Hours
2029 - 2	3	18	54	41	0	13	Y Hours

Recommend name change to: Nuclear Weapons

Recommend scope change to: This course focuses on the operation, effects, and policy of nuclear weapons. Specific topics include special nuclear material, fission mechanics, fusion mechanics, thermal radiation, blast effects, initial and delayed radiation, biological effects, conventional nuclear integration, and nuclear policy. Extension problems with design components apply the concepts presented to the solution of practical problems.

Recommend capping AY 27-2 sections to 3.



NE450—CAT Briefing AY26-2

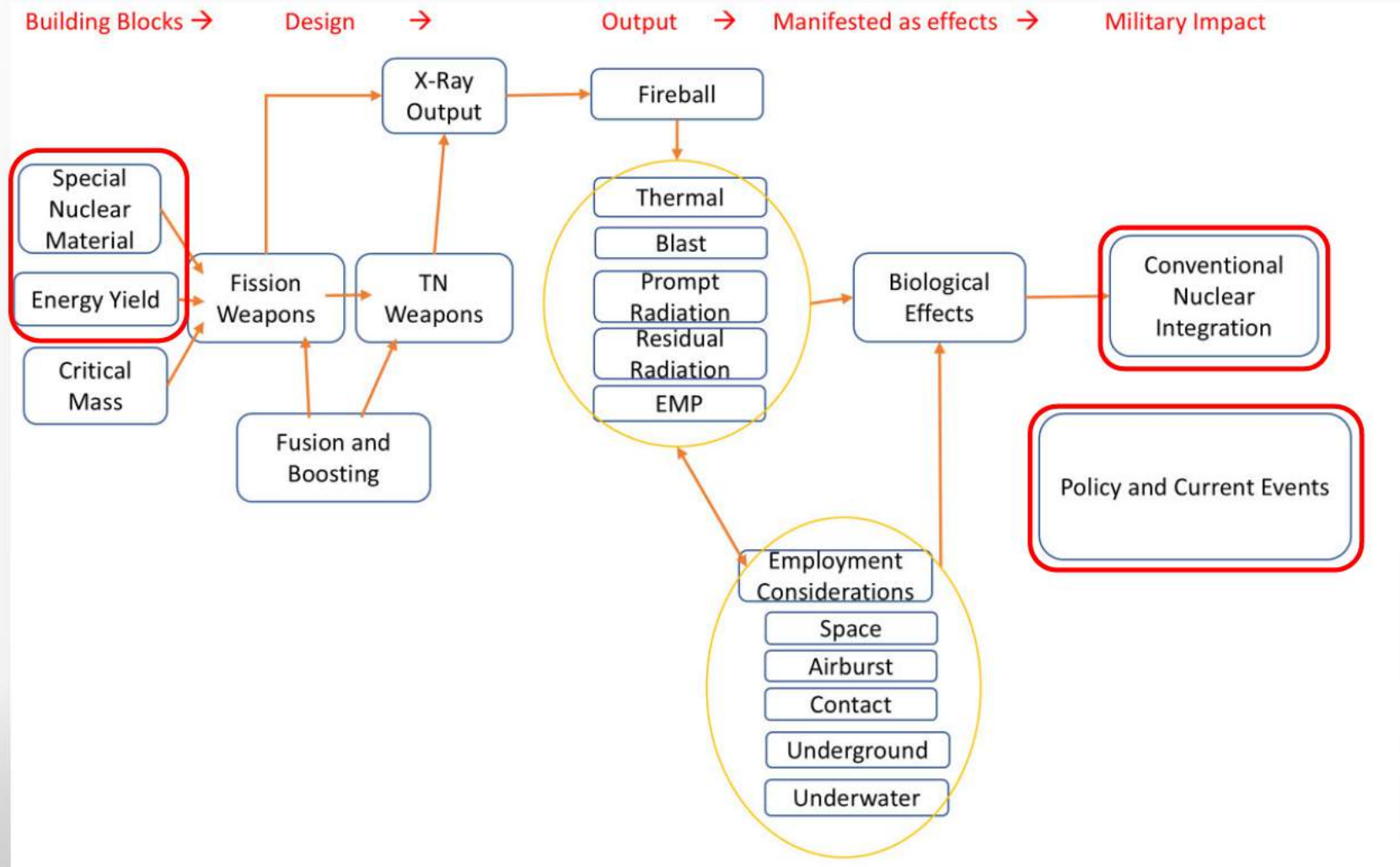
Department of Physics & Nuclear Engineering, USMA

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Map/Topic Coverage



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Indicator	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Problem Sets							
Extension Problems				X			X
FNEK Writs							
WPRs							X
TEE							

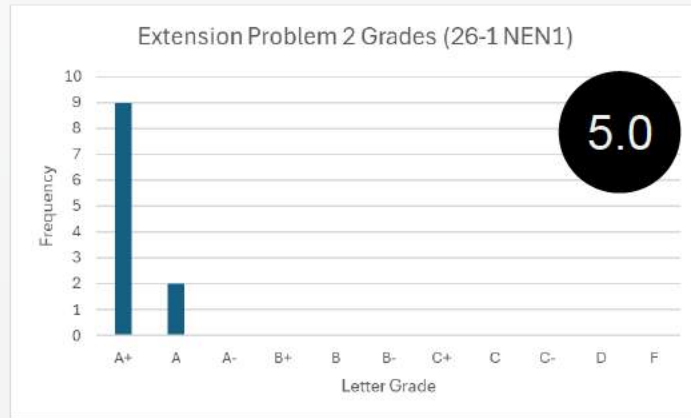
Best Assessment Instrument

- SO 4: EP 2 (150 Points)
- SO 7:
 - EP 1 (150 Points)
 - WPR 1 Thermonuclear Calculation (60 Points)
 - WPR 2 Initial Rad Dose Calculation (30 Points)

Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

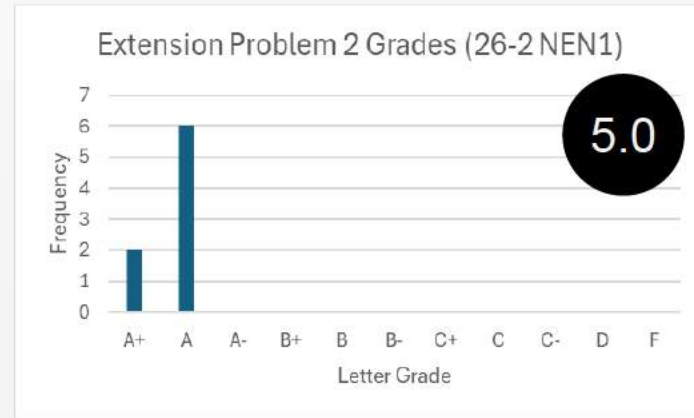
Indicator: Extension Problem 2

AY 26-1



Excellent. NEN1 cadets continue to impress, though they struggled a bit when the MINES tool was unavailable this semester.

AY 26-2

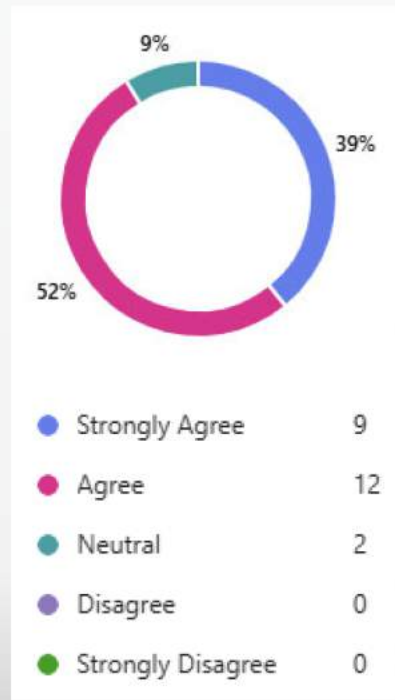


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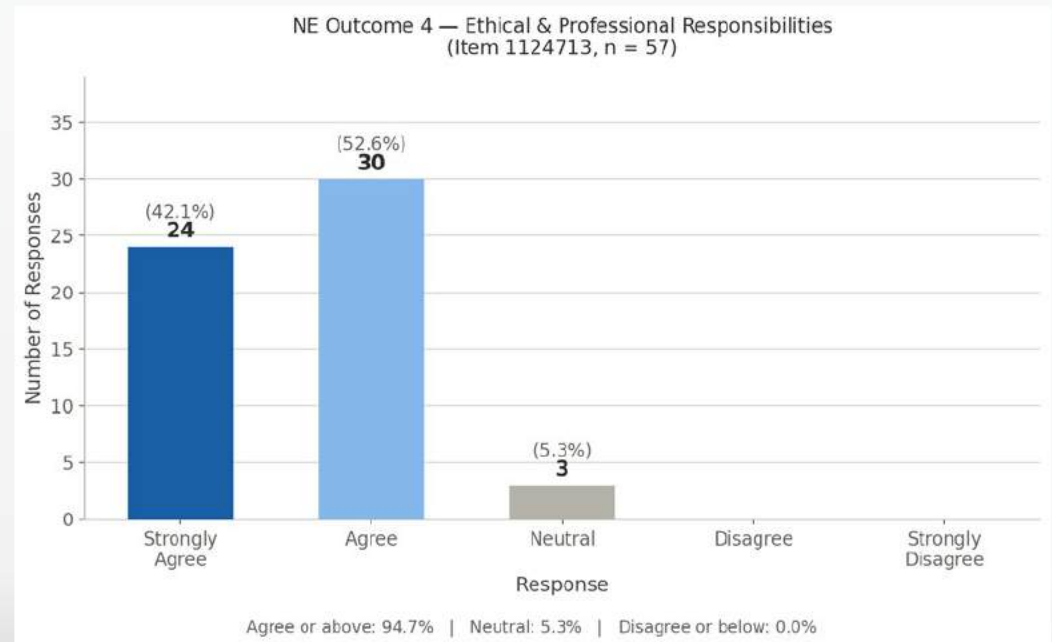
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Course-Level Feedback

AY 26-1



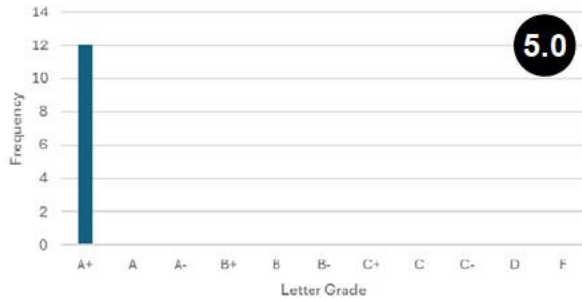
AY 26-2



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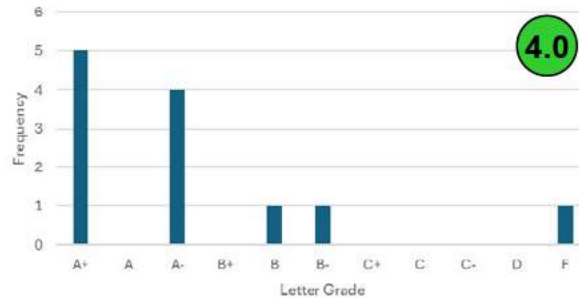
AY 26-1

SO 7 (Extension Problem 1)



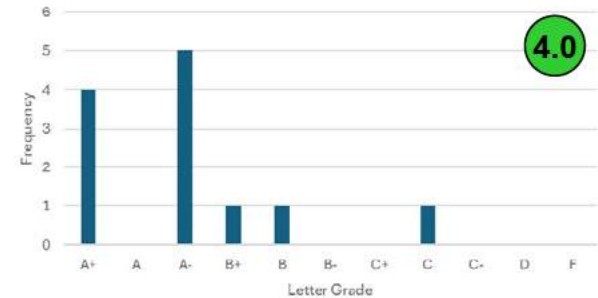
5.0

SO 7 (WPR 1 Fusion)



4.0

SO 7 (WPR 2 Initial Rad Dose)



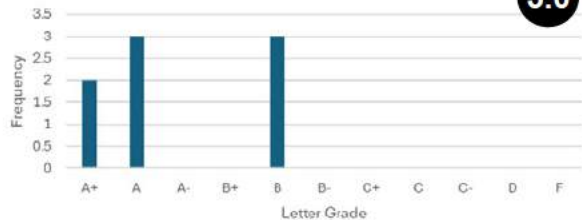
4.0

Block
4.0

Good. The NEN1 cadets excelled at the first extension problem. As in past years, the longer calculations challenge some of them. They struggle with the thermonuclear calculations because it is not an equation directly from the reference card. Initial rad dose has several steps, and many of them will miss one or two steps along the way.

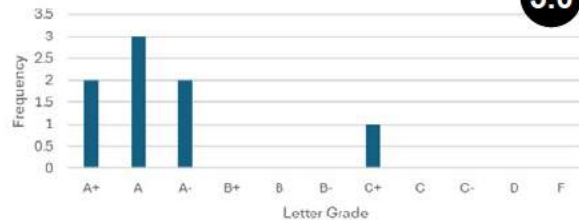
AY 26-2

SO 7 (Extension Problem 1)



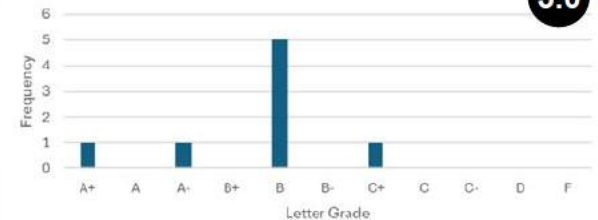
5.0

SO 7 (WPR 1 Fusion)



5.0

SO 7 (WPR 2 Initial Rad Dose)



5.0

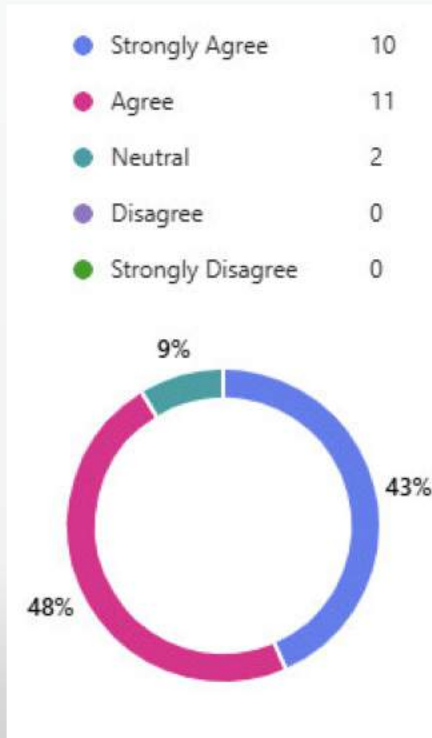
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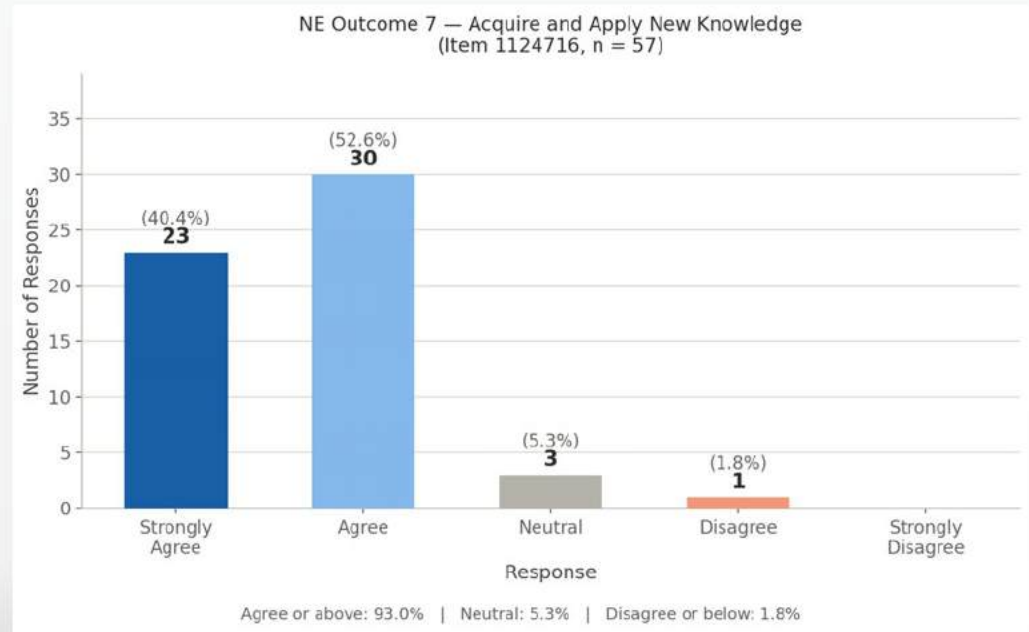
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Course-Level Feedback

AY 26-1



AY 26-2



NE Outcomes Assessment

AY	NE Outcome	Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
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KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

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- 5 ● -- Excellent
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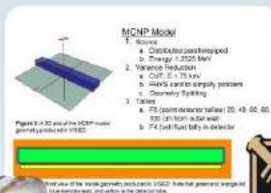
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Current Events & Policy Discussion	Tying nuclear policy to current events was consistently praised for adding real-world context.
Interactive Equation Sheet (IRDC)	Specifically called out as an innovative resource that should be shared across physics/NE courses.
Points Earn-Back AI Policy	Strongly appreciated; cadets felt it encouraged learning from mistakes. Several said all courses should adopt it.
Instructor Energy & Enthusiasm	Near-universal praise. Cadets noted enthusiasm was critical for keeping second-semester firsties engaged.
Pace and Lesson Sequencing	The arc from heavy math in Block 1 to "beer math" at end of semester was viewed as appropriate and well-designed.
Cat Pictures & Humor	Mentioned multiple times as a small but meaningful touch that improved classroom atmosphere.
Improve Theme	Improve Summary
More Board Problems	~10 cadets requested more in-class practice and specifically asked to interleave board problems with content
Earlier Problem Set Feedback	~7 cadets noted that receiving PS grades before the WPR would allow them to correct errors before the exam.
EP1 Coding Guidance	~5 cadets found EP1 coding requirements unclear or inaccessible, especially for those without prior coding experience.
Laptop Policy	~5 cadets advocated for restricting laptop use, arguing it would improve focus, class culture, and responsiveness.
Solution Accessibility	~5 cadets wanted hand-written, step-by-step solutions for suggested problems.
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AI / Gradescope Grading	~4 cadets preferred human grading feedback; found AI grading insufficient for understanding where they went wrong.
Pre-Class Survey Format	~4 cadets noted that pre-class surveys took up too much class time, or that answers were covered in slides anyway.
WPR 2 Timing	1 cadet specifically suggested moving WPR 2 before spring break to avoid the long gap.
Side Tracks / Focus	~3 cadets mentioned frequent tangents to niche topics as diluting study time and making it hard to separate testable content from enrichment.

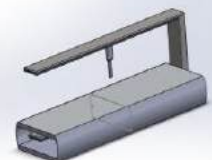
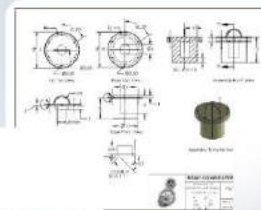
Resource Requirements

- Hoberman Sphere (acquired)
- 1000 mouse traps
- 1000 ping pong ball
- Large open room

Questions?



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TAB C

AY2024 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
AY 26-2 Course Assessment



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MAJ Dave Pert
NE450 Course Director

Agenda

AY 26-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE450 Assessment Process
- AY26-2 Course Assessment
- Recommended Changes for AY27-1

Program Director's Areas of Focus

Academic Year 2025

NE450

- ✓ NE Outcomes assessment are: 4, 7
- ✓ Implement CAT recommendations (complete)
- ✓ Develop sustained solution for text requirements for course (on-going)
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up (complete)

Sustain & Improve from AY26-1 (for 26-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Guest speakers when available.
- Improve:
 - Find a replacement for MINES or redesign EP2. DTRA has made some changes to MINES and it is only available on NIPR. Using the NIPR remote desktop may solve this. This is left as an exercise from the next CD. (Replaced with NukeMap critical analysis)
 - Combine reactors and enrichment into a single lesson focused on the pathways to weapons grade material. (Complete)
 - Consider removal of radius of sustained burned calculation (Considered, now in agreement)

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

AYT	#SECT/SIZE	CPBLTY	ENRLD	WAIT	SEATS	CLOSED	DETAILS
2026 - 2	3	20	60	59	1	1	N Hours
2027 - 1	3	18	54	40	1	14	N Hours
2027 - 2	4	18	72	57	0	15	N Hours
2028 - 1	3	18	54	33	0	21	N Hours
2028 - 2	5	18	90	79	0	11	N Hours
2029 - 1	3	18	54	2	0	52	Y Hours
2029 - 2	3	18	54	41	0	13	Y Hours

Recommend name change to: Nuclear Weapons

Recommend scope change to: This course focuses on the operation, effects, and policy of nuclear weapons. Specific topics include special nuclear material, fission mechanics, fusion mechanics, thermal radiation, blast effects, initial and delayed radiation, biological effects, conventional nuclear integration, and nuclear policy. Extension problems with design components apply the concepts presented to the solution of practical problems.

Recommend capping AY 27-2 sections to 3.



NE450—CAT Briefing AY26-2

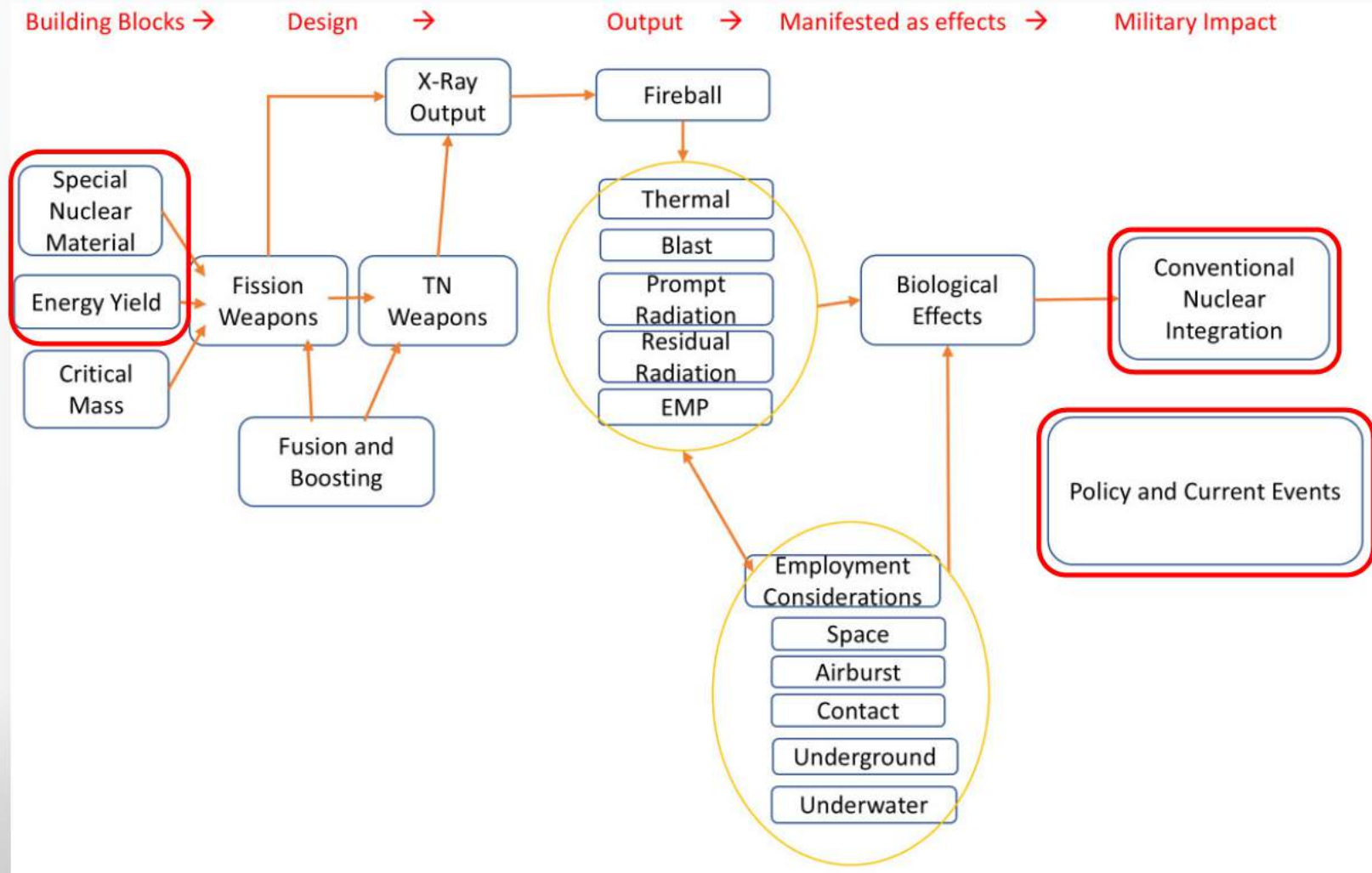
Department of Physics & Nuclear Engineering, USMA

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Map/Topic Coverage



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Indicator	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Problem Sets							
Extension Problems				X			X
FNEK Writs							
WPRs							X
TEE							

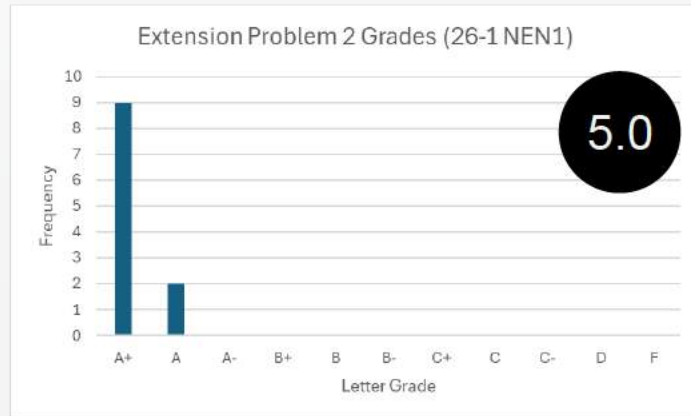
Best Assessment Instrument

- SO 4: EP 2 (150 Points)
- SO 7:
 - EP 1 (150 Points)
 - WPR 1 Thermonuclear Calculation (60 Points)
 - WPR 2 Initial Rad Dose Calculation (30 Points)

Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

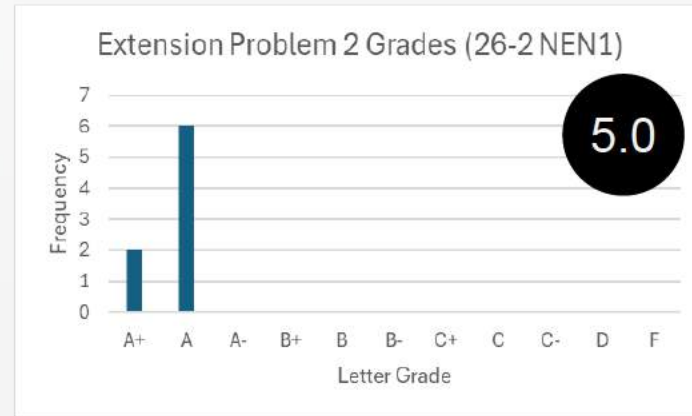
Indicator: Extension Problem 2

AY 26-1



Excellent. NEN1 cadets continue to impress, though they struggled a bit when the MINES tool was unavailable this semester.

AY 26-2

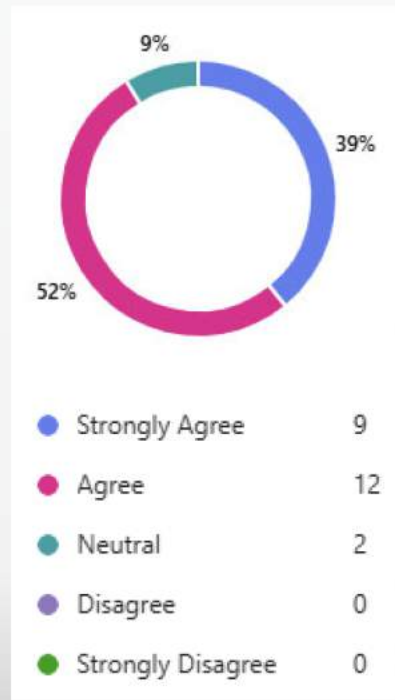


Excellent. EP2 was rewritten for 26-2 due to lack of MINES access. Cadets were tasked with validating NukeMap against NE450 effects calculations and planning maneuver through fallout.

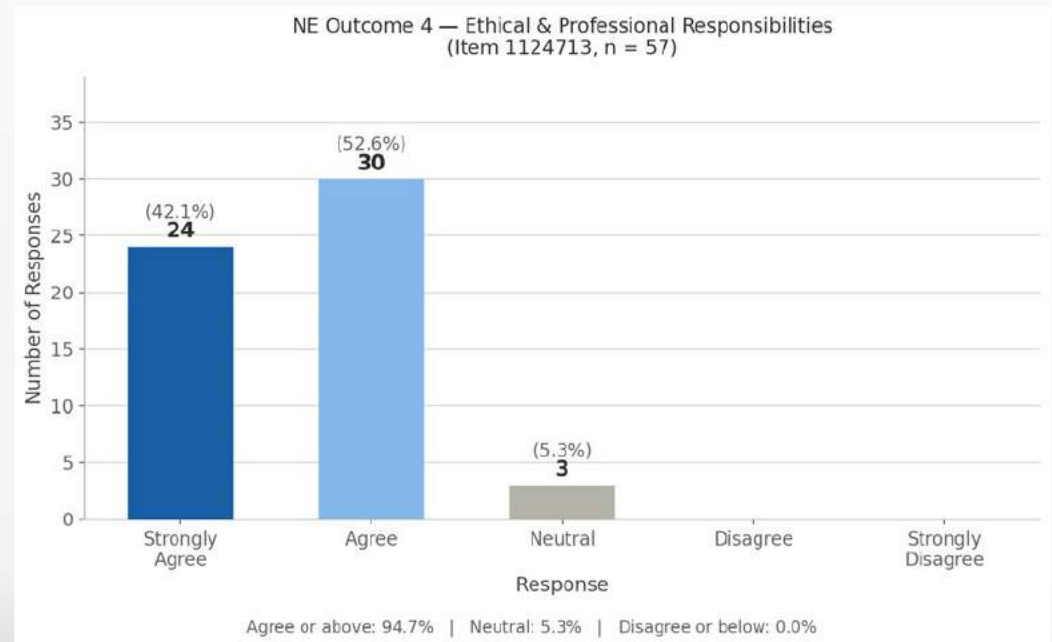
Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

Course-Level Feedback

AY 26-1



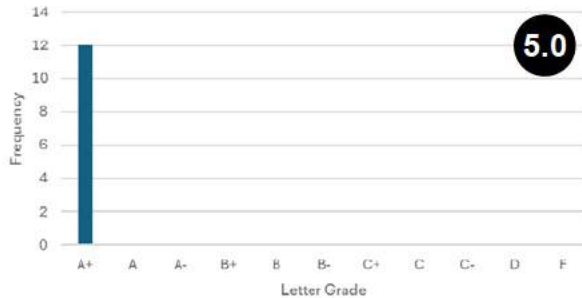
AY 26-2



Cadets must have the ability to acquire and apply new knowledge as needed, using appropriate learning strategies (SO7)

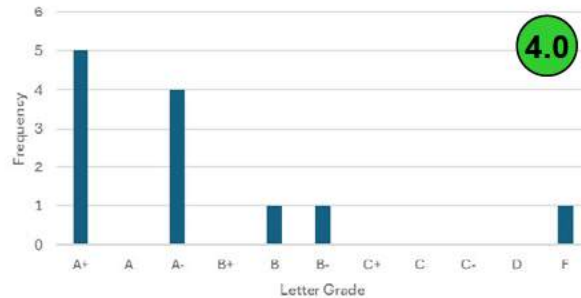
AY 26-1

SO 7 (Extension Problem 1)



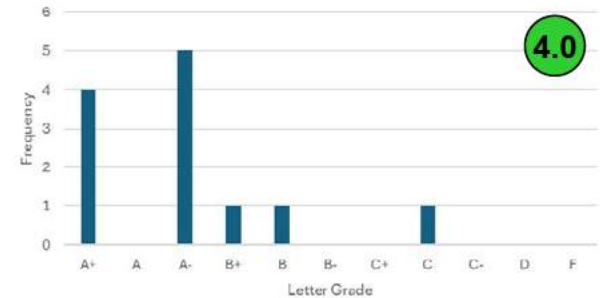
5.0

SO 7 (WPR 1 Fusion)



4.0

SO 7 (WPR 2 Initial Rad Dose)



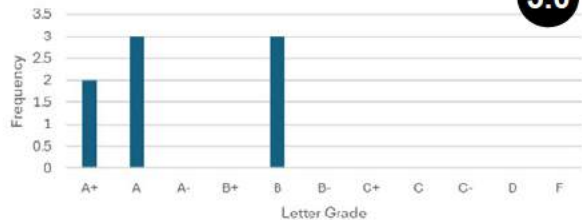
4.0

**Block
4.0**

Good. The NEN1 cadets excelled at the first extension problem. As in past years, the longer calculations challenge some of them. They struggle with the thermonuclear calculations because it is not an equation directly from the reference card. Initial rad dose has several steps, and many of them will miss one or two steps along the way.

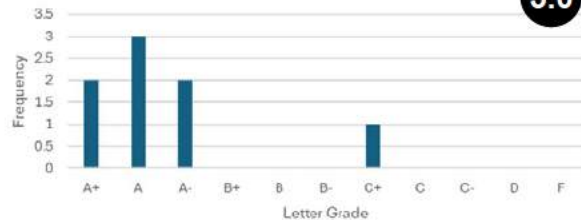
AY 26-2

SO 7 (Extension Problem 1)



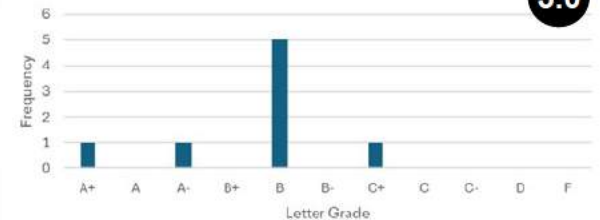
5.0

SO 7 (WPR 1 Fusion)



5.0

SO 7 (WPR 2 Initial Rad Dose)



5.0

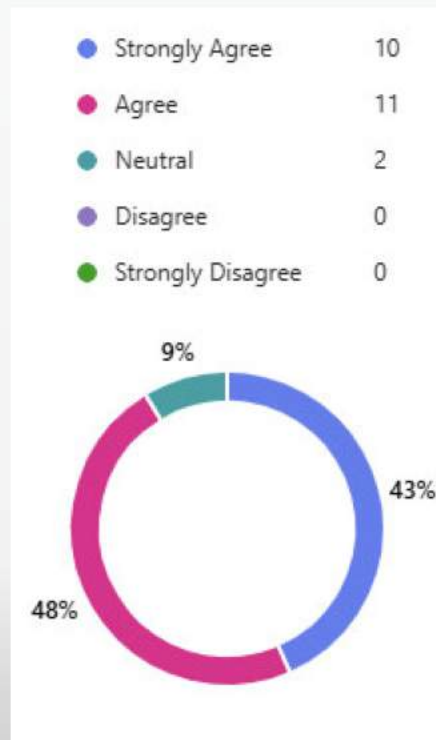
**Block
5.0**

Excellent. The NEN1 cadets were challenged with EP1 and some were intermixed in groups with non-majors. Thermonuclear calculations improved overall. Initial rad dose was worth less points overall, so mistakes may have had a larger impact on letter grade calculations.

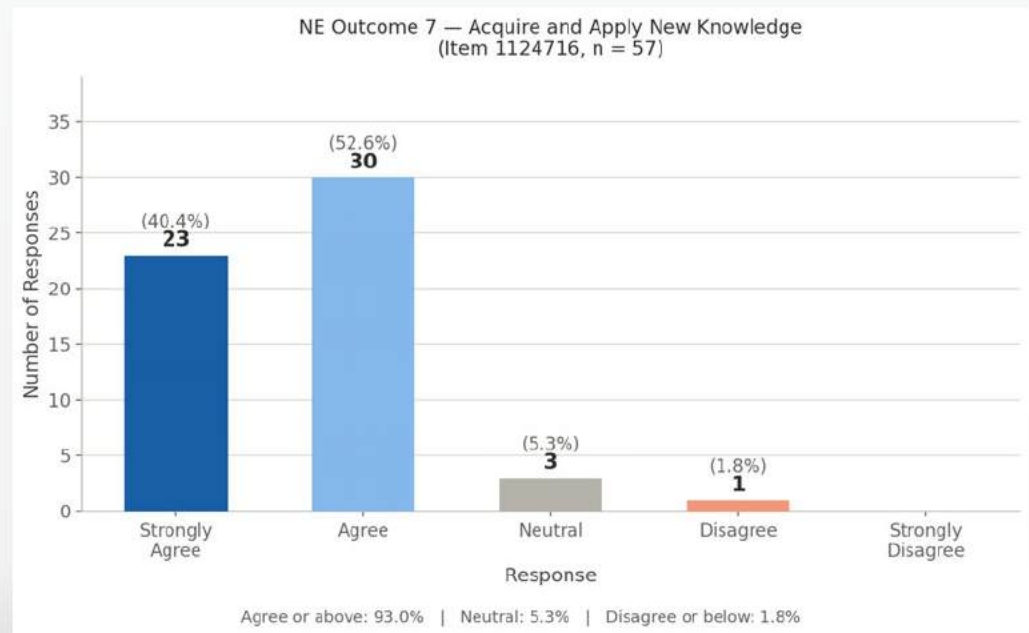
Cadets must have the ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts(SO4)

Course-Level Feedback

AY 26-1



AY 26-2



NE Outcomes Assessment

AY	NE Outcome	Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
26-1	4	4.3	5.0	4.5	EP2 was difficult to assess during AY26-1, as the MINES tool was unavailable which required some paring down of the assignment and limited time to discuss ethical considerations around nuclear use.
26-2		4.4	5.0	4.5	The EP2 rewrite is a good start to bringing in a multi-dimensional approach to weapon effects understanding and situational planning from an engineering and (partially) an ethical perspective. Further refinement will improve this indicator.
26-1	7	4.3	4.0	4.5	The NEN majors performed well on the more complex problems during timed, graded events. There was one F on the WPR1 indicator, but the most frequent grade was A or A+.
26-2		4.3	5.0	4.5	Complex problems were generally well handled. There was a growing apathy as the end of their curriculum approached that will be addressed in following semesters.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5 ● -- Excellent
- 4 ● -- Good
- 3 ● -- Acceptable
- 2 ● -- Marginally Acceptable
- 1 ● -- Unacceptable

Sustain & Improve from AY26-2 (for 27-1)

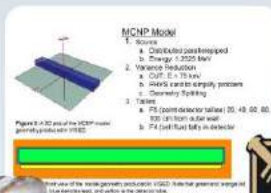
Sustain Theme	Sustain Summary
Extension Problems (EP1 & EP2)	Mentioned by 15+ cadets as top sustain. EP2 received especially strong praise as practical and engaging.
Classified/CNWDI Briefing	Repeatedly cited as a highlight and motivator. Many requested additional classified sessions throughout the semester.
Pre-Class Surveys	Appreciated as a primer for lessons; most cadets found the format useful for engagement.
Problem Sets & Board Problems	Problem sets rated as fair difficulty and well-aligned to WPRs. Board problems cited as essential for learning.
Current Events & Policy Discussion	Tying nuclear policy to current events was consistently praised for adding real-world context.
Interactive Equation Sheet (IRDC)	Specifically called out as an innovative resource that should be shared across physics/NE courses.
Points Earn-Back AI Policy	Strongly appreciated; cadets felt it encouraged learning from mistakes. Several said all courses should adopt it.
Instructor Energy & Enthusiasm	Near-universal praise. Cadets noted enthusiasm was critical for keeping second-semester firsties engaged.
Pace and Lesson Sequencing	The arc from heavy math in Block 1 to "beer math" at end of semester was viewed as appropriate and well-designed.
Cat Pictures & Humor	Mentioned multiple times as a small but meaningful touch that improved classroom atmosphere.

Improve Theme	Improve Summary
More Board Problems	~10 cadets requested more in-class practice and specifically asked to interleave board problems with content
Earlier Problem Set Feedback	~7 cadets noted that receiving PS grades before the WPR would allow them to correct errors before the exam.
EP1 Coding Guidance	~5 cadets found EP1 coding requirements unclear or inaccessible, especially for those without prior coding experience.
Laptop Policy	~5 cadets advocated for restricting laptop use, arguing it would improve focus, class culture, and responsiveness.
Solution Accessibility	~5 cadets wanted hand-written, step-by-step solutions for suggested problems.
More Classified Content	~5 cadets wanted more SCIF/classified lessons throughout the semester, not just one.
AI / Gradescope Grading	~4 cadets preferred human grading feedback; found AI grading insufficient for understanding where they went wrong.
Pre-Class Survey Format	~4 cadets noted that pre-class surveys took up too much class time, or that answers were covered in slides anyway.
WPR 2 Timing	1 cadet specifically suggested moving WPR 2 before spring break to avoid the long gap.
Side Tracks / Focus	~3 cadets mentioned frequent tangents to niche topics as diluting study time and making it hard to separate testable content from enrichment.

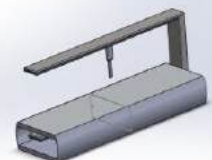
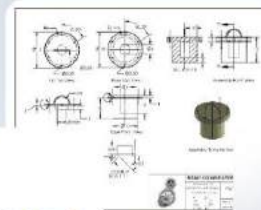
Resource Requirements

- Hoberman Sphere (acquired)
- 1000 mouse traps
- 1000 ping pong ball
- Large open room

Questions?



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TAB D

AY2023 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
Spring 2023 (AY23-2) Course Assessment



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LTC Jon Baker
NE450 Course Director

Agenda

AY 23-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE Assessment Process
- AY23 Course Assessment
- Recommended Changes for AY24

Program Director's Areas of Focus

Academic Year 2023

NE450

- ✓ NE Outcomes assessment are: 4, 7
- ✓ Implement CAT recommendations
- ✓ Develop sustained solution for text requirements for course (Ongoing)
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up (Ongoing)
- ✓ Look at possibly having outside persons receive cadet briefings (DTRA MINES POC)

Sustain & Improve from AY22-1 (for 22-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Removed dedicated computational class (was focused on MATLAB), but instead force the students, on multiple occasions, to perform a problem which would be very tedious by hand. Students choose their computational platform based on their own strengths (MATLAB, Python, Mathematica, Excel) (some just took cut or used brute force)
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Integrate Reactors I and II lessons from NE350 and use those lessons to evaluate NEN1 NE355 retention. (complete, will use NE355 graduates as cadet led class when available)
- Improve:
 - No major changes planned between AY22-1 and AY22-2. Focus will be working with Deputy on slides and CD notes

Sustain & Improve from AY23-1 (for 23-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 using MINES)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others, but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Integrate Reactors I and II lessons from NE350 and use those lessons to evaluate NEN1 NE355 retention. (complete, will use NE355 graduates as cadet led class when available)
 - Student Led Class (Learning Objective Posters)
- Improve:
 - No major course changes. LaTeX the ENW and Kaminski resources into a professional PDF and possibly explore a student funded hardbound copy
 - Get Baker and Allen assessment such to all work to guide next CD

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

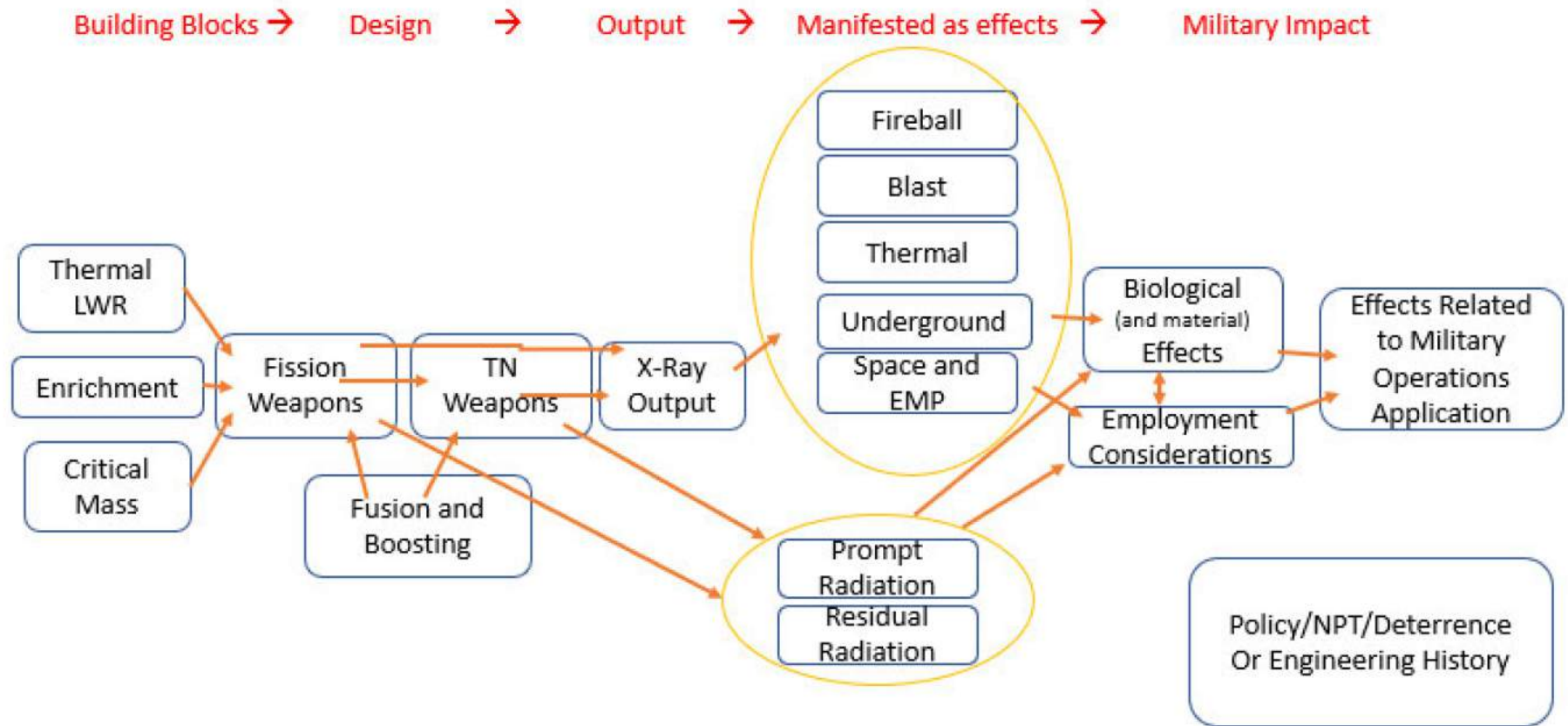
No changes recommended.

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Map/Topic Coverage



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Homework Sets							
Policy Paper				X			
Extension Problems				X			X
Instructor Writs							
Cadet Led Class							X
WPRs							X
TEE							

Best Assessment Instrument

- SO 4 – EP2 and ~~Paper~~ (optional in 23-2)
- SO 7 – EP1 and EP2 and Cadet LO Posters

Outcome 4

EP1 and EP2



CD: 4.5

	A	B	C	D
EP1	20	1	0	0
EP2	21	0	0	0

Outcome 7

LO Poster & EPs & WPRs



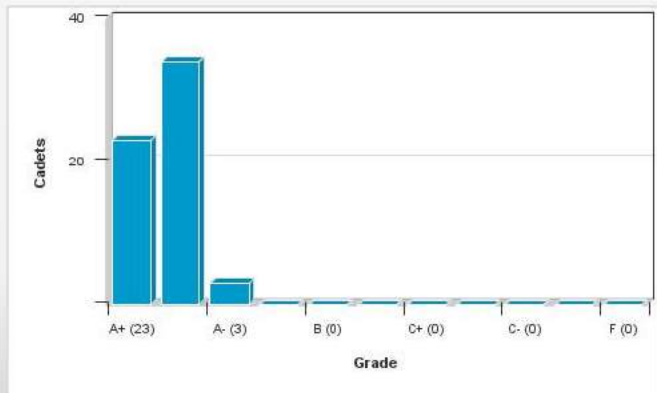
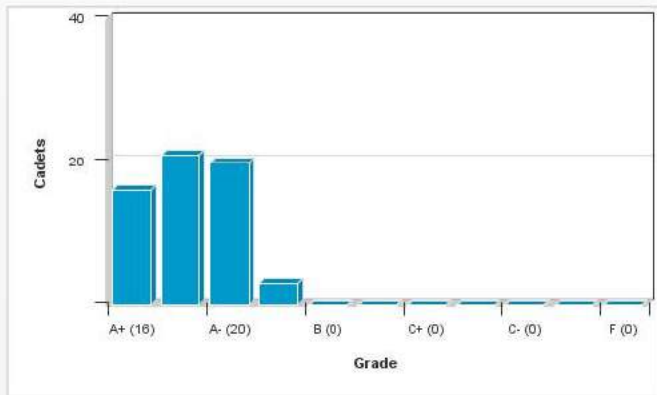
CD: 4.0

	A	B	C	D
EP1	20	1	0	0
EP2	21	0	0	0
LO	19	1	1	0
WPR1	17	4	0	0
WPR2	7	9	3	2

Outcome 4

EP1 and EP2

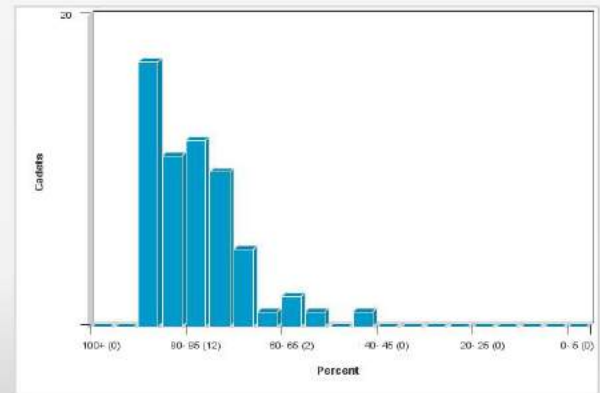
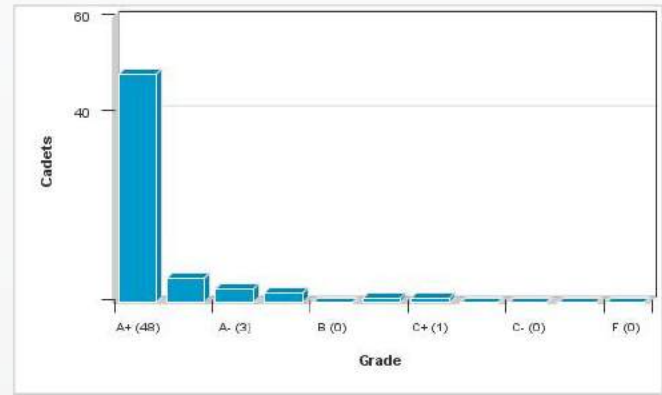
CD: 5.0



Outcome 7

LO Poster & EPs & WPRs

CD: 4.5



Survey Data (AY22-2 Comingled)

- Confident (SO4)
4.4 SD 0.48
 - Generally positive comments on the course
 - EP2 not as useful (being worked)
- Confident (SO7)
4.3 SD 0.68
 - Recommend to nonSTEM??
 - Comments of challenging
 - Comments of concept heavy
- Result of this course (SO4)
4.3 SD 0.61
 - Organization of materials (one wanted email)
 - Only one student + 2 column traditional ENW format
 - important as we go to all digital
- Result of this course (SO7)
4.2 SD 0.58
 - Teachers care about students and learning

NE Outcomes Assessment

NE Outcome	CDT Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
4	4.3	4.5	4.5	Cadets understand they have a professional responsibility to make ethical decisions and showed their ability to present this in written work, but they struggle with understanding the difference between presenting information as opposed to presenting analysis to decision makers.
7	4.2	4.0	4.5	The LO Posters gave the cadets a chance to display their skills in SO7 and they performed well on the indicators. WPR2 was still a poor performance, but they still displayed that they were only acceptable at acquiring and applying new knowledge in the course.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5 ● -- Excellent
- 4 ● -- Good
- 3 ● -- Acceptable
- 2 ● -- Marginally Acceptable
- 1 ● -- Unacceptable

NE CES Assessment: What Graduates Can Do 5.4: Graduates apply an engineering design process to create effective and adaptable solutions.

Supporting Outcomes (SO): The supporting outcomes for STEM Goal 5.4 are outlined below.

- 5.4.1 In an environment of uncertainty and change, identify needs that can be fulfilled via engineered solutions. (Identify) NE350
- 5.4.2 Define a complex technological problem, accounting for its political, social, and economic dimensions. (Define) NE450
- 5.4.3 Develop appropriate alternatives to solve an engineering problem. (Develop) NE350
- 5.4.4 Apply mathematics, basic science, and engineering science to model and analyze potential solutions to engineering problems. (Apply) NE450
- 5.4.5 Communicate an engineering solution to both technical and non-technical audiences and recommend an engineering solution to a decision maker. (Communicate) NE350
- 5.4.6 Demonstrate basic-level technical proficiency in an engineering discipline that is relevant to the needs of the Army. (Demonstrate) NE300

NE CES Assessment

SO	Course	Direct or Indirect Evidence	% of Course	Type of Event
5.4.1	NE350	TEE Question	10	Individual
5.4.2	NE450	WPR 2	10	Individual
5.4.3	NE350	Design Project, IPR 1	10	Group
5.4.4	NE450	WPR 1	10	Individual
5.4.5	NE350	Design Project, Final	10	Group
5.4.6	NE300	TEE	25	Individual

WGCD 5.4 Supporting Outcome	# Cadets	Failed to Meet Standard (%) D or F	Met Standard (%) C	Exceeded Standard (%) B	Greatly Exceeded Standard (%) A
5.4.1					
5.4.2	33	2 (6%)	6 (18%)	15 (45%)	10 (30%)
5.4.3					
5.4.4	33	9 (27%)	10 (30%)	6 (18%)	8 (24%)
5.4.5					
5.4.6					
Overall					
3 Year Average					

Numbers from All of AY23

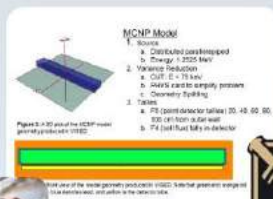
Sustain & Improve from AY23-2 (for 24-1)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others, but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Integrate Reactors I and II lessons from NE350 and use those lessons to evaluate NEN1 NE355 retention. (complete, will use NE355 graduates as cadet led class when available)
- Improve:
 - Extension problems (EP1 will focus on KD and not let BU be a deterrent to student learning and EP2 will be using MINES with DTRA supported access and instruction)
 - No major course changes. LaTeX the ENW and Kaminski resources into a professional PDF and possibly explore a student funded hardbound copy
 - Student led classes become a student created poster focusing on SO7 and technical communication skills.

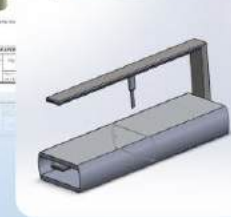
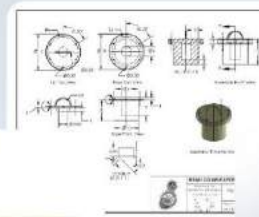
Acquisition Need

- We are entirely PDF this semester. PDF quality will improve by end of the semester (and hopefully just in time for student use this semester).
- If we have a consolidated PDF, do we give the students an option to “purchase” this printed and bound
- DTRA will support with MINES access and instructional support
- Do we need to level the semesters? Trend of 2 vs 4 sections -1 vs -2 due to NEN1?

Questions?



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TAB E

AY2022 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
Spring 2022 (AY22-2) Course Assessment



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LTC Ron Hasz
NE450 Course Director

Agenda

AY 22-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE Assessment Process
- AY22 Course Assessment
- Recommended Changes for AY23

AY21 Program Director's Recommendations (for AY22)

(1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of COVID response for students in follow-on courses
- Assess impact of MC311 taken in STAP for Class 22
- Continue assessment SO (1)-(7)

Standardization of briefings

- Work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Work on use of Overleaf

Standard use of Academic Terms

- Where appropriate use terms standard in academia (i.e. course-wide writ vs. PLW)

Effective use of NE labs and spaces

- Orientation for 2nd class for BH347 as collaborative space
- Orientation for 1st class for BH66B as collaborative space
- Cadets and faculty leverage remote use VMs

AY21 Program Director's Recommendations (for AY22) (2 of 3)

Emphasize assumptions in design process and assess the impact to a design throughout the program.

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- Reestablish trip sections for courses (STERIS)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to develop laboratory tech assigned to the program
- More external review within courses when possible

AY21 Program Director's Recommendations (for AY22)

(3 of 3)

NE Working Group Topics

- Develop funded research to help bring research dollars to department in support of Department Head guidance
- Refine NE program concept map.
- Get accelerator operational and back in program.
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Annual update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus

Academic Year 2022

NE450

- ✓ NE Outcomes assessment are: 4, 7 **complete**
- ✓ Implement CAT recommendations **complete**
- ✓ Prepare for in-person classes for AY22 at 75 min **complete**
- ✓ Maintain ability to use remote format for cadets on quarters **ongoing**
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up **ongoing**
- ✓ Look at possibly having outside persons receive cadet briefings **ongoing**

Sustain & Improve from AY22-1 (for 22-2)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Removed dedicated computational class (was focused on MATLAB), but instead force the students, on multiple occasions, to perform a problem which would be very tedious by hand. Students choose their computational platform based on their own strengths (MATLAB, Python, Mathematica, Excel) (some just took cut or used brute force)
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Integrate Reactors I and II lessons from NE350 and use those lessons to evaluate NEN1 NE355 retention. (complete, will use NE355 graduates as cadet led class when available)
- Improve:
 - No major changes planned between AY22-1 and AY22-2. Focus will be working with Deputy on slides and CD notes

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

No changes recommended.

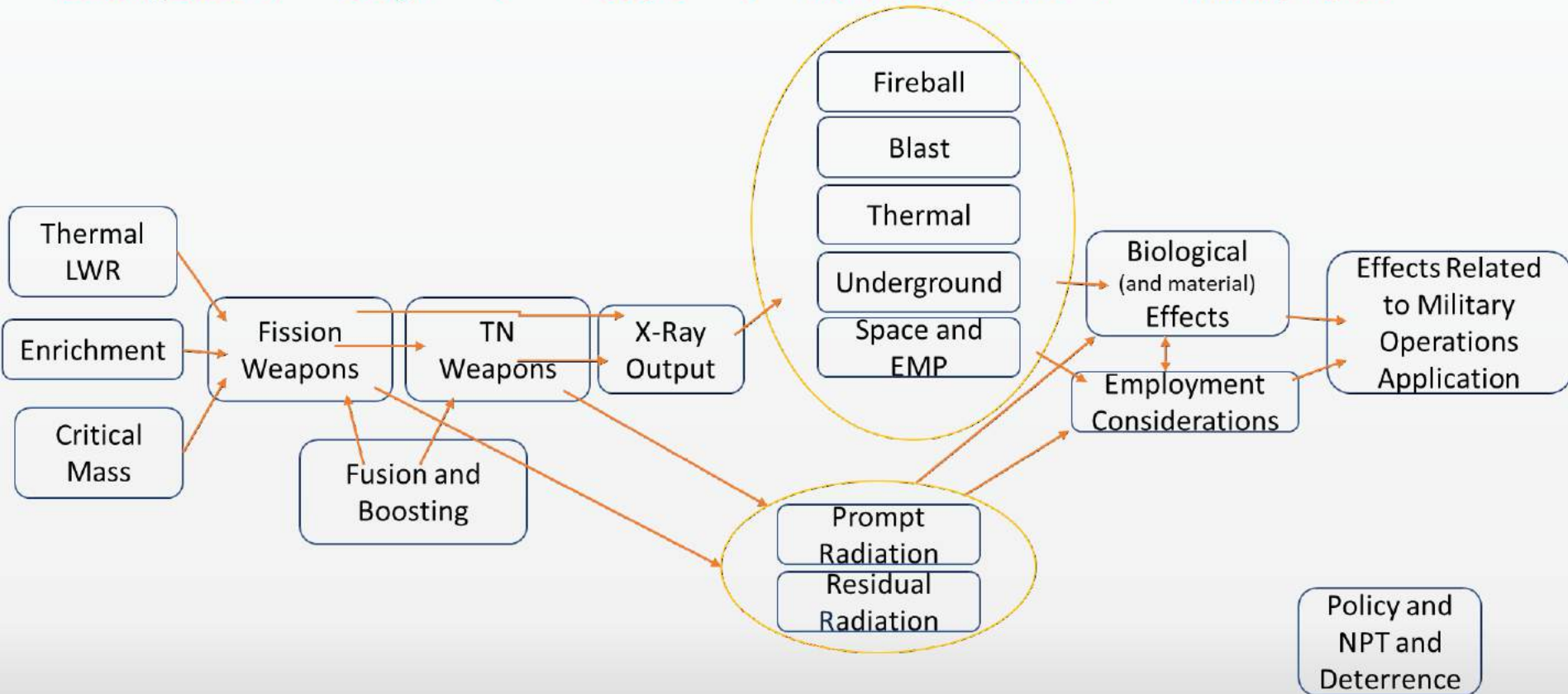
NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Map/Topic Coverage

Building Blocks → Design → Output → Manifested as effects → Military Impact



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Homework Sets							
Policy Paper				X			
Extension Problems				X			X
Instructor Writs							
Cadet Led Class							X
WPRs							X
TEE							

Best Assessment Instrument

- SO 4 – EP2 and **Policy Paper**
- SO 7 – EP1 and EP2 and **Cadet Led Class**

Outcome 4

Policy Paper & EPs

CD: 4.5

	A	B	C	D
EP1	13	3	0	0
EP2	16	0	0	0
Policy	14	2	0	0

Outcome 7

CDT Led & EPs & WPRs

CD: 4.0

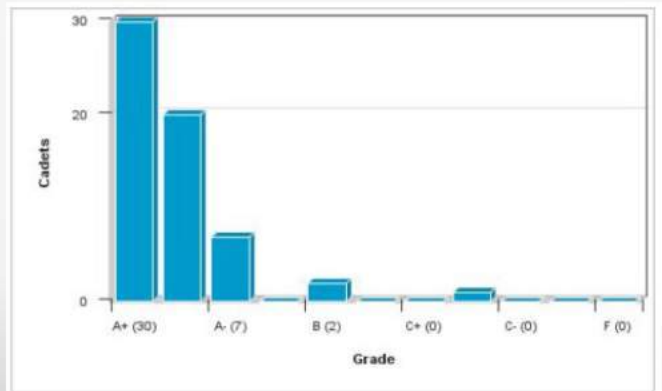
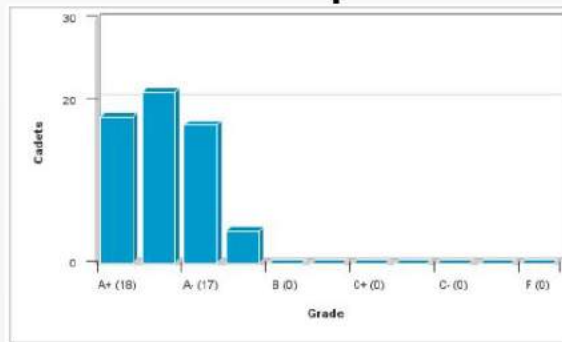
	A	B	C	D/F
EP1	13	3	0	0
EP2	16	0	0	0
CDT Led	12	4	0	0
WPR1&2	10	7	4	10

Outcome 4

Policy Paper & EPs



CD: 4.0

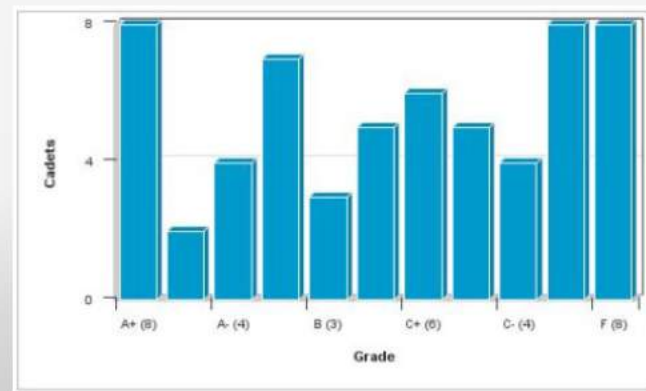
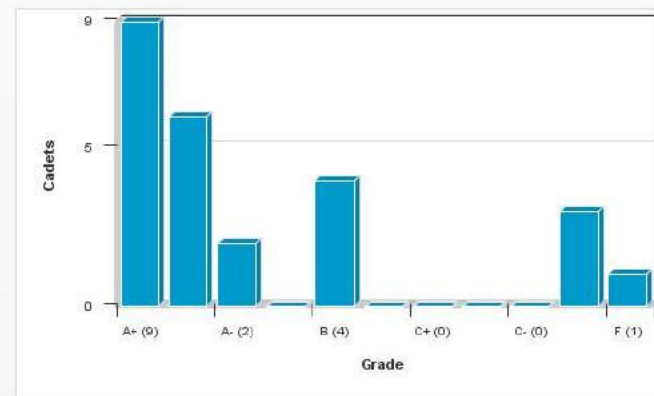


Outcome 7

CDT Led & EPs & WPRs



CD: 3.5



Survey Data (AY22-2 Comingled)

- Confident (SO4)
4.4 SD 0.48
 - Generally positive comments on the course
 - EP2 not as useful (being worked)
- Confident (SO7)
4.3 SD 0.68
 - Recommend to nonSTEM??
 - Comments of challenging
 - Comments of concept heavy
- Result of this course (SO4)
4.3 SD 0.61
 - Organization of materials (one wanted email)
 - Only one student + 2 column traditional ENW format
 - important as we go to all digital
- Result of this course (SO7)
4.2 SD 0.58
 - Teachers care about students and learning

NE Outcomes Assessment

NE Outcome	CDT Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
4	4.3	5	4.5	Cadets understand they have a professional responsibility to make ethical decisions and showed their ability to present this in written work, but they struggle with understanding the difference between presenting information as opposed to presenting analysis to decision makers.
7	4.2	3	3.5	The cadet led class, whether they were leading “review” material on reactors, or new material on weapon effects, gave the cadets a chance to display their skills in SO7 and they performed well on the indicators. I believe the WPR data is an anomaly based on the population and what turned out to be bad timing of the WPR, but they still displayed that they were only acceptable at acquiring and applying new knowledge in the course.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5 ● -- Excellent
- 4 ● -- Good
- 3 ● -- Acceptable
- 2 ● -- Marginally Acceptable
- 1 ● -- Unacceptable

NE CES Assessment: What Graduates Can Do 5.4: Graduates apply an engineering design process to create effective and adaptable solutions.

Supporting Outcomes (SO): The supporting outcomes for STEM Goal 5.4 are outlined below.

- 5.4.1 In an environment of uncertainty and change, identify needs that can be fulfilled via engineered solutions. (Identify) NE350
- 5.4.2 Define a complex technological problem, accounting for its political, social, and economic dimensions. (Define) NE450
- 5.4.3 Develop appropriate alternatives to solve an engineering problem. (Develop) NE350
- 5.4.4 Apply mathematics, basic science, and engineering science to model and analyze potential solutions to engineering problems. (Apply) NE450
- 5.4.5 Communicate an engineering solution to both technical and non-technical audiences and recommend an engineering solution to a decision maker. (Communicate) NE350
- 5.4.6 Demonstrate basic-level technical proficiency in an engineering discipline that is relevant to the needs of the Army. (Demonstrate) NE300

NE CES Assessment

SO	Course	Direct or Indirect Evidence	% of Course	Type of Event
5.4.1	NE350	TEE Question	10	Individual
5.4.2	NE450	WPR 2	10	Individual
5.4.3	NE350	Design Project, IPR 1	10	Group
5.4.4	NE450	WPR 1	10	Individual
5.4.5	NE350	Design Project, Final	10	Group
5.4.6	NE300	TEE	25	Individual

WGCD 5.4 Supporting Outcome	# Cadets	Failed to Meet Standard (%) D or F	Met Standard (%) C	Exceeded Standard (%) B	Greatly Exceeded Standard (%) A
5.4.1					
5.4.2	85	12.9%	11.8%	35.3%	40.0%
5.4.3					
5.4.4	85	28.2%	15.3%	23.5%	32.9%
5.4.5					
5.4.6					
Overall					
3 Year Average					

Numbers from All of AY22

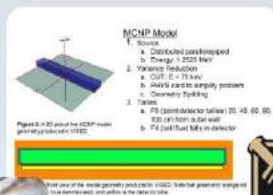
Sustain & Improve from AY22-2 (for 23-1)

- Sustain:
 - Extension problems (EP1 focusing, EP2 looking to learn from wargaming capstone)
 - Forcing the students to do a problem, EP1, which must use a computational tool. Some tools are better than others, but even using excel shows the student how a computational tool aids the learning.
 - Focus on learning at the boards (goal of 1/3 time at boards)
 - Integrate Reactors I and II lessons from NE350 and use those lessons to evaluate NEN1 NE355 retention. (complete, will use NE355 graduates as cadet led class when available)
- Improve:
 - Extension problems (EP1 will focus on KD and not let BU be a deterrent to student learning and EP2 will be using MINES with DTRA supported access and instruction)
 - No major course changes. LaTeX the ENW and Kaminski resources into a professional PDF and possibly explore a student funded hardbound copy
 - Student led classes become a student created poster focusing on SO7 and technical communication skills.

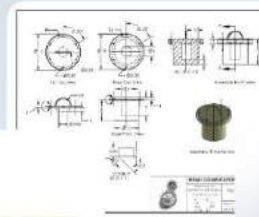
Acquisition Need

- We are entirely PDF this semester. PDF quality will improve by end of the semester (and hopefully just in time for student use this semester).
- If we have a consolidated PDF, do we give the students an option to “purchase” this printed and bound
- DTRA will support with MINES access and instructional support
- Do we need to level the semesters? Trend of 2 vs 4 sections -1 vs -2 due to NEN1?

Questions?



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TAB F

AY2021 COURSE ASSESSMENT

NE 450 Nuclear Weapon Effects
Spring 2021 (AY21-2) Course Assessment



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LTC Ron Hasz
NE450 Course Director

Agenda

AY 21-2 NE450 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE Assessment Process
- AY21 Course Assessment
- Recommended Changes for AY22

AY20 Program Director's Recommendations (for AY21) (1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of remote classes for students in follow-on courses
- Continue assessment SO (1)-(7)

Standard use of Blackboard

- Look to see if there are some of the resources used in the remote environment that will benefit the course content even when taught in person

Upgrades for new BH177B computer center and teaching of NE361 and use in other courses

- Finish upgrade this space NLT beginning of 21-1
- Every NE cadet has a unique IP address for a remote machine
- Cadets and faculty leverage in-class and remote use of BH177B and BH347 to continue to build on computational thread within program

Integrate NE461 as a required course in the NE program

Create a Program Drop Schedule to post to NE SharePoint

AY20 Program Director's Recommendations (for AY21) (2 of 3)

Update Nuclear Engineering Design Supplement

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- DTRA,MSKCC, NDU, STERIS (continue modified trip sections because of 75-min classes)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET visit in Nov remotely
- Develop a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Train and establish a relationship with new laboratory tech assigned to the program

AY20 Program Director's Recommendations (for AY21)

(3 of 3)

NE Working Group Topics

- Refine NE program concept map.
- Look at overlap and redundancy and where NE350 fits in program
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Complete all administrative edits to Redbook
- Complete update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus

Academic Year 2021

NE450

- ✓ NE Outcomes assessment are: 4, 7
- ✓ Implement CAT recommendations
- ✓ Be prepared for ABET accreditation visit in Nov
- ✓ Prepare for in-person, remote or hybrid classes for AY21
- ✓ Assess use of remote technology to enhance course even when taught in-person such as asynchronous video lessons for “flipped classrooms” or video AI for cadets who are on quarters or on trips.
- ✓ Sustain changes to course material to incorporate rigor vs. table look-up
- ✓ Look at possibly having outside persons receive cadet briefings remotely even if classes are taught in person via Teams/BB

Sustain & Improve from AY20 (for AY21)

- Sustain:
 - Extension Problems
 - Removed dedicated computational class (was focused on MATLAB), but instead force the students, on multiple occasions, to perform a problem which would be very tedious by hand. Students choose their computational platform based on their own strengths (MATLAB, Python, Mathematica, Excel)
- Improve:
 - Collaboration with DOE (LANL) project lead as they have taken on an undergraduate, unclassified textbook project. (working)
 - Review of the design portion with our LANL endowed chair for topical coverage and academic rigor and unclassified/classified. (complete)
 - One semester no comingled (based on which semester has heavier NEN1 load) and one semester comingled more focused on CES assessment. (complete -1 LITE vs -2 CAT)
 - Same with NE300 and then NE350 assessment focused on CES assessment.
 - Introduce Space Effects lesson and Reactors I&II (complete)

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators	NE Student Outcomes						
	1.	2.	3.	4.	5.	6.	7.
Homework Sets							
Policy Paper				X			
Extension Problems				X			X
Instructor Writs							
Cadet Led Class							X
WPRs							X
TEE							

Best Assessment Instrument

- SO 4 – EP2 and **Policy Paper**
- SO 7 – EP1 and EP2 and **Cadet Led Class**

Outcome 4

Policy Paper & EPs

CD: 4.5

	A	B	C	D
EP1	10	6	1	0
EP2	12	4	1	0
Policy	10	7	0	0

Outcome 7

CDT Led & EPs & WPRs

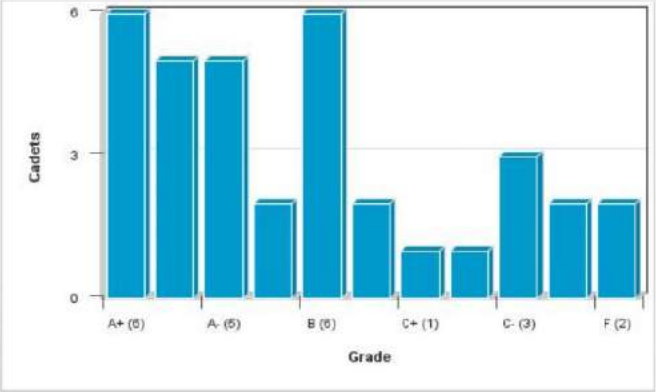
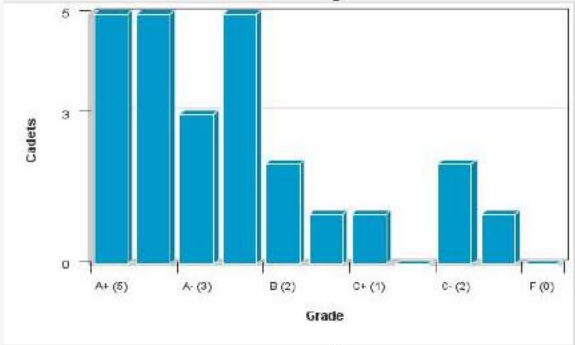
CD: 4.5

	A	B	C	D
EP1	10	6	1	0
EP2	12	4	1	0
CDT Led	13	4	0	0
WPR1&2	24	6	3	1

Outcome 7

Policy Paper & EPs

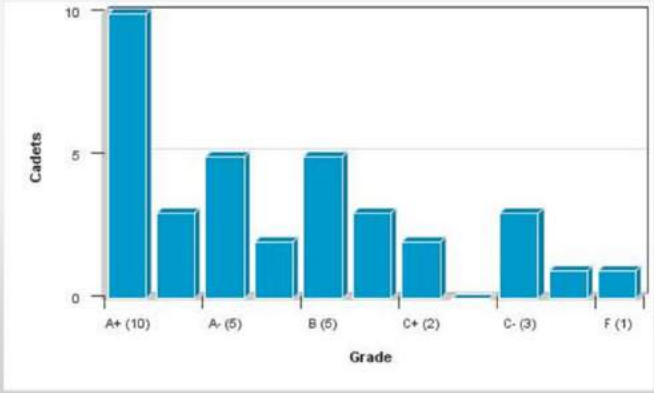
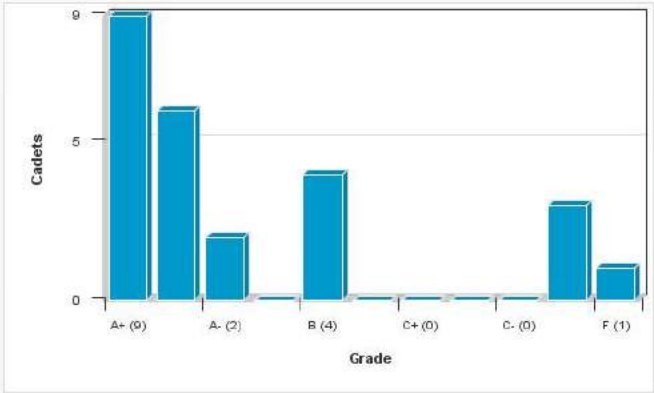
CD: 4.0



Outcome 7

CDT Led & EPs & WPRs

CD: 4.0



Survey Data (AY21-1 Comingled)

- Confident (SO4)
4.2 SD 0.57
- Confident (SO7)
4.3 SD 0.62
- Result of this
course (SO4)
4.2 SD 0.57
- Result of this
course (SO7)
4.4 SD 0.48

NE Outcomes Assessment

NE Outcome	CDT Survey (Comingled)	Embedded Indicator (NEN1 Only)	CD	Course Director Comments
4	4.2	5	4.5	Cadets understand they have a professional responsibility to make ethical decisions and showed their ability to present this in written work, but they struggle with understanding the difference between presenting information as opposed to presenting analysis to decision makers.
7	4.3	5	4.5	The cadet led class, whether they were leading “review” material on reactors, or new material on weapon effects, gave the cadets a chance to display their skills in SO7 and they performed well on the indicators.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5 ● -- Excellent
- 4 ● -- Good
- 3 ● -- Acceptable
- 2 ● -- Marginally Acceptable
- 1 ● -- Unacceptable

Redbook Entry

NE450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE450	NUCLEAR WEAPONS EFFECTS	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

Extension problems with design components.

NE450 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE	NE300	2008	1	1	Y

No changes recommended.

Course Map/Topic Coverage

Foundation work (and reactors)

Design (primary then secondary)

Weapon Output (x-rays)

Manifestation into effects

Application to the battlefield

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
17 R301 7 JAN	18 R301 8	19 N/A	20 N/A	21	22 LSN1 for K1 Hour L1* K1	23
24 JAN	25 TEAMS	26 TEAMS LSN1 for K1 and K1 hours L1* H1/H1	27 Enrichment L2	28 Asynch	29 MDT/Date	30
31 JAN/FEB	1 In Person	2 Critical Mass L4	3 SD	4 Readers I L5	5	6
7 FEB	8 Readers II L6	9	10 WPN Yield L7	11	12 WPN Mechanics L8	13 100th Night
14 FEB	15 Pres. Day 1	16	17 Fusion/Boost L9	18	19 ThermoNuclear L10	20 YWW
21 FEB	22	23 Extension I L11	24 WPR1 SD	25 WPR1 L12	26	27
28 FEB/MAR	1 CMWD Dismiss L13	2	3 EP1 Briefs L14	4	5 Intro to Effects L15	6 Pulse Week
7 MAR	8 NO CLASS	9	10 SD	11 X-Ray L16	12	13 100th Night
14 MAR	15 Blast L17	16	17 Thermal L18	18	19 Underground L19	20
21 MAR	22	23 Particle Interactions L20	24 SD	25 Initial Radiation L21	26	27
28 MAR/APR	29 Residual Radiation L22	30 HLD	31	1 Biological L23	2	3
4 APR	5 NO CLASS	6 EMP L24	7 SD	8 MOD DAY N/A	9	10
11 APR	12 Space L25	13	14 Extension I L26	15	16 MOD DAY N/A	17
18 APR	19	20 WPR2 L27	21 WPR2 SD	22	23 EP2 Intro/HFAC L28	24 A/D
25 APR	26	27 SD	28 EP2 Working Day L29	29 PROJECTS	30	1
2 MAY	3 Effects Talk L30	4	5 EP2 Briefs L31	6 SD	7 History/FNEX L32	8
9 MAY	10	11 N/A	12 TEEs	13 TEEs	14 TEEs	15 TEEs
16 MAY	17 GRAD WK	18 GRAD WK	19 GRAD WK	20 GRAD WK	21 GRAD WK	22 GRAD DAY

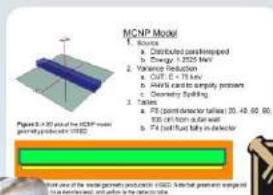
Sustain & Improve from AY21 (for AY22)

- Sustain:
 - Extension problems.
 - Removed dedicated computational class (was focused on MATLAB), but instead force the students, on multiple occasions, to perform a problem which would be very tedious by hand. Students choose their computational platform based on their own strengths (MATLAB, Python, Mathematica, Excel)
 - Focus on learning at the boards
- Improve:
 - Collaboration with DOE (LANL) project lead as they have taken on an undergraduate, unclassified textbook project.
 - Collaborate more with Space Science program for space effects class
 - Provide a hardcopy of all readings in a booklet instead of only available on the course management tool (blackboard or teams.) Cadet complaint
 - Real time use of HPAC (not by cadets) as they are doing boardwork. Cadet Idea
 - If no NEN1 in AY22-1, should there be a CAT LITE?

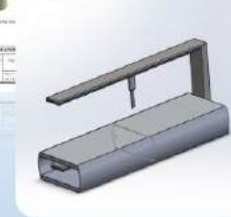
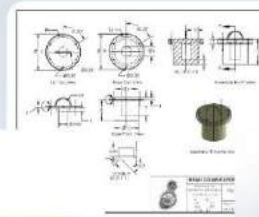
Acquisition Need

- I guess we will make our own survey to obtain and organize the data we want in this process
- DTRIAC continues to send copies of ENW, but we are prepared with PDF if they cease.
- Print plant for course readings.
- Dr Chitanvis has offered to come 1-2 times per semester for CNWDI.

Questions?



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SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
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TAB G

WPR #1

Instructor Solution

Time of completion _____ Cadet _____
11/12 Feb 2026 (First) (Last)

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review I

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 10 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **20 February 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

- 10 1. Which statement correctly identifies the two most common methods for assembling a supercritical mass AND explains why plutonium cannot be used in one of them?
- ☐ Gun-type and compression-type; plutonium cannot be used in gun-type assembly because it has too low of fission cross section for the gun mechanism to work effectively.
 - ☐ Implosion-type and linear-type; plutonium cannot be used in linear assembly because its density is too high to achieve the required assembly speed.
 - ☒ Gun-type and implosion-type; plutonium cannot be used in gun-type assembly because its high spontaneous fission rate would cause pre-initiation before full assembly is achieved.
 - ☐ Gun-type and implosion-type; plutonium cannot be used in implosion-type assembly because the shock wave would cause the plutonium to fragment before reaching criticality.
 - ☐ Boosted-type and implosion-type; plutonium cannot be used in boosted-type because it reacts with tritium in the fusion fuel.
- 10 2. Which statement correctly identifies the two mechanisms that shut down the chain reaction in a fissioning weapon AND accurately describes one?
- ☒ Kinetic disassembly and burn-up; kinetic disassembly occurs when the weapon assembly expands and bomb alpha becomes negative.
 - ☐ Neutron leakage and fuel depletion; neutron leakage occurs when neutrons escape the core faster than fission can replace them to sustain the chain reaction.
 - ☐ Pre-initiation and post-initiation; pre-initiation occurs when spontaneous fission starts the chain reaction too early, causing the assembly to disassemble before achieving maximum yield.
 - ☐ Kinetic disassembly and neutron leakage; neutron leakage occurs when the surface area to volume ratio becomes large enough for bomb alpha to become negative.
 - ☐ Burn-up and thermal expansion; thermal expansion occurs when heat from fission causes the fuel to melt into a non-spherical configuration.

Wt. No.

- 10 3. What is the primary purpose of fusion in a nuclear weapon?
- ☐ To generate more yield than fission with higher Q values.
- ☒ To produce high-energy neutrons that increase fission rate and yield.
- ☐ To boost the outward velocity, u_2 , of the secondary.
- ☐ To initiate the fission chain reaction by providing the initial neutron population.
- 10 4. Gas centrifuge enrichment efficiency is dependent on the rotor's angular velocity, radius, and temperature.
- ☒ TRUE
- ☐ FALSE
- 10 5. Select **all** the components of the six-factor formula that apply to weapons.
- ☐ ϵ ☒ η
- ☐ p ☒ P_{NL}^f
- ☐ f ☐ P_{NL}^{th}
- 10 6. The purpose of a tamper is to push neutrons back into the fissioning core.
- ☐ TRUE
- ☒ FALSE
- 10 7. Select **all** properties of a good reflector in a weapon assembly.
- ☐ Low σ_s ☐ Low σ_f
- ☒ High σ_s ☐ High σ_f
- ☒ Low σ_a ☒ Low Z
- ☐ High σ_a ☐ High Z

Wt. No.

Cadet: _____

- 60 8. Pam, an aspiring nuclear weapons designer, requires 67.0 kg of Highly Enriched Uranium at 93.0% by weight of ^{235}U . Assume a natural feed assay of 0.711%, and a tails assay of 0.200%. For brevity, $M_F = 19.0 \text{ g/mol}$, and $M_U = 235 \text{ g/mol}$ and 238 g/mol . **Report all answers to 3 significant figures.**

- (10) a. Calculate the mass of feed, in kg, Pam needs to complete her enrichment.

$$\frac{F}{P} = \frac{x_p - x_w}{x_f - x_w} \quad F = \frac{0.93 - 0.002}{0.00711 - 0.002} 67 \text{ kg}$$

$$F = \frac{x_p - x_w}{x_f - x_w} P \quad F = 1.2168 \times 10^4 \text{ kg}$$

$$1.22 \times 10^4 \text{ kg}$$

Report your answer in this box.

- (30) b. Calculate the total separative work Pam needs to produce her product.

$$W = F - P$$

$$W = 1.2168 \times 10^4 - 67$$

$$W = 1.2100 \times 10^4 \text{ kg}$$

$$SWU = [PV(x_p) + WV(x_w) - FV(x_f)]\tau$$

$$\tau = \text{unity}$$

$$SWU = 1.5782 \times 10^4 \text{ kg SWU}$$

$$V(x_i) = (2x_i - 1) \ln \left(\frac{x_i}{1 - x_i} \right)$$

$$V(x_p) = 2.2246$$

$$V(x_f) = 4.8689$$

$$V(x_w) = 6.1878$$

$$1.58 \times 10^4 \text{ kg SWU}$$

Report your answer in this box.

- (20) c. Calculate the number of enrichment stages Pam needs if she only has access to a gaseous diffusion facility using UF_6 .

$$\alpha = \sqrt{\frac{m_H}{m_L}}$$

$$N = \frac{2}{\ln(\alpha)} \ln \left(\frac{x_p(1 - x_w)}{x_w(1 - x_p)} \right) - 1$$

$$\alpha = \sqrt{\frac{352}{349}}$$

$$N = \frac{2}{\ln(1.0043)} \ln \left(\frac{0.93(1 - 0.002)}{0.002(1 - 0.93)} \right) - 1$$

$$N = 4.1112 \times 10^3$$

$$\alpha = 1.0043$$

$$4.11 \times 10^3 \text{ stages}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 60 9. Sheldon, an OG nuclear weapons engineer, assembles a bare sphere of ^{239}Pu with a density of 67.0 g/cm^3 and initiates it with 1.67×10^4 neutrons. The non-leakage factor for his assembly is 0.670. **Report all answers to 3 significant figures.**

- (10) a. Calculate the multiplication factor of Sheldon's assembly.

$$k = \frac{\nu \sigma_f}{\sigma_a} L$$

$$k = \frac{5.61}{1.98} 0.67$$

$$k = 1.8983$$

1.90

Report your answer in this box.

- (10) b. Calculate the average neutron lifetime of Sheldon's assembly.

$$l = \frac{ML}{\rho N_A \sigma_a v}$$

$$l = 1.856 \times 10^{-9} \text{ s}$$

$1.86 \times 10^{-9} \text{ s}$

Report your answer in this box.

- (10) c. Calculate the "bomb alpha" of Sheldon's assembly in gen/shake.

$$\alpha = \frac{k - 1}{l}$$

$$\alpha = 4.8403 \times 10^8 \text{ sec}^{-1}$$

$$\alpha = 4.84 \text{ shake}^{-1}$$

$4.84 \frac{\text{gen}}{\text{shake}}$

Report your answer in this box.

- (30) d. Calculate the yield after 50 generations, neglecting any shutdown mechanisms.

$$Y = \frac{\Sigma_f \nu Q n_0}{\alpha} (e^{\alpha t} - 1)$$

$$t = gl$$

$$Y = \frac{\rho N_A \sigma_f \nu Q n_0}{M \alpha} (e^{\alpha gl} - 1)$$

$$Y = \frac{0.3178(1.08 \times 10^9)(7.08 \times 10^{-24})(1.67 \times 10^4)}{4.84 \times 10^8} (e^{4.84 \times 10^8 (50)(1.86 \times 10^{-9})} - 1)$$

$$Y = 2.6948 \text{ kT}$$

2.70 kT

Report your answer in this box.

Wt. No.

Cadet: _____

- 60 10. Pam and Sheldon team up to make a thermonuclear device. Pam's primary is designed to yield 6.70 kT. She places it next to Sheldon's secondary, which contains 31.5 g of deuterium surrounded by a ^{238}U tamper and is designed to compress to a radius of 1.67 cm. **Report all answers to 3 significant figures.**

- (30) a. Calculate the 100% burn yield from fusion of Sheldon's secondary in kT.

$$V = \frac{4}{3}\pi r^3$$

$$Y = V n_{tar} Q$$

$$V = 19.5091 \text{ cm}^3$$

$$Y = V \frac{n_{tot}}{2} Q$$

$$n_{tot} = \frac{\rho N_A}{M}$$

$$n_{tot} = \frac{m N_A}{VM}$$

$$Y = 19.5091 \text{ cm}^3 \frac{4.8616 \times 10^{23} \frac{\text{atoms}}{\text{cm}^3}}{2} \frac{9.96 \text{ MeV}}{2.613 \times 10^{25} \frac{\text{MeV}}{\text{kT}}}$$

$$Y = 1.8076$$

$$n_{tot} = 4.8616 \times 10^{23} \frac{\text{atoms}}{\text{cm}^3}$$

$$1.81 \text{ kT}$$

Report your answer in this box.

- (20) b. Calculate the fission yield in kT of Sheldon's secondary if fusion neutrons are 50% efficient at ^{238}U fission. Assume the same fission Q-values for all U isotopes.

$$Y = \text{Num}_{reac} Q$$

$$Y = 0.5 \frac{V n_{tot}}{5} \frac{180 \text{ MeV}}{2.613 \times 10^{25} \frac{\text{MeV}}{\text{kT}}}$$

$$Y = 6.5336 \text{ kT}$$

$$6.53 \text{ kT}$$

Report your answer in this box.

- (10) c. Calculate the total yield of the device in kT.

$$Y = Y_p + Y_{fus} + Y_{fiz}$$

$$Y = 6.7 + 1.81 + 6.53$$

$$Y = 15.0413 \text{ kT}$$

$$15.0 \text{ kT}$$

Report your answer in this box.

Wt. No.

Cadet: _____

10

B

Bonus: Pam thinks she can make a better secondary than Sheldon. She fills her's with D-T fuel at a 1:1 atom ratio. Upon compression, it is squeezed to a total number density of 1.25×10^{25} atoms/cm³ at a temperature of 3.75×10^8 K and a volume of 67.0 cm³. **Report all answers to 3 significant figures.**

Calculate the reaction rate of Pam's secondary. Note: There is no partial credit for bonus problems, it is all or nothing.

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} E^{-\frac{2}{3}} e^{-19.94 E^{-\frac{1}{3}}}$$

$$E = 3.75 \times 10^8 \text{ K} \left(\frac{1 \text{ eV}}{1.1605 \times 10^4 \text{ K}} \right) \left(\frac{1 \text{ keV}}{1000 \text{ eV}} \right)$$

$$E = 32.3137 \text{ keV}$$

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} (32.3^{-\frac{2}{3}}) e^{-19.94 (32.3^{-\frac{1}{3}})}$$

$$(\overline{\sigma v})_{DT} = 6.9306 \times 10^{-16} \frac{\text{cm}^3}{\text{sec}}$$

$$N_{\text{reac}} = (\overline{\sigma v}) n_{\text{tar}} n_{\text{bomb}}$$

$$N_{\text{reac}} = (\overline{\sigma v}) \left(\frac{n_{\text{tot}}}{2} \right)^2$$

$$N_{\text{reac}} = 6.9306 \times 10^{-16} \frac{\text{cm}^3}{\text{sec}} \left(\frac{1.25 \times 10^{25} \text{ cm}^{-3}}{2} \right)^2$$

$$N_{\text{reac}} = 2.7073 \times 10^{34} \text{ cm}^{-3} \text{ s}^{-1}$$

$$R_{\text{rate}} = N_{\text{reac}} V$$

$$R_{\text{rate}} = 1.8139 \times 10^{36} \text{ s}^{-1}$$

$$1.81 \times 10^{36} \text{ s}^{-1}$$

Report your answer in this box.

Outstanding Student Work

Student

Total Points
248 / 260 pts

Question 1

Assembly

10 / 10 pts

✓ - 0 pts C (Correct)

Question 2

Shut down

10 / 10 pts

✓ - 0 pts A (Correct)

Question 3

Fusion

10 / 10 pts

✓ - 0 pts B (Correct)

Question 4

Centrifuge

10 / 10 pts

✓ - 0 pts True (Correct)

Question 5

Six-factor formula

8 / 10 pts

✓ - 2 pts €

Question 6

Tamper

10 / 10 pts

✓ - 0 pts False (Correct)

Question 7

Reflector

4 / 10 pts

✓ - 4 pts Doesn't select Low σ_a

✓ - 2 pts High σ_a

Question 8

Enrichment

58 / 60 pts

8.1 — **Feed mass**

10 / 10 pts

✓ - 0 pts Correct 1.22×10^4 kg8.2 — **SWU**

30 / 30 pts

✓ - 0 pts Correct 1.58×10^4 kg SWU8.3 — **Stages**

18 / 20 pts

✓ - 2 pts Incorrect answer

Question 9

Primary

58 / 60 pts

9.1 — **Multiplication factor**

10 / 10 pts

✓ - 0 pts Correct (1.90)

9.2 — **Neutron lifetime**

10 / 10 pts

✓ - 0 pts Correct (1.86×10^{-9} s)9.3 — **Bomb alpha**

8 / 10 pts

✓ - 1 pt Doesn't convert to shakes

✓ - 1 pt Incorrect answer

9.4 — **Yield**

30 / 30 pts

✓ - 0 pts Correct/follows (2.70 kT)

Question 10

Secondary

60 / 60 pts

10.1 — **Fusion yield**

30 / 30 pts

✓ - 0 pts Correct (1.81 kT)

10.2 — **Fission yield**

20 / 20 pts

✓ - 0 pts Correct/follows (6.53 kT)

10.3 — **Total yield**

10 / 10 pts

✓ - 0 pts Correct/follows (15.0 kT)

Question 11**Bonus: Rxn rate**

10 / 10 pts

✓ - 0 pts Correct 1.81×10^{36} s⁻¹

Time of completion 1510 Cadet [REDACTED] Section
11/12 Feb 2026 (First) (Last) Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review I

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 10 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **20 February 2026 at 1700.**
- b. Cadet Signature: [REDACTED]
- c. Cadet Name (print) [REDACTED]

- 10 1. Which statement correctly identifies the two most common methods for assembling a supercritical mass AND explains why plutonium cannot be used in one of them?
- ☐ Gun-type and compression-type; plutonium cannot be used in gun-type assembly because it has too low of fission cross section for the gun mechanism to work effectively.
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 - ☐ Gun-type and implosion-type; plutonium cannot be used in implosion-type assembly because the shock wave would cause the plutonium to fragment before reaching criticality.
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 - ☐ Kinetic disassembly and neutron leakage; neutron leakage occurs when the surface area to volume ratio becomes large enough for bomb alpha to become negative.
 - ☐ Burn-up and thermal expansion; thermal expansion occurs when heat from fission causes the fuel to melt into a non-spherical configuration.

- 10 3. What is the primary purpose of fusion in a nuclear weapon?
- ☐ To generate more yield than fission with higher Q values.
 - ☒ To produce high-energy neutrons that increase fission rate and yield.
 - ☐ To boost the outward velocity, u_2 , of the secondary.
 - ☐ To initiate the fission chain reaction by providing the initial neutron population.
- 10 4. Gas centrifuge enrichment efficiency is dependent on the rotor's angular velocity, radius, and temperature.
- ☒ TRUE
 - ☐ FALSE
- 10 5. Select **all** the components of the six-factor formula that apply to weapons.
- | | |
|---|---|
| ☒ ϵ | ☒ η |
| ☐ k | ☒ P_{NL}^f |
| ☐ f | ☐ ρ |
- 10 6. The purpose of a tamper is to push neutrons back into the fissioning core.
- ☐ TRUE
 - ☒ FALSE
- 10 7. Select **all** properties of a good reflector in a weapon assembly.
- | | |
|--|--|
| ☐ Low σ_s | ☐ Low σ_f |
| ☒ High σ_s | ☐ High σ_f |
| ☐ Low σ_a | ☒ Low Z |
| ☒ High σ_a | ☐ High Z |

Wt. No.

60

8.

Cadet: _____

Pam, an aspiring nuclear weapons designer, requires 67.0 kg of Highly Enriched Uranium at 93.0% by weight of ^{235}U . Assume a natural feed assay of 0.711%, and a tails assay of 0.200%. For brevity, $M_F = 19.0$ g/mol, and $M_U = 235$ g/mol and 238 g/mol. Report all answers to 3 significant figures.

- (10) a. Calculate the mass of feed, in kg, Pam needs to complete her enrichment.

$$X_f = 0.00711 \quad \frac{F}{P} = \frac{X_p - X_w}{X_f - X_w} = 12,167.5 \text{ kg}$$

$$X_p = 0.93$$

$$X_w = 0.002$$

$$F = \left(\frac{X_p - X_w}{X_f - X_w} \right) P = 12,200 \text{ kg}$$

$$= \left(\frac{0.93 - 0.002}{0.00711 - 0.002} \right) 67 \text{ kg}$$

$$F = 12,200 \text{ kg}$$

Report your answer in this box.

- (30) b. Calculate the total separative work Pam needs to produce her product.

$$SWU = \left[V(X_p) + \frac{W}{P} V(X_w) - \frac{F}{P} V(X_f) \right] \text{TP} \quad V(X_p) = (2X_p - 1) \ln \left(\frac{X_p}{1 - X_p} \right) = 2.225$$

$$= \left[V(X_p) + \frac{X_p - X_f}{X_f - X_w} V(X_w) - \frac{X_p - X_w}{X_f - X_w} V(X_f) \right] \text{TP}$$

$$V(X_f) = (2X_f - 1) \ln \left(\frac{X_f}{1 - X_f} \right) = 4.869$$

$$V(X_w) = (2X_w - 1) \ln \left(\frac{X_w}{1 - X_w} \right) = 6.188$$

$$= \left[2.225 + 180.6(6.188) - 181.6(4.869) \right] \text{TP} \quad \frac{X_p - X_f}{X_f - X_w} = 180.6$$

$$= 235 \text{ SWU/kg} \cdot P$$

$$\frac{X_p - X_w}{X_f - X_w} = 181.6$$

$$= 15,783 \text{ SWU} = 15,800 \text{ SWU}$$

$$15,800 \text{ SWU}$$

Report your answer in this box.

- (20) c. Calculate the number of enrichment stages Pam needs if she only has access to a gaseous diffusion facility using UF_6 .

$$\alpha = \sqrt{\frac{m_H}{m_L}} = \sqrt{\frac{352}{349}} = 1.004$$

$$M(\text{UF}^{238}) = 352 \text{ Amu}$$

$$M(\text{UF}^{235}) = 349 \text{ Amu}$$

$$N = \frac{2}{\ln(\alpha)} \ln \left[\frac{X_p(1 - X_w)}{X_w(1 - X_p)} \right] - 1$$

$$= \frac{2}{\ln(1.004)} \ln \left[\frac{0.93(1 - 0.002)}{0.002(1 - 0.93)} \right] - 1$$

$$= 4407.4 \Rightarrow 4410 \text{ stages}$$

$$4410 \text{ stages}$$

Report your answer in this box.

60

9.

Sheldon, an OG nuclear weapons engineer, assembles a bare sphere of ^{239}Pu with a density of 67.0 g/cm^3 and initiates it with 1.67×10^4 neutrons. The non-leakage factor for his assembly is 0.670. Report all answers to 3 significant figures.

- (10) a. Calculate the multiplication factor of Sheldon's assembly.

$$k_{\infty} = \frac{\nu \sigma_f}{\sigma_a} = \frac{5.615}{1.986} \cdot 0.670$$

$$k = k_{\infty} L = 1.898$$

$$k = \frac{\nu \sigma_f L}{\sigma_a} = 1.90$$

$$k = 1.90$$

Report your answer in this box.

- (10) b. Calculate the average neutron lifetime of Sheldon's assembly.

$$l = \frac{L}{\Sigma_a v} = \frac{(0.670)(239 \text{ g/mol})}{(67.0 \text{ g/cm}^3)(N_A \text{ #/mol})(1.98 \times 10^{-24} \text{ cm}^2)(1.08 \times 10^9 \text{ cm/s})}$$

$$= \frac{L M}{\rho N_A \sigma_a v} = 1.856 \times 10^{-9} \text{ s}$$

$$= 1.86 \times 10^{-9} \text{ s}$$

$$l = 1.86 \times 10^{-9} \text{ s}$$

Report your answer in this box.

- (10) c. Calculate the "bomb alpha" of Sheldon's assembly in gen/shake.

$$\alpha = \frac{k-1}{l}$$

$$= \frac{(1.90-1)}{1.86 \times 10^{-9} \text{ s}} = 4.839 \times 10^8 \text{ s}^{-1}$$

$$\alpha = 4.84 \times 10^8 \text{ s}^{-1}$$

Report your answer in this box.

- (30) d. Calculate the yield after 50 generations, neglecting any shutdown mechanisms.

$$t = gl$$

$$Y(g) = \frac{\Sigma_f v Q n_0}{\alpha} (\exp(\alpha gl) - 1)$$

$$\Sigma_f = \sigma_f \frac{\rho N_A}{M} = (1.88 \times 10^{-24} \text{ cm}^2) \frac{67.0 \text{ g/cm}^3 \cdot N_A \text{ #/mol}}{239 \text{ g/mol}} = 0.317 \text{ cm}^{-1}$$

$$Q = 185 \text{ MeV} \cdot \frac{1 \text{ kT}}{8.613 \times 10^{-5} \text{ MeV}} = 7.679 \times 10^{24}$$

$$Y(50) = \frac{(0.317 \text{ cm}^{-1})(1.08 \times 10^9 \text{ cm/s})(7.679 \times 10^{24} \text{ kT})(1.67 \times 10^4 \text{ neutrons})}{(4.84 \times 10^8 \text{ s}^{-1}) (\exp(50 \cdot 1.86 \times 10^{-9} \text{ s} \cdot 4.84 \times 10^8 \text{ s}^{-1}) - 1)}$$

$$= 2.957 \text{ kT}$$

$$Y(50) = 2.96 \text{ kT}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 60 10. Pam and Sheldon team up to make a thermonuclear device. Pam's primary is designed to yield 6.70 kT. She places it next to Sheldon's secondary, which contains 31.5 g of deuterium surrounded by a ^{238}U tamper and is designed to compress to a radius of 1.67 cm. **Report all answers to 3 significant figures.**

- (30) a. Calculate the 100% burn yield from fusion of Sheldon's secondary in kT.

$$N = \frac{m N_A}{M} = \frac{(31.5 \text{ g})(N_A \text{ #/mol})}{(2.01410 \text{ g/mol})} = 9.418 \times 10^{24} \text{ D atoms}$$

$$Y = N \cdot Q = \text{atoms} \cdot \frac{1 \text{ DD fusion}}{2 \text{ atoms}} \cdot \frac{9.96 \text{ MeV}}{1 \text{ fusion}} \cdot \frac{1 \text{ kT}}{2.613 \times 10^{25} \text{ MeV}} = 1.795 \text{ kT}$$

$$Y = \frac{N_{\text{tot}}}{2} Q$$

$$= \frac{m N_A Q_{\text{DD}}}{M} \checkmark$$

$$Y_s = 1.80 \text{ kT}$$

Report your answer in this box.

- (20) b. Calculate the fission yield in kT of Sheldon's secondary if fusion neutrons are 50% efficient at ^{238}U fission. Assume the same fission Q-values for all U isotopes.

$1/5$ D
fuse w/
T, carrying
14.1 MeV
for ~~fission~~
fission of
 ^{238}U

$$Y = \frac{N_D}{5} \cdot \frac{1}{2} \cdot Q_{238}$$

$$= \frac{N_D Q_{238}}{10}$$

$$= \frac{9.418 \times 10^{24} \text{ atoms} \cdot 180 \text{ MeV}}{(5)(10)} \cdot \frac{1 \text{ kT}}{2.613 \times 10^{25} \text{ MeV}}$$

$$= 6.488 = 6.49 \text{ kT}$$

$$Y_{\text{SF}} = 6.49 \text{ kT}$$

Report your answer in this box.

- (10) c. Calculate the total yield of the device in kT.

$$Y = Y_{\text{PF}} + Y_s + Y_{\text{SF}}$$

$$= 6.70 \text{ kT} + 1.80 \text{ kT} + 6.49 \text{ kT}$$

$$= 14.99 = 15.0 \text{ kT}$$

$$Y = 15.0 \text{ kT}$$

Report your answer in this box.

Wt. No.

Cadet: _____

10

B

Bonus: Pam thinks she can make a better secondary than Sheldon. She fills her's with D-T fuel at a 1:1 atom ratio. Upon compression, it is squeezed to a total number density of 1.25×10^{25} atoms/cm³ at a temperature of 3.75×10^8 K and a volume of 67.0 cm³. Report all answers to 3 significant figures.

Calculate the reaction rate of Pam's secondary. Note: There is no partial credit for bonus problems, it is all or nothing.

$$N_{\text{reac}} = (\sigma v)_{\text{DT}} n_{\text{D}} n_{\text{T}} V$$

$$N_{\text{reac}} = (\sigma v)_{\text{DT}} \frac{n_{\text{tot}}^2}{4} V$$

$$(\sigma v)_{\text{DT}} = 3.68 \times 10^{-22} \text{ E}^{-2/3} \exp(-19.94 \text{ E}^{-1/3})$$

$$T = 3.75 \times 10^8 \text{ K} \cdot \frac{1 \text{ eV}}{1.1605 \times 10^4 \text{ K}} \cdot \frac{1 \text{ keV}}{1000 \text{ eV}}$$

$$= 32.3 \text{ keV}$$

$$N_{\text{reac}} = \frac{(6.941 \times 10^{-16} \text{ cm}^3/\text{s})(1.25 \times 10^{25} \text{ atoms/cm}^3)^2}{4} \cdot (67.0 \text{ cm}^3)$$

$$= 1.816 \times 10^{36} \text{ \#}/\text{s}$$

* on iRDC it says that $(\sigma v)_{\text{DT}}$ is only valid for $T < 23 \text{ keV}$ but,

$$N_{\text{reac}} = 1.82$$

$$(\sigma v)_{\text{DT}} = 3.68 \times 10^{-12} (32.3)^{-2/3} \exp(-19.94 (32.3)^{-1/3})$$

$$= 6.941 \times 10^{-16} \text{ cm}^3/\text{s}$$

$$N_{\text{reac}} = 1.82 \times 10^{36} \text{ s}^{-1}$$

Report your answer in this box.

Poor Student Work

Student

Total Points
219 / 260 pts

Question 1

Assembly

10 / 10 pts

✓ - 0 pts C (Correct)

Question 2

Shut down

10 / 10 pts

✓ - 0 pts A (Correct)

Question 3

Fusion

10 / 10 pts

✓ - 0 pts B (Correct)

Question 4

Centrifuge

10 / 10 pts

✓ - 0 pts True (Correct)

Question 5

Six-factor formula

3 / 10 pts

✓ - 2 pts ϵ

✓ - 5 pts Doesn't Select η

Question 6

Tamper

10 / 10 pts

✓ - 0 pts False (Correct)

Question 7

Reflector

10 / 10 pts

✓ - 0 pts Selects only High σ_s Low σ_a and Low Z

Question 8

Enrichment

57 / 60 pts

8.1 — **Feed mass**

10 / 10 pts

✓ - 0 pts Correct 1.22×10^4 kg8.2 — **SWU**

27 / 30 pts

✓ - 3 pts Incorrect answer

8.3 — **Stages**

20 / 20 pts

✓ - 0 pts Correct (4.11×10^3 stages)**Question 9**

Primary

55 / 60 pts

9.1 — **Multiplication factor**

10 / 10 pts

✓ - 0 pts Correct (1.90)

9.2 — **Neutron lifetime**

10 / 10 pts

✓ - 0 pts Correct (1.86×10^{-9} s)9.3 — **Bomb alpha**

10 / 10 pts

✓ - 0 pts Correct/follows (4.84 gen/shake)

9.4 — **Yield**

25 / 30 pts

✓ - 2 pts Doesn't convert time to gl

✓ - 3 pts Incorrect answer

Question 10

Secondary

44 / 60 pts

10.1 — **Fusion yield**

24 / 30 pts

✓ - 3 pts Incorrect subs

✓ - 3 pts Incorrect answer

10.2 — **Fission yield**

14 / 20 pts

✓ - 4 pts Doesn't divide n total by 5

✓ - 2 pts Incorrect answer

10.3 — **Total yield**

6 / 10 pts

✓ - 3 pts Incorrect subs

✓ - 1 pt Incorrect answer

Question 11

Bonus: Rxn rate

0 / 10 pts

✓ - 10 pts Incorrect answer

Time of completion 1525
11/12 Feb 2026

Cadet [REDACTED]
(First) (Last)

Section 5
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review I

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 10 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **20 February 2026 at 1700.**
- b. Cadet Signature: [REDACTED]
- c. Cadet Name (print) [REDACTED]

- 10 1. Which statement correctly identifies the two most common methods for assembling a supercritical mass AND explains why plutonium cannot be used in one of them?
- ☐ Gun-type and compression-type; plutonium cannot be used in gun-type assembly because it has too low of fission cross section for the gun mechanism to work effectively.
 - ☐ Implosion-type and linear-type; plutonium cannot be used in linear assembly because its density is too high to achieve the required assembly speed.
 - ☒ Gun-type and implosion-type; plutonium cannot be used in gun-type assembly because its high spontaneous fission rate would cause pre-initiation before full assembly is achieved.
 - ☐ Gun-type and implosion-type; plutonium cannot be used in implosion-type assembly because the shock wave would cause the plutonium to fragment before reaching criticality.
 - ☐ Boosted-type and implosion-type; plutonium cannot be used in boosted-type because it reacts with tritium in the fusion fuel.
- 10 2. Which statement correctly identifies the two mechanisms that shut down the chain reaction in a fissioning weapon AND accurately describes one?
- ☒ Kinetic disassembly and burn-up; kinetic disassembly occurs when the weapon assembly expands and bomb alpha becomes negative.
 - ☐ Neutron leakage and fuel depletion; neutron leakage occurs when neutrons escape the core faster than fission can replace them to sustain the chain reaction.
 - ☐ Pre-initiation and post-initiation; pre-initiation occurs when spontaneous fission starts the chain reaction too early, causing the assembly to disassemble before achieving maximum yield.
 - ☐ Kinetic disassembly and neutron leakage; neutron leakage occurs when the surface area to volume ratio becomes large enough for bomb alpha to become negative.
 - ☐ Burn-up and thermal expansion; thermal expansion occurs when heat from fission causes the fuel to melt into a non-spherical configuration.

- 10 3. What is the primary purpose of fusion in a nuclear weapon?
- ☐ To generate more yield than fission with higher Q values.
 - ☒ To produce high-energy neutrons that increase fission rate and yield.
 - ☐ To boost the outward velocity, u_2 , of the secondary.
 - ☐ To initiate the fission chain reaction by providing the initial neutron population.
- 10 4. Gas centrifuge enrichment efficiency is dependent on the rotor's angular velocity, radius, and temperature.
- ☒ TRUE
 - ☐ FALSE
- 10 5. Select **all** the components of the six-factor formula that apply to weapons.
- | | |
|---|--|
| ☒ ϵ fast fission factor | ☐ η thermal fission factor |
| ☐ p resonance escape probability | ☒ P_{NL}^f |
| ☐ f | ☐ P_{NL}^{th} thermal non-leakage factor |
- 10 6. The purpose of a tamper is to push neutrons back into the fissioning core.
→ Delay - kinetic disassembly
- ☐ TRUE
 - ☒ FALSE
- 10 7. Select **all** properties of a good reflector in a weapon assembly.
- | | |
|--|--|
| ☐ Low σ_s | ☐ Low σ_f |
| ☒ High σ_s | ☐ High σ_f |
| ☒ Low σ_a | ☒ Low Z |
| ☐ High σ_a | ☐ High Z |

Wt. No.

60

8.

Pam, an aspiring nuclear weapons designer, requires 67.0 kg of Highly Enriched Uranium at 93.0% by weight of ^{235}U . Assume a natural feed assay of 0.711%, and a tails assay of 0.200%. For brevity, $M_F = 19.0$ g/mol, and $M_U = 235$ g/mol and 238 g/mol. Report all answers to 3 significant figures.

Cadet: _____

- (10) a. Calculate the mass of feed, in kg, Pam needs to complete her enrichment.

$$x_F = 0.9311$$

$$x_W = 0.002$$

$$F = ?$$

$$F = P \left(\frac{x_F - x_W}{x_F - x_W} \right)$$

$$F = (67 \text{ kg}) \left(\frac{0.93 - 0.002}{0.00711 - 0.002} \right)$$

$$F = 12,167.5 \text{ kg}$$

natural U

$$F = 12,200 \text{ kg}$$

natural U.

Report your answer in this box.

- (30) b. Calculate the total separative work Pam needs to produce her product.

$$P = 67 \text{ kg}$$

$$F = 12,167.5 \text{ kg}$$

$$W = 12,100.5 \text{ kg}$$

$$V(x_i) = (2x_i - 1) \ln \left(\frac{x_i}{1-x_i} \right)$$

$$\begin{aligned} & \times \left\{ \begin{aligned} V(x_F) &= 4.8689 \text{ SWU} \\ V(x_P) &= 2.22455 \text{ (SWU)} \\ V(x_W) &= 6.18776 \text{ SWU} \end{aligned} \right. \end{aligned}$$

$$\text{SWU} = [P V(x_P) + W V(x_W) - F V(x_F)]$$

$$= 134,266 \text{ SWU} \cdot \text{kg}$$

$$\text{SWU} =$$

$$134,000 \text{ SWU} \cdot \text{kg}$$

Report your answer in this box.

- (20) c. Calculate the number of enrichment stages Pam needs if she only has access to a gaseous diffusion facility using UF_6 .

$$\alpha = \sqrt{\frac{M_H}{M_L}}$$

$$M_H = 6(19) + 238 = 352 \text{ amu}$$

$$M_L = 6(19) + 235 = 349 \text{ amu}$$

$$\alpha = \sqrt{\frac{352}{349}} = 1.0043$$

$$N = \frac{2}{\ln \alpha} \ln \left(\frac{x_P(1-x_W)}{x_W(1-x_P)} \right) - 1$$

$$N = 4111.17$$

$$\hookrightarrow 4112 \text{ stages} \quad (\text{Rounding})$$

$$4110 \text{ stages}$$

Report your answer in this box.

$$\rho = 67.0 \text{ g/cm}^3 \quad n_0 = 1.67 \times 10^4 \text{ n/s}$$

$$L = 0.67$$

$$259 \text{ p.m.}$$

Wt. No.

Cadet: _____

- 60 9. Sheldon, an OG nuclear weapons engineer, assembles a bare sphere of ^{239}Pu with a density of 67.0 g/cm^3 and initiates it with 1.67×10^4 neutrons. The non-leakage factor for his assembly is 0.670 . Report all answers to 3 significant figures.

- (10) a. Calculate the multiplication factor of Sheldon's assembly.

$$k = k_{\infty} L \quad k_{\infty} = \frac{\sigma_F}{\sigma_a}$$

$$k_{\infty} = (2.98) \left(\frac{(5.61/2.98)}{1.98} \right) = 2.833; \quad L = 0.670$$

$$k = 1.8983$$

$$k = 1.90$$

Report your answer in this box.

- (10) b. Calculate the average neutron lifetime of Sheldon's assembly.

$$l_{\infty} = \frac{1}{\Sigma_a v}; \quad \Sigma_a = \sigma_a \left(\frac{\rho N_a}{m} \right) = 1.986 \left(\frac{(67.0 \text{ g/cm}^3)(6.0221419 \times 10^{23} \text{ 1/mol})}{(185 \text{ g/mol})} \right)$$

$$\Sigma_a = 0.33427 \text{ 1/cm}$$

$$l_{\infty} L = (0.67) / (0.334266 \text{ cm}^{-1} \cdot 1.08 \times 10^9 \text{ cm/s}) = l$$

$$l = 1.856 \times 10^{-9} \text{ s} \text{ or } 1.856 \text{ ns}$$

$$l = 1.856 \times 10^{-9} \text{ s}$$

Report your answer in this box.

- (10) c. Calculate the "bomb alpha" of Sheldon's assembly in gen/shake.

$$\alpha = \frac{k-1}{l} = \frac{1.8983-1}{1.856 \times 10^{-9} \text{ s}} = 4.84 \text{ gen/shake}$$

$$\text{or}$$

$$4.84 \times 10^{-8} \text{ gen/s}$$

$$\alpha = 4.84 \text{ gen/shake}$$

Report your answer in this box.

- (30) d. Calculate the yield after 50 generations, neglecting any shutdown mechanisms.

$$Y(t) = \left(\frac{\Sigma_F v Q n_0}{\alpha} \right) (e^{\alpha t} - 1)$$

$$Y(t) = \left(\frac{(0.317815072 \text{ 1/cm})(1.08 \times 10^9 \text{ cm/s})(185 \text{ MeV})(1.67 \times 10^4 \text{ n/s})}{(4.840381028 \times 10^{-8} \text{ gen/s})} \right) (e^{50} - 1) \left(\frac{1 \text{ kT}}{2.63 \times 10^7 \text{ MeV}} \right)$$

$$Y(t) = 4.347019 \times 10^{18} \text{ kT}$$

(made a unit mistake somewhere)

$$\alpha t = \frac{50}{1} = 50 \text{ gen}$$

$$Y = 4.35 \times 10^{18} \text{ kT}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 60 10. Pam and Sheldon team up to make a thermonuclear device. Pam's primary is designed to yield 6.70 kT. She places it next to Sheldon's secondary, which contains 31.5 g of deuterium surrounded by a ^{238}U tamper and is designed to compress to a radius of 1.67 cm. **Report all answers to 3 significant figures.**

- (30) a. Calculate the 100% burn yield from fusion of Sheldon's secondary in kT.

$$\begin{array}{|c|c|c|c|c|c|}
 \hline
 31.5 \text{ g} & \text{D} & 1 \text{ mol} & 6.024119 \times 10^{23} & 1 \text{ rxn} & 3.27 \text{ MeV} & 1 \text{ kT} \\
 \hline
 1 & 2 \text{ g} & \text{mol} & 2 & 1 \text{ rxn} & 2.613 \times 10^{25} \text{ MeV} & \\
 \hline
 \end{array}$$

$= 0.593485 \text{ kT}$

$4.742436817 \times 10^{24}$ neutrons
 or
 rxns

$$Y = 0.593 \text{ kT}$$

Report your answer in this box.

- (20) b. Calculate the fission yield in kT of Sheldon's secondary if fusion neutrons are 50% efficient at ^{238}U fission. Assume the same fission Q-values for all U isotopes.

$$\begin{array}{|c|c|c|c|}
 \hline
 4.742436817 \times 10^{24} \text{ neutrons} & 185 \text{ MeV} & 1 \text{ kT} & \\
 \hline
 2 \text{ (efficiency)} & 1 \text{ neutron} & 2.613 \times 10^{25} \text{ MeV} & \\
 \hline
 \end{array}$$

$\approx Y = 16.79$
 $\approx 16.8 \text{ kT}$

$$Y = 16.8 \text{ kT}$$

Report your answer in this box.

- (10) c. Calculate the total yield of the device in kT.

$$16.79 \text{ kT} + 0.593 \text{ kT} + 4.35 \times 10^8 \text{ kT}$$

$= 4.35 \times 10^8 \text{ kT}$

$$4.35 \times 10^8 \text{ kT}$$

(it's correct but follows from Qd)

Report your answer in this box.

Wt. No.

Cadet: _____

10

B

Bonus: Pam thinks she can make a better secondary than Sheldon. She fills her's with D-T fuel at a 1:1 atom ratio. Upon compression, it is squeezed to a total number density of 1.25×10^{25} atoms/cm³ at a temperature of 3.75×10^8 K and a volume of 67.0 cm³. Report all answers to 3 significant figures.

Calculate the reaction rate of Pam's secondary. Note: There is no partial credit for bonus problems, it is all or nothing.

$$E = \frac{3.75 \times 10^8 \text{ K} \cdot 1 \text{ eV} \cdot 1 \text{ keV}}{1 \cdot 1.160564 \times 10^{-4} \text{ eV}} = 32.31365 \text{ keV}$$

$$N_{\text{reac}}(\text{vol}) = V(\sigma v) n_{\text{tar}} n_{\text{bomb}}$$

$$(67 \text{ cm}^3)(1.25 \times 10^{25} \text{ atoms/cm}^3) = 8.375 \times 10^{26} \text{ atoms}$$

$$= n_{\text{tar}}$$

$$(67 \text{ cm}^3)(8.375 \times 10^{26} / 2)^2 (6.93 \times 10^{-16})$$

$$\hookrightarrow N_{\text{reac}}(\text{vol}) = 8.14 \times 10^{39}$$

$$(\sigma v)_{\text{DT}} = 3.68 \times 10^{-12} (32.31365 \text{ keV})^{-2/3} e^{-19.94 (32.3)^{-1/3}} = 6.73056299 \times 10^{-16}$$

$$8.14 \times 10^{39} \text{ rxn/s}$$

Report your answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

WPR #2

Instructor Solution

Time of completion _____ Cadet _____
13/14 Apr 2026 (First) (Last)

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review II

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 12 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **17 April 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

- 10 1. Which answer best describes the double flash phenomenon that occurs during a nuclear detonation?
- ☐ During burn-out, the weapon case re-emits x-rays creating the first flash. During diffusion growth, thermal radiation escapes creating the second flash.
 - ☐ At diffusion growth, the expanding fireball creates the first flash. At hydrodynamic growth, compression of the shock front creates a brighter second flash.
 - ☐ The first flash occurs during burn-out when the fissioning core releases energy. The second flash occurs at hydrodynamic separation when the blast wave shock heats the surrounding air.
 - ☐ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic growth, the shock front creates the second flash.
 - ☒ At hydrodynamic separation, the blast wave shrouds the x-ray fireball, causing it to dim. At breakaway, the fireball shines through, creating the second flash.
- 10 2. Which answer best describes the formation of an electromagnetic pulse (EMP) in a high-altitude burst?
- ☒ Prompt gamma rays escaping the weapon case ionize the upper atmosphere via Compton scattering, producing free electrons. Earth's magnetic field accelerates these electrons, and the accelerating charges emit broad-spectrum electromagnetic waves.
 - ☐ X-rays from the weapon case heat the upper atmosphere, creating a plasma that expands rapidly. The expanding plasma generates a magnetic field that induces an electromagnetic pulse in nearby conductors.
 - ☐ Neutrons from the weapon interact with atmospheric nitrogen, producing secondary gamma rays. These gamma rays create an ionized channel that acts as an antenna, radiating electromagnetic energy.
 - ☐ The fireball's thermal radiation ionizes the atmosphere, creating a conductive path. The weapon's residual magnetic field couples with this path, generating a pulse that propagates along Earth's magnetic field lines.
 - ☐ Beta particles from fission products spiral along Earth's magnetic field lines, creating a time-varying current. This current generates electromagnetic waves that propagate to the surface as an EMP.

Wt. No.

- 10 3. What is meant by a "clean" versus "dirty" nuclear weapon?
- ☒ A "clean" weapon derives more yield from fusion reactions, while a "dirty" weapon derives more yield from fission reactions.
 - ☐ A "clean" weapon is detonated at high altitude to minimize fallout, while a "dirty" weapon is detonated at ground level.
 - ☐ A "clean" weapon uses highly enriched uranium (HEU), while a "dirty" weapon uses plutonium as its fissile material.
 - ☐ A "clean" weapon has a tamper that reflects neutrons efficiently, while a "dirty" weapon lacks a neutron reflector.
- 10 4. Which effect does hydrodynamic enhancement enhance?
- ☐ Thermal
 - ☐ Blast and Shock
 - ☒ Initial Radiation
 - ☐ Residual Radiation
 - ☐ EMP
- 10 5. Dynamic overpressure imparts a crushing force on objects.
- ☐ TRUE
 - ☒ FALSE
- 10 6. What mechanism primarily determines the partition of energy into different nuclear weapon effects (blast, thermal, x-rays, etc.)?
- ☐ The ratio of fission to fusion yield in the weapon design
 - ☐ The total yield of the weapon
 - ☐ The type of nuclear fuel used
 - ☒ The interactions with the surrounding medium
- 10 7. Blackbody radiation and Bremsstrahlung radiation are the primary mechanisms for the energy release of a fissioning sphere.
- ☒ TRUE
 - ☐ FALSE

Wt. No.

Cadet: _____

- 50 8. SAT GUS is a selfie taking satellite built by youtuber Mark Rober and named after a squirrel. Pam, an aspiring nuclear targeteer, doesn't like squirrels. How close to SAT GUS does Pam need to detonate a 500. kT weapon to make 4.00 J/cm³ of pressure 0.369 cm into it's copper shield? Assume the Planckian distribution of energy can be approximated as a single energy of 0.800 MeV photons. Report your answer in **km** to **three** significant figures.

$$p(x) = E_{dep}(x)\rho_0$$

$$E_{dep}(x) = F_0 e^{-\mu x} \left(\frac{\mu_a}{\rho} \right)$$

$$E_{dep}(x) = \frac{p(x)}{\rho_0}$$

$$F_0 = \frac{E_{dep}(x)}{e^{-\frac{\mu}{\rho}\rho x} \left(\frac{\mu_a}{\rho} \right)}$$

$$E_{dep}(x) = \frac{4 \frac{J}{cm^3}}{8.96 \frac{g}{cm^3}}$$

$$F_0 = \frac{0.4464 \frac{J}{g}}{e^{-0.0654 \frac{cm^2}{g}} 8.96 \frac{g}{cm^3} 0.369 cm \left(0.0271 \frac{cm^2}{g} \right)}$$

$$E_{dep}(x) = 0.4464 \frac{J}{g}$$

$$F_0 = 20.4485 \frac{J}{cm^2}$$

$$F(r) = \frac{Y_x}{4\pi r^2}$$

$$r^2 = \frac{0.7Y}{4\pi F}$$

$$r^2 = \frac{0.7(500 kT) \left(4.186 \times 10^{12} \frac{J}{kT} \right)}{4\pi 20.4485 \frac{J}{cm^2}}$$

$$r^2 = 5.7016 \times 10^{12} cm^2$$

$$r = 2.3878 \times 10^6 cm$$

$$r = 23.9 km$$

23.9 km

Report your answer in this box.

Wt. No.

Cadet: _____

- 40 9. Sheldon, a cold cat, sabotages Pam's squirrel hunting mission to catch some thermal energy for warmth. Pam's 500. kT weapon detonates at a height of just 4.00 kft. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

- (20) a. Calculate the amount of thermal energy emitted 2.00 seconds after the explosion to **three** significant figures.

$$t_{max} = 0.0417W^{0.44}$$

$$t_{max} = 0.0417(500)^{0.44}$$

$$t_{max} = 0.6422 \text{ s}$$

$$\frac{t}{t_{max}} = \frac{2}{0.6422} = 3.11$$

Fig 5-15 (7.84): In 3.11 Out 0.60

$$E_{tot} = fW$$

$$\frac{E}{E_{tot}} = 0.60$$

$$\frac{E}{fW} = 0.60$$

$$E = 0.60fW$$

$$E = 0.60(0.35)(500 \text{ kT})$$

$$E = 105 \text{ kT}$$

105 kT

Report your answer in this box.

- (20) b. Sheldon is located 50.0 kft from ground zero. Calculate his radiant exposure in **cal/cm²** to **three** significant figures.

$$D^2 = d^2 + h^2$$

$$D^2 = (50 \text{ kft})^2 + (4 \text{ kft})^2$$

$$D^2 = 2516 \text{ kft}^2$$

$$Q = \frac{85.6fW\tau}{D^2}$$

$$f = 0.35$$

$$\tau = 0.4$$

$$Q = \frac{85.6(0.35)(500)(0.4)}{2516 \text{ kft}^2}$$

$$Q = 2.3816 \frac{\text{cal}}{\text{cm}^2}$$

2.38 $\frac{\text{cal}}{\text{cm}^2}$

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 10. Pam, disappointed with Sheldon's sabotage, placed Sheldon's favorite dangle toy 8,731 ft from ground zero. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

- (20) a. Calculate the peak overpressure the toy will experience from the 500. kT burst at an altitude of 4,000 ft in **psi** to **two** significant figures.

$$h = h_1 W^{\frac{1}{3}}$$

$$d = d_1 W^{\frac{1}{3}}$$

Fig 5-6 (3.73b): In h 500, d 1100

Out 15 psi

$$h_1 = \frac{h}{W^{\frac{1}{3}}}$$

$$d_1 = \frac{d}{W^{\frac{1}{3}}}$$

$$h_1 = \frac{4000 \text{ ft}}{500^{\frac{1}{3}}}$$

$$d_1 = \frac{8731 \text{ ft}}{500^{\frac{1}{3}}}$$

$$h_1 = 503.97 \text{ ft}$$

$$d_1 = 1100 \text{ ft}$$

15 psi

Report your answer in this box.

- (10) b. Calculate how long Pam has to run away before the blast wave arrives to **two** significant figures.

Fig 5-10 (3.73b): In h 500, d 1100

Out 0.6 sec

$$t = t_1 W^{\frac{1}{3}}$$

$$t = 0.6 \text{ sec} (500^{\frac{1}{3}})$$

$$t = 4.76 \text{ sec}$$

4.8 sec

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 11. MAJ Pert is taking a picture of this cat mischief to post on Jodel. He is standing at a slant range of 4,000 yards from the 500 kT burst. Assuming a 60% fission thermonuclear device, calculate the total initial radiation absorbed dose MAJ Pert will receive in **rads** to **three** significant figures. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

$$D = D_n + D_{\gamma s} + D_{\gamma f}$$

Fig 5-18 (8.123b): In 4000 Out 1.2×10^{-4}

$$D_n = 1.2 \times 10^{-4} \frac{\text{rads}}{\text{kT}} (500 \text{ kT})$$

$$D_n = 0.06 \text{ rads}$$

Fig 5-20 (8.127b): In 4000 Out 1.9×10^{-3}

$$D_{\gamma s} = 1.9 \times 10^{-3} \frac{\text{rads}}{\text{kT}} (500 \text{ kT})$$

$$D_{\gamma s} = 0.95 \text{ rads}$$

Fig 5-21 (8.130a): In 4000 Out 3.5×10^{-4}

Fig 5-22 (8.130b): In 500 Out 1×10^4

$$D_{\gamma f} = 3.5 \times 10^{-4} \frac{\text{rads}}{\text{kT}} (1 \times 10^4 \text{ kT})(0.6)$$

$$D_{\gamma f} = 2.1 \text{ rads}$$

$$D = 0.06 + 0.95 + 2.1$$

$$D = 3.11 \text{ rads}$$

3.11 rads

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 12. Sheldon fell asleep after receiving his thermal rays, but the fallout cloud passed right overhead. You are tasked for his recovery mission. If you measure the dose rate on entry at 3 hours after detonation as 7.00 rads/hr, calculate the total dose you'll receive if you spend 2 hours recovering Sheldon. Report your answer in **rads** to **three** significant figures. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

$$\text{Fig 5 - 24 (9.16a): } 3 \text{ hrs} \rightarrow 0.27 \frac{DR}{UTRD}$$

$$\text{Fig 5 - 25 (9.20): } 3 \text{ hrs} \rightarrow 6.2 \frac{D_T}{UTRD}$$

$$\frac{DR}{UTRDR} = 0.27$$

$$\text{Fig 5 - 25 (9.20): } 5 \text{ hrs} \rightarrow 6.6 \frac{D_T}{UTRD}$$

$$R_1 = \frac{DR}{0.27}$$

$$\frac{TD}{UTRDR} = 6.5 - 6.2 = 0.4$$

$$R_1 = \frac{7 \frac{\text{rad}}{\text{hr}}}{0.27}$$

$$TD = 0.4 \text{ UTRDR}$$

$$TD = 0.4 (25.93)$$

$$R_1 = 25.93 \frac{\text{rad}}{\text{hr}}$$

$$TD = 10.37 \text{ rad}$$

Alternate:

Alternate:

$$R = R_1 t^{-1.2}$$

$$D_T = 5R_1(t_a^{-0.2} - t_b^{-0.2})$$

$$R_1 = \frac{R}{t^{-1.2}}$$

$$D_T = 5(26.16)(3^{-0.2} - 5^{-0.2})$$

$$D_T = 10.197 \text{ rad}$$

$$R_1 = \frac{7 \frac{\text{rad}}{\text{hr}}}{3^{-1.2}}$$

$$R_1 = 26.16 \frac{\text{rad}}{\text{hr}}$$

10.2 rads

Report your answer in this box.

Outstanding Student Work

Student



Total Points
223 / 250 pts

Question 1

Double Flash

0 / 10 pts

✓ - 10 pts A

Question 2

EMP

10 / 10 pts

✓ - 0 pts A Correct

Question 3

Clean/Dirty

10 / 10 pts

✓ - 0 pts A Correct

Question 4

Hydrodynamic Enhancement

10 / 10 pts

✓ - 0 pts Initial Radiation Correct

Question 5

Dynamic Overpressure

10 / 10 pts

✓ - 0 pts False Correct

Question 6

Energy Partitions

10 / 10 pts

✓ - 0 pts D Correct

Question 7

Energy Release

0 / 10 pts

✓ - 10 pts False

Question 8

X-ray Effects

45 / 50 pts

✓ - 5 pts Incorrect answer

Question 9

Thermal Effects

40 / 40 pts

9.1 Thermal Energy Emission

20 / 20 pts

✓ - 0 pts Correct 96-114 kT

9.2 Radiant Exposure

20 / 20 pts

✓ - 0 pts Correct 1.78-2.98 cal/cm²

Question 10

Blast ans Shock

28 / 30 pts

10.1 Peak Overpressure

18 / 20 pts

✓ - 2 pts Incorrect answer

10.2 Time of Arrival

10 / 10 pts

✓ - 0 pts Correct 4.36-5.16 sec

Question 11

Initial Radiation

30 / 30 pts

✓ - 0 pts Correct ~3.11 rad

Question 12

Delayed Radiation

30 / 30 pts

✓ - 0 pts Correct ~10.2 rad

Time of completion 1521
13/14 Apr 2026

Cadet

(First)

(Last)

Section 31
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review II

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 12 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **17 April 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

- 10 1. Which answer best describes the double flash phenomenon that occurs during a nuclear detonation?
- ☒ During burn-out, the weapon case re-emits x-rays creating the first flash. During diffusion growth, thermal radiation escapes creating the second flash.
 - ☐ At diffusion growth, the expanding fireball creates the first flash. At hydrodynamic growth, compression of the shock front creates a brighter second flash.
 - ☐ The first flash occurs during burn-out when the fissioning core releases energy. The second flash occurs at hydrodynamic separation when the blast wave shock heats the surrounding air.
 - ☐ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic growth, the shock front creates the second flash.
 - ☐ At hydrodynamic separation, the blast wave shrouds the x-ray fireball, causing it to dim. At breakaway, the fireball shines through, creating the second flash.
- 10 2. Which answer best describes the formation of an electromagnetic pulse (EMP) in a high-altitude burst?
- ☒ Prompt gamma rays escaping the weapon case ionize the upper atmosphere via Compton scattering, producing free electrons. Earth's magnetic field accelerates these electrons, and the accelerating charges emit broad-spectrum electromagnetic waves.
 - ☐ X-rays from the weapon case heat the upper atmosphere, creating a plasma that expands rapidly. The expanding plasma generates a magnetic field that induces an electromagnetic pulse in nearby conductors.
 - ☐ Neutrons from the weapon interact with atmospheric nitrogen, producing secondary gamma rays. These gamma rays create an ionized channel that acts as an antenna, radiating electromagnetic energy.
 - ☐ The fireball's thermal radiation ionizes the atmosphere, creating a conductive path. The weapon's residual magnetic field couples with this path, generating a pulse that propagates along Earth's magnetic field lines.
 - ☐ Beta particles from fission products spiral along Earth's magnetic field lines, creating a time-varying current. This current generates electromagnetic waves that propagate to the surface as an EMP.

Wt. **No.**

Cadet: _____

- 10 3. What is meant by a "clean" versus "dirty" nuclear weapon?
- ☒ A "clean" weapon derives more yield from fusion reactions, while a "dirty" weapon derives more yield from fission reactions.
 - ☐ A "clean" weapon is detonated at high altitude to minimize fallout, while a "dirty" weapon is detonated at ground level.
 - ☐ A "clean" weapon uses highly enriched uranium (HEU), while a "dirty" weapon uses plutonium as its fissile material.
 - ☐ A "clean" weapon has a tamper that reflects neutrons efficiently, while a "dirty" weapon lacks a neutron reflector.
- 10 4. Which effect does hydrodynamic enhancement enhance?
- ☐ Thermal
 - ☐ Residual Radiation
 - ☐ Blast and Shock
 - ☐ EMP
 - ☒ Initial Radiation
- 10 5. Dynamic overpressure imparts a crushing force on objects.
- ☐ TRUE
 - ☒ FALSE
- 10 6. What mechanism primarily determines the partition of energy into different nuclear weapon effects (blast, thermal, x-rays, etc.)?
- ☐ The ratio of fission to fusion yield in the weapon design
 - ☐ The total yield of the weapon
 - ☐ The type of nuclear fuel used
 - ☒ The interactions with the surrounding medium
- 10 7. Blackbody radiation and Bremsstrahlung radiation are the primary mechanisms for the energy release of a fissioning sphere.
- ☐ TRUE
 - ☒ FALSE

Wt. **No.**

Cadet: _____

50

8.

SAT GUS is a selfie taking satellite built by youtuber Mark Rober and named after a squirrel. Pam, an aspiring nuclear targeteer, doesn't like squirrels. How close to SAT GUS does Pam need to detonate a 500. kT weapon to make 4.00 J/cm^3 of pressure 0.369 cm into it's copper shield? Assume the Planckian distribution of energy can be approximated as a single energy of 0.800 MeV photons. Report your answer in km to three significant figures.

$$\frac{\text{J}}{\text{m}^2} \left[\frac{\text{m}^2}{\text{g}} \right]$$

$$\frac{\text{J}}{\text{g}} \left[\frac{\text{g}}{\text{cm}^3} \right]$$

$$\rho_0 = 8.96 \text{ g/cm}^3$$

$$P(x) = E_{\text{dep}}(x) P_0$$

$$E_{\text{dep}} = f_0 e^{-\mu_g x} \left(\frac{\mu_a}{\rho} \right)_g$$

$$f_0 = \frac{0.7 Y}{4\pi r^2}$$

$$P(x) = E_{\text{dep}} P_0$$

$$= \left[\frac{0.7 Y}{4\pi r^2} e^{-\mu_g x} \left(\frac{\mu_a}{\rho} \right)_{\text{air}} \right] P_{0,\text{Cu}}$$

$$r = \left(\left[\frac{0.7 Y}{4\pi P(x)} e^{-\mu_g x} \left(\frac{\mu_a}{\rho} \right)_{\text{air}} \right] P_{0,\text{Cu}} \right)^{1/2}$$

$$= \left[\frac{0.7 (2.09 \times 10^{15} \text{ J})}{4\pi (4.00 \text{ J/cm}^3)} e^{-(0.586 \text{ cm}^{-1} \cdot 0.369 \text{ cm})} \left(0.0289 \text{ cm}^2/\text{g} \right) (8.96 \text{ g/cm}^3) \right]^{1/2} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \cdot \frac{1 \text{ km}}{1000 \text{ m}}$$

$$r = d = 24.6 \text{ km}$$

$$Y = 500 \text{ kT} \cdot \frac{4.186 \times 10^{12} \text{ J}}{\text{kT}} = 2.09 \times 10^{15} \text{ J}$$

$$P(x) = 4.00 \text{ J/cm}^3$$

$$\mu_g = \left(\frac{\mu}{\rho} \right) P_0 = (0.0654 \text{ cm}^2/\text{g}) (8.96 \text{ g/cm}^3)$$

$$\left(\frac{\mu_a}{\rho} \right)_{\text{air}} = 0.0289 \text{ cm}^2/\text{g}$$

$$x = 0.369 \text{ cm}$$

$$d = 24.6 \text{ km}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 40 9. Sheldon, a cold cat, sabotages Pam's squirrel hunting mission to catch some thermal energy for warmth. Pam's 500. kT weapon detonates at a height of just 4.00 kft. If you need to reference a graph, ensure to indicate starting values and specify the figure number used.

- (20) a. Calculate the amount of thermal energy emitted 2.00 seconds after the explosion to three significant figures.

$$4 \text{ kft} = 4,000 \text{ ft}$$

$$\therefore P_{\max} \approx 3.18 \text{ W}^{0.56}$$

$$t_{\max} \approx 0.0417 \text{ W}^{0.44}$$

$$t_{\max} \approx 0.0417 (500 \text{ kT})^{0.44}$$

$$\approx 0.6422 \text{ s}$$

$$t/t_{\max} = 2.00 / 0.6422 \text{ s} = 3.115$$

fig: 5-15

$$\ln \approx 3.1 \text{ } t/t_{\max}$$

$$\text{out: } P/P_{\max} \approx 0.2$$

TE $\approx 60\%$

$$E_{\text{tot}} = f W$$

from 7.88 $f = 0.35$

$$= 0.35 (500 \text{ kT}) (0.60)$$

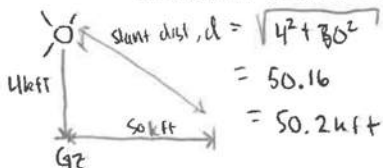
$$= 105 \text{ kT}$$

Beat Navy

$$E_{\text{tot}, t} = 105 \text{ kT}$$

Report your answer in this box.

- (20) b. Sheldon is located 50.0 kft from ground zero. Calculate his radiant exposure in cal/cm^2 to three significant figures.



$$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) \approx \frac{85.6 f W \uparrow}{D^2 [\text{kft}]}$$

$$f = 0.35$$

$\uparrow$: figure 5-16

In: d from GZ, 50 kft

WOB $\approx 4 \text{ kft}$

$$\text{out } \uparrow \approx 0.45$$

$$Q = \frac{85.6 (0.35) (500 \text{ kT}) (0.45)}{(50.2 \text{ kft})^2}$$

$$\approx 2.67$$

AAAAA

Beat Navy

$$Q \approx 2.67 \text{ cal/cm}^2$$

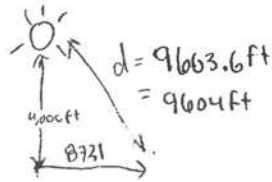
Report your answer in this box.

Wt. No.

Cadet: _____

- 30 10. Pam, disappointed with Sheldon's sabotage, placed Sheldon's favorite dangle toy 8,731 ft from ground zero. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

- (20) a. Calculate the peak overpressure the toy will experience from the 500. kT burst at an altitude of 4,000 ft in **psi** to **two** significant figures.



$$d_1 = \frac{d}{W^{1/3}}$$
$$h_1 = \frac{4000\text{ft}}{(500\text{kT})^{1/3}}$$
$$\approx 503.9$$
$$\approx 504$$
$$d_1 = \frac{8731\text{ft}}{(500)^{1/3}} \approx 1100\text{ft}$$

figure 5-6 in: $h_1 = 500$ $d_1 = 1100$

out: $P_1 = 15$ psi

$$P = P_1 W^{1/3}$$
$$= (15 \text{ psi}) (500\text{kT})^{1/3}$$
$$= 119 \text{ psi}$$

Beat Navy

$$P = 120 \text{ psi}$$

Report your answer in this box.

- (10) b. Calculate how long Pam has to run away before the blast wave arrives to **two** significant figures.

figure 5-10: in: $h_1 = 500\text{ft}$
 $d_1 = 1100\text{ft}$

out: 0.6 Sec

$$t = t_1 \times W^{1/3}$$

$$t = (0.6\text{s}) (500\text{kT})^{1/3}$$
$$= 4.76\text{s}$$

Beat Navy

$$t = 4.8\text{s}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 11. MAJ Pert is taking a picture of this cat mischief to post on Jodel. He is standing at a slant range of 4,000 yards from the 500 kT burst. Assuming a 60% fission thermonuclear device, calculate the total initial radiation absorbed dose MAJ Pert will receive in rads to three significant figures. If you need to reference a graph, ensure to indicate starting values and specify the figure number used.

$$\text{Slant range} = d = 4,000 \text{ yd}$$

$$Y = 500 \text{ kT}$$

Thermonuclear

$$D = D_n + D_{rs} + D_{rf}$$

$$D_n: \text{fig 5-18: in: 4,000 yds} \\ \text{out: } 1.5 \times 10^{-4} \text{ rad/kT}$$

$$D_n = (500 \text{ kT})(1.5 \times 10^{-4} \text{ rad/kT}) \\ = 0.075 \text{ rad}$$

$$D_{rs}: \text{fig 5-20} \\ \text{in } \approx 4,000 \text{ yds} \\ \text{out } \approx 1.9 \times 10^{-3} \text{ rad/kT}$$

$$D_{rs} = (1.9 \times 10^{-3})(500 \text{ kT}) \\ = 0.95 \text{ rad}$$

$$D_{rf} = \text{figure 5-22: in: 4000 yds} \\ \text{500 kT} \\ \text{out: } 10^4 \text{ kT}$$

$$= \text{figure 5-21: in: 4000 yds} \\ \text{out: } 3.5 \times 10^{-4} \text{ rad/kT}$$

$$D_{rf} = (10^4 \text{ kT})(3.5 \times 10^{-4} \text{ rad/kT})(60\%) \\ = 2.1 \text{ rad}$$

$$D = D_n + D_{rs} + D_{rf}$$

$$= 0.075 + 0.95 + 2.1 \text{ rad}$$

$$= 3.125$$

$$= 3.13 \text{ rad}$$

$$D = 3.13 \text{ rad}$$

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 30 12. Sheldon fell asleep after receiving his thermal rays, but the fallout cloud passed right overhead. You are tasked for his recovery mission. If you measure the dose rate on entry at 3 hours after detonation as 7.00 rads/hr, calculate the total dose you'll receive if you spend 2 hours recovering Sheldon. Report your answer in **rads** to three significant figures. If you need to reference a graph, ensure to indicate starting values and specify the figure number used.

$$R_t = R_i t^{-1.2}$$

$$R_i = \frac{R_t}{t^{-1.2}}$$

$$= \frac{7.00 \text{ rad/hr}}{(3 \text{ hr})^{-1.2}}$$

$$= 26.16 \text{ rad/hr}$$

$$t_a = 3 \text{ hr}$$

$$t_b = 5 \text{ hr}$$

$$D \approx 5 R_i (t_a^{-0.2} - t_b^{-0.2})$$

$$\approx 5 (26.16) \left((3 \text{ hr})^{-0.2} - (5 \text{ hr})^{-0.2} \right)$$

$$\approx 10.19 \text{ rad}$$

$$D = 10.2 \text{ rad}$$

Report your answer in this box.

Poor Student Work

Student

Total Points
179 / 250 pts

Question 1

Double Flash

10 / 10 pts

✓ - 0 pts E Correct

Question 2

EMP

10 / 10 pts

✓ - 0 pts A Correct

Question 3

Clean/Dirty

0 / 10 pts

✓ - 10 pts B

Question 4

Hydrodynamic Enhancement

0 / 10 pts

✓ - 10 pts Blast and Shock

Question 5

Dynamic Overpressure

0 / 10 pts

✓ - 10 pts True

Question 6

Energy Partitions

10 / 10 pts

✓ - 0 pts D Correct

Question 7

Energy Release

0 / 10 pts

✓ - 10 pts False

Question 8

X-ray Effects

50 / 50 pts

✓ - 0 pts Correct 23.9 km

Question 9

Thermal Effects

20 / 40 pts

9.1 Thermal Energy Emission

0 / 20 pts

✓ - 4 pts Doesn't use Eq 25 (t max)

✓ - 2 pts Doesn't solve t / tmax

✓ - 4 pts Doesn't use Fig 5-15

✓ - 2 pts Doesn't get E / Etot (0.55-0.65)

✓ - 4 pts Doesn't use Eq 27 (E total)

✓ - 2 pts Incorrect subs in Eq 27

✓ - 2 pts Incorrect answer

9.2 Radiant Exposure

20 / 20 pts

✓ - 0 pts Correct 1.78-2.98 cal/cm²

Question 10

Blast and Shock

24 / 30 pts

10.1 Peak Overpressure

14 / 20 pts

✓ - 4 pts Doesn't use Fig 5-6

✓ - 2 pts Incorrect answer

10.2 Time of Arrival

10 / 10 pts

✓ - 0 pts Correct 4.36-5.16 sec

Question 11

Initial Radiation

25 / 30 pts

✓ - 2 pts Incorrect subs in ff gamma dose (2.5 - 3.5 E-4)

✓ - 3 pts Incorrect answer

Question 12

Delayed Radiation

30 / 30 pts

✓ - 0 pts Correct ~10.2 rad

Time of completion 1525
13/14 Apr 2026

Cadet [REDACTED]
(First) (Last)

Section J1
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Written Partial Review II

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 12 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **75 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **17 April 2026 at 1700**.
- b. Cadet Signature: [REDACTED]
- c. Cadet Name (print): [REDACTED]

- 10 1. Which answer best describes the double flash phenomenon that occurs during a nuclear detonation?
- ☐ During burn-out, the weapon case re-emits x-rays creating the first flash. During diffusion growth, thermal radiation escapes creating the second flash.
 - ☒ At diffusion growth, the expanding fireball creates the first flash. At hydrodynamic growth, compression of the shock front creates a brighter second flash.
 - ☒ The first flash occurs during burn-out when the fissioning core releases energy. The second flash occurs at hydrodynamic separation when the blast wave shock heats the surrounding air.
 - ☒ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic growth, the shock front creates the second flash.
 - ☒ At hydrodynamic separation, the blast wave shrouds the x-ray fireball, causing it to dim. At breakaway, the fireball shines through, creating the second flash.
- 10 2. Which answer best describes the formation of an electromagnetic pulse (EMP) in a high-altitude burst?
- ☒ Prompt gamma rays escaping the weapon case ionize the upper atmosphere via Compton scattering, producing free electrons. Earth's magnetic field accelerates these electrons, and the accelerating charges emit broad-spectrum electromagnetic waves.
 - ☒ X-rays from the weapon case heat the upper atmosphere, creating a plasma that expands rapidly. The expanding plasma generates a magnetic field that induces an electromagnetic pulse in nearby conductors.
 - ☒ Neutrons from the weapon interact with atmospheric nitrogen, producing secondary gamma rays. These gamma rays create an ionized channel that acts as an antenna, radiating electromagnetic energy.
 - ☒ The fireball's thermal radiation ionizes the atmosphere, creating a conductive path. The weapon's residual magnetic field couples with this path, generating a pulse that propagates along Earth's magnetic field lines.
 - ☒ Beta particles from fission products spiral along Earth's magnetic field lines, creating a time-varying current. This current generates electromagnetic waves that propagate to the surface as an EMP.

Wt. No.

Cadet: _____

10

3.

What is meant by a "clean" versus "dirty" nuclear weapon?

☒ A "clean" weapon derives more yield from fusion reactions, while a "dirty" weapon derives more yield from fission reactions.

☒ A "clean" weapon is detonated at high altitude to minimize fallout, while a "dirty" weapon is detonated at ground level.

☒ A "clean" weapon uses highly enriched uranium (HEU), while a "dirty" weapon uses plutonium as its fissile material.

☐ A "clean" weapon has a tamper that reflects neutrons efficiently, while a "dirty" weapon lacks a neutron reflector.

10

(4)

Which effect does hydrodynamic enhancement enhance?

☐ Thermal

☒ Residual Radiation

☒ Blast and Shock

☒ EMP

☒ Initial Radiation

10

(5)

Dynamic overpressure imparts a crushing force on objects.

☒ TRUE

☐ FALSE

10

6.

What mechanism primarily determines the partition of energy into different nuclear weapon effects (blast, thermal, x-rays, etc.)?

☐ The ratio of fission to fusion yield in the weapon design

☐ The total yield of the weapon

☐ The type of nuclear fuel used

☒ The interactions with the surrounding medium

10

7.

Blackbody radiation and Bremsstrahlung radiation are the primary mechanisms for the energy release of a fissioning sphere.

☐ TRUE

☒ FALSE

Wt. No.

Cadet: _____

50

8. SAT GUS is a selfie taking satellite built by youtuber Mark Rober and named after a squirrel. Pam, an aspiring nuclear targeteer, doesn't like squirrels. How close to SAT GUS does Pam need to detonate a 500. kT weapon to make 4.00 J/cm³ of pressure 0.369 cm into it's copper shield? Assume the Planckian distribution of energy can be approximated as a single energy of 0.800 MeV photons. Report your answer in km to three significant figures.

$$Y_x = 0.7 Y, \quad F(r) = \frac{Y_x}{4\pi r^2}; \quad E_{dep}(x) = \int_0^x F e^{-\mu_g x} \left(\frac{\mu_a}{\rho} \right)$$

$$p(x) = E_{dep}(x) \rho_0$$

$$p(x) = \left(\frac{0.7 Y}{4\pi r^2} \right) e^{-\left(\mu_t/\rho \right)(x)(\rho) \left(\frac{\mu_a}{\rho} \right) \rho_0}$$

$$\frac{p(x)(4\pi r^2)}{0.7 Y \left(\frac{\mu_a}{\rho} \right) e^{-\left(\mu_t/\rho \right)(x)(\rho) \left(\frac{\mu_a}{\rho} \right) \rho_0}} = \frac{1}{2} \quad \begin{aligned} p(x) &= 4 \text{ J/cm}^3 \\ Y &= 500 \text{ kT} = 2.093 \times 10^{15} \text{ J} \\ \frac{\mu_a}{\rho} &= .0271 \text{ cm}^2/\text{g} \\ \frac{\mu_t}{\rho} &= 0.0654 \text{ cm}^2/\text{g} \\ \rho &= 8.96 \text{ g/cm}^3 \end{aligned}$$

$$x = 0.369 \text{ cm}$$

$$\frac{1}{r^2} = \frac{(4 \text{ J/cm}^3)(4\pi)}{0.7(2.093 \times 10^{15} \text{ J}) \left(.0271 \frac{\text{cm}^2}{\text{g}} \right) \left(e^{-\left(.0654 \frac{\text{cm}^2}{\text{g}} \right) (0.369 \text{ cm}) (8.96 \text{ g/cm}^3)} \right) (8.96 \text{ g/cm}^3)}$$

$$\frac{1}{r^2} = 1.754 \times 10^{-13} \frac{1}{\text{cm}^2};$$

$$r = 2387723.4 \text{ cm}$$

$$r = 23.88 \text{ km}$$

$$r = 23.9 \text{ km}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 40 9. Sheldon, a cold cat, sabotages Pam's squirrel hunting mission to catch some thermal energy for warmth. Pam's 500. kT weapon detonates at a height of just 4.00 kft. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

- (20) a. Calculate the amount of thermal energy emitted 2.00 seconds after the explosion to three significant figures.

~~$P_{max} = \frac{3.56 W^{0.59}}{[P(h)/P_0]^{0.45}} \text{ kT/s}$~~

~~$\frac{P(4.0 \text{ kft})}{P_0} \approx 0.86$~~

$P_{max} = \left(\frac{3.56 W^{0.59}}{[0.86]^{0.45}} \right) (2) \left(\frac{1.999812 \text{ cal}}{\text{kT}} \right)$

$P_{max} = 2.9788 \text{ E14 cal}$

$F = 0.35 \rightarrow \text{TBL 7-88}$

$W = 500 \text{ kT}$

$\epsilon \approx 0.9 \rightarrow \text{Fig 5-16}$

$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) \approx \frac{85.6 F W \epsilon}{D^2 (\text{kft})}$

$Q = \frac{(85.6)(0.35)(500 \text{ kT})(0.9)}{(4)^2} = 842.625 \text{ cal/cm}^2$

~~$Q = 2.98 \times 10^{14} \text{ cal}$~~
Report your answer in this box.

- (20) b. Sheldon is located 50.0 kft from ground zero. Calculate his radiant exposure in cal/cm² to three significant figures.

$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) \approx \frac{(85.6) F W \epsilon}{D^2 (\text{kft})}$

$D = 50 (\text{kft})$

$\epsilon = 0.9$

$F = 0.35$

$Q = \frac{(85.6)(0.35)(500)(0.9)}{(50)^2}$

$Q = 2.398 \rightarrow 2.40$

$Q = 2.40 \text{ cal/cm}^2$
Report your answer in this box.

Wt. No.

30

10.

Bazinga

Cadet: _____

Pam, disappointed with Sheldon's sabotage, placed Sheldon's favorite dangle toy 8,731 ft from ground zero. **If you need to reference a graph, ensure to indicate starting values and specify the figure number used.**

- (20) a. Calculate the peak overpressure the toy will experience from the 500. kT burst at an altitude of 4,000 ft in **psi** to **two** significant figures.

$$p = p_0 S_p \quad S_p = \frac{p}{p_0} \quad S_d = \left(\frac{p_0}{p} \right)^{1/3} \quad d = d_1 \times W^{1/3} \quad h = h_1 \times W^{1/3}$$

$$d_1 = 1055 \text{ ft} \quad h_1 = 504 \text{ ft}$$

$$\frac{p}{p_0} = (S_d)^{-3} \quad \text{using FS-8, } p_1 \text{ is between 2 and 4 psi, } p_1 \approx 3 \text{ psi}$$

$$d = d_1 W^{1/3} S_d$$

$$S_d = \frac{d}{d_1 W^{1/3}} = \frac{(8731 \text{ ft})}{(1055 \text{ ft})(500 \text{ kT})^{1/3}} = 1.043$$

$$p = p_0 S_p = (3 \text{ psi}) (0.882) = 2.646 \text{ psi}$$

$$\frac{p}{p_0} = S_d^{-3} = 0.882 = S_p$$

$$p = p_0 S_p = (0.882)(3 \text{ psi}) = 2.646 \text{ psi}$$

2.6 psi

Report your answer in this box.

- (10) b. Calculate how long Pam has to run away before the blast wave arrives to **two** significant figures.

$$d_1 = 1055 \text{ ft}$$

$$h_1 = 504 \text{ ft}$$

Using FS-11,

$$t_1 \approx 0.55 \text{ s}$$

$$t = (0.55 \text{ s}) \times (500 \text{ kT})^{1/3} = 4.36 \text{ s}$$

t = 4.36 s

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 11. MAJ Pert is taking a picture of this cat mischief to post on Jodel. He is standing at a slant range of 4,000 yards from the 500 kT burst. Assuming a 60% fission thermonuclear device, calculate the total initial radiation absorbed dose MAJ Pert will receive in rads to three significant figures. If you need to reference a graph, ensure to indicate starting values and specify the figure number used.

$$D_n = (500 \text{ kT}) (1.2 \text{ E-}4 \text{ rad/kT}) = 0.06 \text{ rad}$$

↑
FS-17, 4000 yd_s

$$D_{1/2} = (500 \text{ kT}) (1.8 \text{ E-}3 \text{ rad/kT}) = 0.9 \text{ rad}$$

↑
FS-20, 4000 yd

$$D_{7F} = (0.60 \times 8000 \text{ kT}) (.003 \text{ rad/kT}) = 14.4 \text{ rad}$$

↑
effective
yield
From
FS-22,
started
4000 yd,
500 kT

$$\Sigma D = 15.36 \text{ rad}$$

$$D = 15.4 \text{ rad}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 12. Sheldon fell asleep after receiving his thermal rays, but the fallout cloud passed right overhead. You are tasked for his recovery mission. If you measure the dose rate on entry at 3 hours after detonation as 7.00 rads/hr, calculate the total dose you'll receive if you spend 2 hours recovering Sheldon. Report your answer in **rads** to **three significant figures**. If you need to reference a graph, ensure to indicate starting values and specify the figure number used.

Bazinga

$$D \approx 5R_1(t_a^{-0.2} - t_b^{-0.2})$$

$$D = 5(26.16 \text{ rad/hr})((3 \text{ hr})^{-0.2} - (5 \text{ hr})^{-0.2})$$
$$= 10.2 \text{ rad}$$

$$R_3 = 7 \text{ rad/hr}$$

$$R_t = R_1 t^{-1.2}$$

$$R_1 = \frac{7 \text{ rad/hr}}{(3 \text{ hr})^{-1.2}} = 26.16 \text{ rad/hr at hr 1}$$

10.2 rads

Report your answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I TERM END EXAMINATION

Instructor Solution

Time of completion _____ Cadet _____
12/13 May 2026 (First) (Last)

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Term End Exam

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 10 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **210 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **20 February 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

- 10 1. Which statement best explains how each method of weapon assembly increases bomb alpha?
- ☐ Both gun-type and implosion-type methods rapidly increase density of the fissile material
 - ☐ Both gun-type and implosion-type methods methods rapidly increase mass-to-surface-area ratio of the fissile material
 - ☐ Gun-type rapidly increases the density of fissile material, while implosion-type rapidly increases mass-to-surface-area ratio of fissile material.
 - ☒ Gun-type rapidly increases the mass-to-surface-area ratio of fissile material, while implosion-type rapidly increases density of fissile material.
 - ☐ Gun-type shoots highly compressed D-T gas into the fissile material for fusion neutrons, while implosion-type compresses D-T gas for fusion neutrons.
- 10 2. Which answer best describes why a nuclear detonation initially appears bright, then dark, then bright again?
- ☐ During burn-out, x-rays fully ionize the atmosphere, creating the x-ray fireball and first flash. At hydrodynamic separation, the blast wave shock heats the air, causing a second flash.
 - ☐ During diffusion growth, the expanding x-ray fireball creates the first flash. During hydrodynamic growth, the expanding shock front creates a second flash.
 - ☒ At hydrodynamic separation, the shock front hides the x-ray fireball. At breakaway, the shock front stops hiding the x-ray fireball.
 - ☐ During burnout, the weapon x-rays super heat the atmosphere, causing it to produce visible light. During hydrodynamic growth, the shock front re-heats their atmosphere, causing a second flash.
 - ☐ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic separation, the shock front creates the second flash.

Wt. No.

- 10 3. What is the primary purpose of fusion in a nuclear weapon?
- ☐ To generate more yield than fission with higher Q values.
 - ☒ To produce high-energy neutrons that increase fission rate and yield.
 - ☐ To boost the outward velocity, u_2 , of the secondary.
 - ☐ To initiate the fission chain reaction by providing the initial neutron population.
- 10 4. Gas centrifuge enrichment efficiency is dependent on the rotor's angular velocity, radius, and temperature.
- ☒ TRUE
 - ☐ FALSE
- 10 5. Select **all** the components of the six-factor formula that apply to weapons.
- | | |
|----------------------------------|---|
| ☐ ϵ | ☒ η |
| ☐ p | ☒ P_{NL}^f |
| ☐ f | ☐ P_{NL}^{th} |
- 10 6. Blackbody radiation and Bremsstrahlung radiation are the primary mechanisms for the energy release of a fissioning sphere.
- ☐ TRUE
 - ☒
 - ☐ FALSE
- 10 7. Fallout free HOB.
- | | |
|--|--|
| ☐ Low σ_s | ☐ Low σ_f |
| ☒ High σ_s | ☐ High σ_f |
| ☒ Low σ_a | ☒ Low Z |
| ☐ High σ_a | ☐ High Z |

Wt. No.

Cadet: _____

- 10 8. Which answer best describes hydrodynamic enhancement?
- ☐ Prompt gamma rays escaping the weapon case ionize the upper atmosphere via Compton scattering, producing free electrons. Earth's magnetic field accelerates these electrons, and the accelerating charges emit electromagnetic waves.
 - ☐ X-rays from the weapon case heat the upper atmosphere, creating a plasma that expands rapidly. The expanding plasma generates a magnetic field that induces an electromagnetic pulse in nearby conductors.
 - ☐ Neutrons from the weapon interact with atmospheric nitrogen, producing secondary gamma rays. These gamma rays create an ionized channel that acts as an antenna, radiating electromagnetic energy.
 - ☐ The fireball's thermal radiation ionizes the atmosphere, creating a conductive path. The weapon's residual magnetic field couples with this path, generating a pulse that propagates along Earth's magnetic field lines.
 - ☐ Beta particles from fission products spiral along Earth's magnetic field lines, creating a time-varying current. This current generates electromagnetic waves that propagate to the surface as an EMP.
-
- 10 4. UTRDR
- ☐ Thermal
 - ☐ Blast and Shock
 - ☐ Initial Radiation
 - ☐ Residual Radiation
 - ☐ EMP
-
- 10 5. Mach Stem.
- ☐ TRUE
 - ☐ FALSE

Wt. No.

Cadet: _____

- 50 8. A country desires 23.0 kg of uranium enriched to 82.0%. The feed material UF_6 has a natural enrichment of 0.711%, and the tail assay (waste material) is depleted to 0.200%. Calculate the number of gaseous diffusion enrichment stages required for this desire. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$\alpha = \sqrt{\frac{m_H}{m_L}}$$

$$N = \frac{2}{\ln(\alpha)} \ln \left(\frac{x_p(1 - x_w)}{x_w(1 - w_p)} \right) - 1$$

$$\alpha = \sqrt{\frac{352}{349}}$$

$$N = \frac{2}{\ln(1.0043)} \ln \left(\frac{0.82(1 - 0.002)}{0.002(1 - 0.82)} \right) - 1$$

$$N = 3604$$

$$\alpha = 1.0043$$

3604 stages

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 9. A notional spherical weapon assembly composed of pure U-235 has a radius of 6.00 cm and a density of 35.0 g/cm³. Assume the non-leakage probability is 0.750. Calculate “bomb alpha” of the assembly. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$k = k_{\infty}L$$

$$l = l_{\infty}L$$

$$k_{\infty} = \frac{\nu\sigma_f}{\sigma_a}$$

$$l_{\infty} = \frac{1}{\Sigma_a \nu}$$

$$k = \frac{\nu\sigma_f}{\sigma_a}L$$

$$\Sigma_a = N\sigma_a$$

$$N = \frac{\rho N_A}{M}$$

$$k = \frac{3.32}{1.44} 0.75$$

$$\Sigma_a = \frac{\rho N_A \sigma_a}{M}$$

$$k = 1.73$$

$$l = \frac{ML}{\rho N_A \sigma_a \nu}$$

$$l = \frac{235(0.75)}{19.1 \frac{g}{cm^3} (6.022 \times 10^{23} \frac{atoms}{g}) (1.44 \times 10^{-24} cm^2) (1.08 \times 10^9 \frac{cm}{s})}$$

$$l = 5.38 \times 10^{-9} s$$

$$\alpha = \frac{k - 1}{l}$$

$$\alpha = \frac{1.73 - 1}{5.38 \times 10^{-9} s}$$

$$\alpha = 1.36 \times 10^{-8} sec^{-1}$$

$$1.36 \times 10^{-8} s^{-1}$$

Report your answer in this box.

Wt. No.

Cadet: _____

50 10. A pure U-235 fission primary device is boosted with 4.00 grams of Deuterium and 6.00 grams of Tritium. Assume this to be 2.00 mol of each gas. Assume this boosting gas to achieve 100% efficiency fusion.

(30) a. Calculate the 100% fusion yield (in kT) resulting from this fusion. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$Y = Num_{reac}Q$$

$$Y = 2 \text{ mol} \left(6.022 \times 10^{23} \frac{\text{atoms}}{\text{mol}} \right) (17.6 \text{ MeV}) \left(\frac{1 \text{ kT}}{2.61 \times 10^{25} \text{ MeV}} \right)$$

$$Y = 0.811 \text{ kT}$$

0.811 kT

Report your answer in this box.

(20) b. Calculate the energy yield (in kT) resulting from the fission reactions caused by the neutrons produced by the D-T fusion reactions. Assume 100% efficiency (ie every fusion produced neutron causes a fission.) **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$Y = Num_{reac}Q$$

$$Y = 2 \text{ mol} \left(6.022 \times 10^{23} \frac{\text{atoms}}{\text{mol}} \right) (180 \text{ MeV}) \left(\frac{1 \text{ kT}}{2.61 \times 10^{25} \text{ MeV}} \right)$$

$$Y = 8.30 \text{ kT}$$

8.30 kT

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 8. A 250.0 kT nuclear weapon is detonated at an altitude of 45.0 km. A target is located at an altitude of 25.0 km with a ground range of zero (i.e. target is located directly under the detonation). Assume the Planckian distribution of x-rays can be approximated as a single energy group of 0.150 MeV photons. Assume 70% of the weapon yield is released as x-rays. Accounting for only geometric attenuation (ignoring atmospheric scattering and attenuation), calculate the total x-ray fluence, in J/m², at the target. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$F(r) = \frac{Y_x}{4\pi r^2}$$

$$F(r) = \frac{0.7(250 \text{ kT})}{4\pi(20,000 \text{ m})^2} 4.186 \times 10^{12} \frac{\text{J}}{\text{kT}}$$

$$F(r) = 1.457 \times 10^5 \frac{\text{J}}{\text{m}^2}$$

$$1.46 \times 10^5 \frac{\text{J}}{\text{m}^2}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 9. A 700.0 kT nuclear weapon is detonated at an altitude of 20,000.0 ft. Calculate the radiant exposure on the ground at a distance of 48,000.0 ft from ground zero. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$D^2 = d^2 + h^2$$

$$D^2 = (48 \text{ kft})^2 + (20 \text{ kft})^2$$

$$D^2 = 2516 \text{ kft}^2$$

$$Q = \frac{85.6 f W \tau}{D^2}$$

$$f = 0.41 \text{ Table 9}$$

$$\tau = 0.7 \text{ Figure 16}$$

$$Q = \frac{85.6(0.41)(700)(0.7)}{2704 \text{ kft}^2}$$

$$Q = 6.30 \frac{\text{cal}}{\text{cm}^2}$$

$$6.30 \frac{\text{cal}}{\text{cm}^2}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 10. A 100.0 kT nuclear weapon is detonated on the ground. Calculate the peak overpressure at your location (on the ground) 2500 ft from ground zero. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$d = d_1 W^{\frac{1}{3}}$$

Fig 5-6 (3.73b): In h 0, d 538

Out 40 psi

$$d_1 = \frac{d}{W^{\frac{1}{3}}}$$

$$d_1 = \frac{2500 \text{ ft}}{100^{\frac{1}{3}}}$$

$$d_1 = 538 \text{ ft}$$

40 psi

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 11. A 1.25 MT thermonuclear weapon is detonated at 1,000.0 ft above the surface. Calculate the total equivalent neutron dose a soldier will receive located at a slant range of 2,000.0 yards. Only determine the neutron equivalent dose. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

Fig 5-18 (8.123b): In 2000 Out 1.2

$$D_n = 1.2 \frac{\text{rads}}{\text{kT}} (1250 \text{ kT})$$

$$D_n = 1500 \text{ rad}$$

$$H_n = 1500 \text{ rad (10)}$$

$$H_n = 15000 \text{ rem}$$

15,000 rem

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 12. The dose rate in the fallout area is measured to be 7.00 rad/hr at 4.0 hours after the nuclear detonation. Determine the dose rate at 10.0 hours after the nuclear detonation. **If you need to reference a graph/table, ensure to indicate starting values and specify the figure number used.**

$$\text{Fig 5 - 24 (9.16a): } 4 \text{ hrs} \rightarrow 0.2 \frac{DR}{UTRD}$$

$$\frac{DR}{UTRDR} = 0.2$$

$$R_1 = \frac{DR}{0.2}$$

$$R_1 = \frac{7 \frac{\text{rad}}{\text{hr}}}{0.2}$$

$$R_1 = 35 \frac{\text{rad}}{\text{hr}}$$

$$\text{Fig 5 - 24 (9.16a): } 10 \text{ hrs} \rightarrow 0.07 \frac{DR}{UTRD}$$

$$\frac{DR}{UTRDR} = 0.07$$

$$DR = 0.07 R_1$$

$$DR = 0.07 \left(35 \frac{\text{rad}}{\text{hr}} \right)$$

$$DR = 2.45 \frac{\text{rad}}{\text{hr}}$$

Alternate:

$$R = R_1 t^{-1.2}$$

$$R_1 = \frac{R}{t^{-1.2}}$$

$$R_1 = \frac{7 \frac{\text{rad}}{\text{hr}}}{4^{-1.2}}$$

$$R_1 = 36.95 \frac{\text{rad}}{\text{hr}}$$

Alternate:

$$R = R_1 t^{-1.2}$$

$$R = 36.95 \frac{\text{rad}}{\text{hr}} 10^{-1.2}$$

$$R = 2.33 \frac{\text{rad}}{\text{hr}}$$

$$2.33 \frac{\text{rad}}{\text{hr}}$$

Report your answer in this box.

Outstanding Student Work

Student



Total Points

485 / 510 pts

Question 1

Assembly Methods

10 / 10 pts

✓ - 0 pts D

Question 2

Double Flash

10 / 10 pts

✓ - 0 pts C

Question 3

Fallout-free HOB

10 / 10 pts

✓ - 0 pts Correct B, D, F, H

Question 4

UTRDR

10 / 10 pts

✓ - 0 pts True

Question 5

Mach Stem

10 / 10 pts

✓ - 0 pts B

Question 6

EMP

10 / 10 pts

✓ - 0 pts E

Question 7

Bomb Alpha

10 / 10 pts

✓ - 0 pts True

Question 8

Fusion

10 / 10 pts

✓ - 0 pts Correct (All)

Question 9

PUREX

0 / 10 pts

✓ - 10 pts True

Question 10

Spontaneous Fission

0 / 10 pts

✓ - 10 pts D

Question 11

Bonus

10 / 10 pts

✓ - 0 pts B

Question 12

Enrichment Stages

45 / 50 pts

✓ - 5 pts Incorrect answer

Question 13

Bomb Alpha

50 / 50 pts

✓ - 0 pts Correct 1.36×10^8 gen/sec

Question 14

Fusion

50 / 50 pts

14.1 Fusion Yield

30 / 30 pts

✓ - 0 pts Correct 0.811 kT

14.2 Secondary Fission Yield

20 / 20 pts

✓ - 0 pts Correct 8.30 kT

Question 15

X-ray Fluence

50 / 50 pts

✓ - 0 pts Correct 1.46×10^5 J/m²

Question 16

Radiant Exposure

50 / 50 pts

✓ - 0 pts Correct 6.4 cal/cm²

Question 17

Peak Overpressure

50 / 50 pts

✓ - 0 pts Correct 40 psi

Question 18

Neutron Dose

50 / 50 pts

✓ - 0 pts Correct 15,000 rem

Question 19

Dose Rate

50 / 50 pts

✓ - 0 pts Correct 2.3 rad/hr

Time of completion 1910
12/13 May 2026

Cadet

(First)

(Last)

Section

51

Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Term End Exam

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 18 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **210 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **14 May 2026 at 1700**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

- 10 1. Which statement **best** explains how each method of weapon assembly increases bomb alpha?
- ☐ Both gun-type and implosion-type methods rapidly increase density of the fissile material
 - ☐ Both gun-type and implosion-type methods rapidly increase mass-to-surface-area ratio of the fissile material
 - ☐ Gun-type rapidly increases the density of fissile material, while implosion-type rapidly increases mass-to-surface-area ratio of fissile material.
 - ☒ Gun-type rapidly increases the mass-to-surface-area ratio of fissile material, while implosion-type rapidly increases density of fissile material.
 - ☐ Gun-type shoots highly compressed D-T gas into the fissile material for fusion neutrons, while implosion-type compresses D-T gas for fusion neutrons.
- 10 2. Which statement **best** describes why a nuclear detonation initially appears bright, then dark, then bright again?
- ☐ During burn-out, x-rays fully ionize the atmosphere, creating the x-ray fireball and first flash. At hydrodynamic separation, the blast-wave shock heats the air, causing a second flash.
 - ☐ During diffusion growth, the expanding x-ray fireball creates the first flash. During hydrodynamic growth, the expanding shock front creates a second flash.
 - ☒ At hydrodynamic separation, the shock front hides the x-ray fireball. At breakaway, the shock front stops hiding the x-ray fireball.
 - ☐ During burnout, the weapon x-rays superheat the atmosphere, causing it to produce visible light. During hydrodynamic growth, the shock front reheats their atmosphere, causing a second flash.
 - ☐ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic separation, the shock front creates the second flash.

Wt. No.

Cadet: _____

- 10 3. Select **all** of the following answers that are true about a fallout-free height of burst:
- ☐ The fireball touches the ground
 - ☒ The fireball does not touch the ground
 - ☐ Earth is lofted into the mushroom cloud
 - ☒ Earth is not lofted into the mushroom cloud
 - ☐ Fractionation occurs
 - ☒ Fractionation won't occur
 - ☐ $HOB < 180W^{0.4}$
 - ☒ $HOB > 180W^{0.4}$
- 10 4. For a given position in a fallout field, the unit time reference dose rate is unchanging.
- ☒ TRUE R_1
→ unchanged
- ☐ FALSE
- 10 5. The reinforced shock front where ground reflected shock catches up to and reinforces the primary shock along the ground is known as:
- ☐ A triple point
 - ☒ A Mach stem
 - ☐ Hydrodynamic enhancement
 - ☐ Dynamic overpressure
- 10 6. Select which step is **not** a part of EMP formation:
- ☐ Earth's magnetic field turns electrons
 - ☐ Compton scatter events liberate electrons in the upper atmosphere
 - ☐ Emitted EM fields form a downward pulse
 - ☐ Prompt gammas escape the weapon case
 - ☒ Gammas couple with antenna-like objects on the ground
- 10 7. Bomb alpha is the neutron population multiplication rate.
- ☒ TRUE $k ?$
- ☐ FALSE

Wt. No.

Cadet: _____

- 10 8. Select **all** the advantages of using fusion in a weapon assembly:
- ☒ Higher ν (nu) value
 - ☒ Added yield
 - ☒ Added neutrons
 - ☒ Fissionable isotopes can fission
- 10 9. PUREX is a plutonium enrichment process for producing weapons-grade plutonium from spent reactor fuel.
- ☒ TRUE
 - ☐ FALSE
- 10 10. Select the property of Pu-240 that makes it a contaminant for use in nuclear weapons:
- ☐ Low fission energy yield
 - ☐ Low macroscopic fission cross section
 - ☐ High spontaneous fission rate *→ maybe they too.*
 - ☒ Fissionable isotope
 - ☐ Short half-life
- 10 B BONUS: Which treaty legally prohibits the US from various types of nuclear weapons testing?
- ☐ NPT
 - ☒ PTBT
 - ☐ CTBT
 - ☐ SALT I
 - ☐ START I

Wt. No.

Cadet: _____

- 50 11. A country desires 23.0 kg of uranium enriched to 82.0%. The feed material UF_6 has a natural enrichment of 0.711%, and the tail assay (waste material) is depleted to 0.200%. Calculate the number of gaseous diffusion enrichment stages required for this desire. **If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.**

$$X_P = 0.82$$

$$X_F = 0.00711$$

$$X_W = 0.002$$

$$\alpha = \sqrt{\frac{m_H}{m_L}}$$

$$= 1.0043$$

$$^{238}\text{M} = 238 \text{ Amu}$$

$$m^H = 238 + 6(19) = 352 \text{ Amu}$$

$$^{235}\text{M} = 235 \text{ Amu}$$

$$m^L = 235 + 6(19) = 249$$

$$^F\text{M} \approx 19 \text{ Amu}$$

$$N = \frac{2}{\ln \alpha} \left[\frac{X_P(1-X_W)}{X_W(1-X_P)} \right] - 1$$

$$= \frac{2}{\ln(1.0043)} \left[\frac{0.82(1-0.002)}{0.002(1-0.82)} \right] - 1$$

$$= 1.06 \times 10^6 \text{ stages}$$

$$N = 1.06 \times 10^6 \text{ stages}$$

Report your answer in this box.

Wt. No.

50 12.

Cadet: _____

A notional spherical weapon assembly composed of pure U-235 has a radius of 6.00 cm and a density of 35.0 g/cm³. Assume the non-leakage probability is 0.750. Calculate "bomb alpha" of the assembly to three significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

100% U-235

G: $\rho = 35 \text{ g/cm}^3$

$r = 6 \text{ cm}$

$L = 0.750$

$v = 1.08 \times 10^9 \text{ cm/s}$

$N^{235} = \frac{P N_a}{m}$

$= \frac{35 \text{ g/cm}^3 \cdot 6.02214 \times 10^{23} \text{ #/mol}}{235 \text{ g/mol}}$

$= 8.97 \times 10^{22} \text{ #/cm}^3$

$\Sigma_a = N \sigma_a$

$= (8.97 \times 10^{22} \text{ #/cm}^3) (1.44 \times 10^{-24} \text{ cm}^2)$

$= 0.129 \text{ cm}^{-1}$

F: α

$k_{\infty} = \frac{\nu \sigma_f}{\sigma_a} = 2.306$

$K = k_{\infty} \cdot L$
 $= 1.729$

$\alpha = \frac{K-1}{K}$

$= \frac{(1.729 - 1)}{(5.38 \times 10^{-9} \text{ s})}$

$= 1.36 \times 10^8 \text{ s}^{-1}$

$l_{\infty} = \frac{1}{\Sigma_a v}$

$= \frac{1}{(0.129 \text{ cm}^{-1})(1.08 \times 10^9 \text{ cm/s})}$

$= l_{\infty} \cdot L$

$= 5.38 \times 10^{-9} \text{ s}$

$\alpha = 1.36 \times 10^8 \text{ s}^{-1}$

Report your answer in this box.

Wt. No.

Cadet: _____

50 13. A pure U-235 fission primary device is boosted with 4.00 grams of deuterium and 6.00 grams of tritium.

- (30) a. Calculate the 100% fusion yield (in kT) to **three significant figures**. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$N = \frac{m \cdot N_A}{M} \quad \frac{4}{2 \text{amu}} = \frac{6}{3 \text{amu}} \quad \therefore 1:1 \text{ ratio}$$

$$N^D = \frac{4.0 \text{ g} \cdot N_A}{2 \text{amu}}$$

$$Y = \frac{4.0 N_A}{2 \text{amu}} \cdot Q$$

$$= \frac{4.0 \text{ g} \cdot N_A}{2 \text{amu}} (17.6 \text{ MeV}) \cdot \frac{1 \text{ kT}}{2.613 \times 10^{25} \text{ MeV}}$$

$$= 0.811 \text{ kT}$$

$$Y = 0.811 \text{ kT}$$

Report your answer in this box.

- (20) b. Calculate the energy yield (in kT) resulting from the fission reactions caused by the neutrons produced by the D-T fusion reactions to **three significant figures**. Assume 100% efficiency. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$N_{\text{neut}} = N_{\text{fusion}} \rightarrow \frac{4}{2} N_A = 2 N_A$$

$$Y = (2 N_A) Q^{235}$$

$$= 2 N_A (180 \text{ MeV}) \cdot \frac{1}{(2.613 \times 10^{25} \text{ MeV})}$$

$$= 8.30 \text{ kT}$$

$$8.30 \text{ kT}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 14. A 250.0-kT nuclear weapon is detonated at an altitude of 45.0 km. A target is located at an altitude of 25.0 km with a ground range of zero (i.e., target is located directly under the detonation). Assume the Planckian distribution of x-rays can be approximated as a single energy group of 0.150-MeV photons. Accounting for only geometric attenuation (ignoring atmospheric scattering and attenuation), calculate the total x-ray fluence in J/m² at the target to three significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$G: Y = 250 \text{ kT} \\ = 1.0465 \times 10^{15} \text{ J}$$

$$HOB = 45.0 \text{ km} \\ = 45,000 \text{ m}$$

$$loc = 25 \text{ km} \\ = 25,000 \text{ m}$$

$$r = HOB - loc \\ = 20,000 \text{ m}$$

find $F(r)$

$$F(r) = \frac{Y_x}{4\pi r^2}$$

$$= \frac{0.7 Y}{4\pi r^2}$$

$$= \frac{0.7 (1.0465 \times 10^{15} \text{ J})}{4\pi (20,000 \text{ m})^2}$$

$$F(r) = 1.46 \times 10^5 \text{ J/m}^2$$

$$1.46 \times 10^5 \text{ J/m}^2$$

Report your answer in this box.

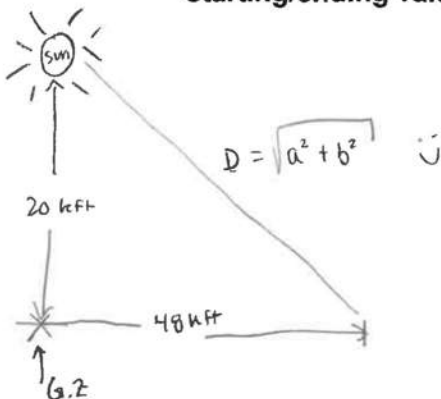
Wt. **No.**

50

15.

Cadet: _____

A 700.0-kT nuclear weapon is detonated at an altitude of 20.0 kft. Calculate the radiant exposure on the ground at a distance of 48.0 kft from ground zero to **two** significant figures. **If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.**



$$D = 52 \text{ kft}$$

$$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) = \frac{85.6 \text{ FW}^{\frac{1}{3}}}{D^2 [\text{kft}]}$$

F: Table 5-9

$$F \approx 0.40 \text{ (20 kft HOB)}$$

figure 5-16; IN: $d \approx 48 \text{ kft}$ OUT: $r = 0.7$
 $h \approx 20 \text{ kft}$

$$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) = \frac{85.6(0.4)(700\text{kT})(0.70)}{[52 \text{ kft}]^2}$$

$$\approx 6.204 \text{ cal/cm}^2$$

$$\approx 6.20 \text{ cal/cm}^2$$

6.20 cal/cm²

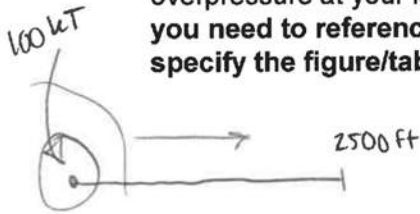
Report your answer in this box.

Wt. No.

Cadet: _____

50 16.

A 100.0-kT nuclear weapon is detonated on the ground. Calculate the peak overpressure at your location 2500 ft from ground zero to **two significant figures**. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.



$$\begin{aligned} D_1 &= \frac{D}{W^{1/3}} \\ &= \frac{2500 \text{ ft}}{(100 \text{ kT})^{1/3}} \\ &= 538.6 \text{ ft} \end{aligned}$$

figure 5-6 In: $d \approx 538.6 \text{ ft}$
 $h \approx 0 \text{ ft}$
 Out: $P_i \approx 40 \text{ PSI}$

for ground blast: Table 5-6 $\frac{P}{P_0} = S_P = 1$

$$P = P_i \cancel{S_P}^1$$

$$P = P_i$$

$$P = 40 \text{ PSI}$$

$$P = 40 \text{ PSI}$$

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 30 17. A 1.25-MT thermonuclear weapon is detonated at 1,000 ft above the surface. Calculate the equivalent neutron dose a soldier will receive located at a slant range of 2,000. yards to **two** significant figures. Only determine the neutron equivalent dose. **If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.**

G:

$$\begin{aligned} Y &= 1.25 \text{ MT} \\ &= 1250 \text{ kT} \end{aligned}$$

$$Q_n = 10 \text{ for unspecified E}$$

Figure 5-18: In: $R = 2,000 \text{ yds}$

Out: $D \approx 1.4 \text{ rad/kT}$

$$H_{OB} = 1,000 \text{ ft}$$

$$SR = 2,000 \text{ yds}$$

F: $H_n = D_n Q_n$

$$\begin{aligned} H_n &= Y D_n Q_n \\ &= (1250 \text{ kT})(1.4 \text{ rad/kT})(10 \text{ rem/rad}) \end{aligned}$$

$$\begin{aligned} H_n &= 17,500 \text{ rem} \\ &= 18,000 \text{ rem} \quad \text{ouch.} \end{aligned}$$

$$H_n = 18,000 \text{ rem}$$

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 50 18. The dose rate in the fallout area is measured to be 7.00 rad/hr at 4.0 hours after the nuclear detonation. Determine the dose rate at 10.0 hours after the nuclear detonation to **two** significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$R_t \approx R_i t^{-1.2}$$

$$R_i \approx \frac{R_t}{t^{-1.2}}$$

$$\approx \frac{7.00 \text{ rad/hr}}{(4.0 \text{ hr})^{-1.2}}$$

$$\approx 36.9 \text{ rad/hr}$$

$$R_t \approx R_i t^{-1.2}$$

$$\approx 36.9 \text{ rad/hr} (10 \text{ hr})^{-1.2}$$

$$R_t \approx 2.33 \text{ rad/hr}$$

$$\approx 2.3 \text{ rad/hr}$$

$$R_{10} \approx 2.3 \text{ rad/hr}$$

Report your answer in this box.

Poor Student Work

Student



Total Points

370 / 510 pts

Question 1

Assembly Methods

5 / 10 pts

✓ - 5 pts B

Question 2

Double Flash

0 / 10 pts

✓ - 10 pts B

Question 3

Fallout-free HOB

8 / 10 pts

✓ - 2 pts Doesn't select F

Question 4

UTRDR

10 / 10 pts

✓ - 0 pts True

Question 5

Mach Stem

10 / 10 pts

✓ - 0 pts B

Question 6

EMP

10 / 10 pts

✓ - 0 pts E

Question 7

Bomb Alpha

0 / 10 pts

✓ - 10 pts False

Question 8

Fusion

8 / 10 pts

✓ - 2 pts Doesn't select A

Question 9

PUREX

0 / 10 pts

✓ - 10 pts True

Question 10

Spontaneous Fission

0 / 10 pts

✓ - 10 pts D

Question 11

Bonus

0 / 10 pts

✓ - 10 pts A

Question 12

Enrichment Stages

35 / 50 pts

✓ - 5 pts Incorrect heavy mass (352)

✓ - 5 pts Incorrect light mass (349)

✓ - 5 pts Incorrect answer

Question 13

Bomb Alpha

50 / 50 pts

✓ - 0 pts Correct 1.36×10^8 gen/sec

Question 14

Fusion

39 / 50 pts

14.1 Fusion Yield

23 / 30 pts

✓ - 4 pts Doesn't find number of D or T atoms/rxns

✓ - 3 pts Incorrect answer

14.2 Secondary Fission Yield

16 / 20 pts

✓ - 2 pts Doesn't sub 235 fission Q-value

✓ - 2 pts Incorrect answer

Question 15

X-ray Fluence

40 / 50 pts

✓ - 5 pts Incorrect r (20,000 m)

✓ - 5 pts Incorrect answer

Question 16

Radiant Exposure

5 / 50 pts

✓ - 30 pts Doesn't use Eq 29 or other Q Eq

✓ - 5 pts Doesn't find f and/or tau

✓ - 5 pts Incorrect subs for Q Eq

✓ - 5 pts Incorrect answer

Question 17

Peak Overpressure

50 / 50 pts

✓ - 0 pts Correct 40 psi

Question 18

Neutron Dose

50 / 50 pts

✓ - 0 pts Correct 15,000 rem

Question 19

Dose Rate

50 / 50 pts

✓ - 0 pts Correct 2.3 rad/hr

Time of completion 1455
12/13 May 2026

Cadet

(First)

(Last)

Section I 1
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Term End Exam

1. Instructions:

- a. Bring only your FEE approved calculator, straight edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices are allowed in the examination room.**
- b. Check your exam for 18 problems. Check for the Reference Data Card (attached) and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question **unless otherwise instructed** by the problem.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to **210 minutes** to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

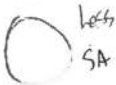
2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **14 May 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

10

1.

Which statement **best** explains how each method of weapon assembly increases bomb alpha?



☒ Both gun-type and implosion-type methods rapidly increase density of the fissile material

☒ Both gun-type and implosion-type methods rapidly increase mass-to-surface-area ratio of the fissile material

☒ Gun-type rapidly increases the density of fissile material, while implosion-type rapidly increases mass-to-surface-area ratio of fissile material.

☐ Gun-type rapidly increases the mass-to-surface-area ratio of fissile material, while implosion-type rapidly increases density of fissile material.

☒ Gun-type shoots highly compressed D-T gas into the fissile material for fusion neutrons, while implosion-type compresses D-T gas for fusion neutrons.

10

2.

Which statement **best** describes why a nuclear detonation initially appears bright, then dark, then bright again?

☐ During burn-out, x-rays fully ionize the atmosphere, creating the x-ray fireball and first flash. At hydrodynamic separation, the blast-wave shock heats the air, causing a second flash.

☒ During diffusion growth, the expanding x-ray fireball creates the first flash. During hydrodynamic growth, the expanding shock front creates a second flash.

☐ At hydrodynamic separation, the shock front hides the x-ray fireball. At breakaway, the shock front stops hiding the x-ray fireball.

☐ During burnout, the weapon x-rays superheat the atmosphere, causing it to produce visible light. During hydrodynamic growth, the shock front reheats their atmosphere, causing a second flash.

☐ During breakaway, the fireball separates from the bomb case creating the first flash. During hydrodynamic separation, the shock front creates the second flash.

Wt. **No.**

Cadet: _____

10 3. Select **all** of the following answers that are true about a fallout-free height of burst:

- ☒ The fireball touches the ground ☒ Fractionation occurs
☒ The fireball does not touch the ground ☐ Fractionation won't occur
☒ Earth is lofted into the mushroom cloud ☐ $HOB < 180W^{0.4}$
☒ Earth is not lofted into the mushroom cloud ☒ $HOB > 180W^{0.4}$

10 4. For a given position in a fallout field, the unit time reference dose rate is unchanging.

☒ TRUE

*R_1 stays constant $C_1 3^9$
 $R_2 = R_1 t^{-1.2}$*

☐ FALSE

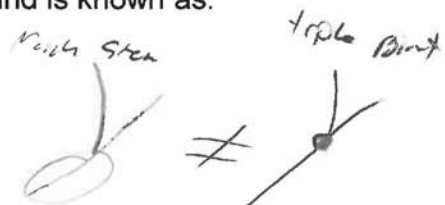
10 5. The reinforced shock front where ground reflected shock catches up to and reinforces the primary shock along the ground is known as:

☐ A triple point

☒ A Mach stem

☒ Hydrodynamic enhancement

☒ Dynamic overpressure



10 6. Select which step is **not** a part of EMP formation:

☐ Earth's magnetic field turns electrons

☐ Compton scatter events liberate electrons in the upper atmosphere

☐ Emitted EM fields form a downward pulse

☒ Prompt gammas escape the weapon case

☒ ^{Electrons} Gammas couple with antenna-like objects on the ground

10 7. Bomb alpha is the neutron population multiplication rate.

☐ TRUE

No that is η

☒ FALSE

Wt. No.

Cadet: _____

- 10 8. Select **all** the advantages of using fusion in a weapon assembly:
- ☒ ☐ Higher ν (nu) value *higher in fission material*
 - ☒ Added yield
 - ☒ Added neutrons
 - ☒ Fissionable isotopes can fission
- 10 9. PUREX is a plutonium enrichment process for producing weapons-grade plutonium from spent reactor fuel.
- ☒ TRUE
 - ☐ FALSE
- 10 10. Select the property of Pu-240 that makes it a contaminant for use in nuclear weapons:
- ☒ ☐ Low fission energy yield
 - ☒ ☐ Low macroscopic fission cross section
 - ☐ High spontaneous fission rate *Pu-239*
 - ☒ Fissionable isotope
 - ☐ Short half-life *No*
- 10 B BONUS: Which treaty legally prohibits the US from various types of nuclear weapons testing?
- ☒ NPT
 - ☐ PTBT
 - ☐ CTBT
 - ☐ SALT I
 - ☐ START I

Wt. No.

Cadet: _____

50

11.

A country desires 23.0 kg of uranium enriched to 82.0%. The feed material UF_6 has a natural enrichment of 0.711%, and the tail assay (waste material) is depleted to 0.200%. Calculate the number of gaseous diffusion enrichment stages required for this desire. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$\frac{F}{P} = \frac{x_p - x_w}{x_f - x_w}$$

$$x_p = 0.82$$

$$x_f = 0.00711$$

$$x_w = 0.002$$

$$\alpha = \sqrt{\frac{m_H}{m_L}}$$

$$N = \frac{2}{\ln \alpha} \ln \left[\frac{x_p(1-x_w)}{x_w(1-x_p)} \right] - 1$$

$$m_H = 1(238) + 6(18) = 346 \text{ g/mol}$$

$$m_L = 1(235) + 6(18) = 343 \text{ g/mol}$$

$$\alpha = \sqrt{\frac{346}{343}} = 1.004363657$$

$$N = \frac{2}{\ln(1.00436)} \ln \left[\frac{0.82(1-0.002)}{0.002(1-0.82)} \right] - 1$$

$$N = 459.33(7.72895) - 1 = 3550 \text{ stages}$$

3550 stages

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 12. A notional spherical weapon assembly composed of pure U-235 has a radius of 6.00 cm and a density of 35.0 g/cm³. Assume the non-leakage probability is 0.750. Calculate "bomb alpha" of the assembly to three significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$\alpha = \frac{K-1}{K} \quad L = 0.75 \quad \rho = 35.0 \text{ g/cm}^3$$

$$l = l_{\infty} L$$

$$\Sigma_a = N \sigma_a$$

$$N = \frac{\rho N_A}{M} = \frac{35.0 \text{ g/cm}^3 \cdot 6.02214144 \times 10^{23} \text{ mol}^{-1}}{235 \text{ g/mol}}$$

$$K = k_{\infty} L$$

$$\sigma_a = 1.44 \times 10^{-24} \text{ cm}^2$$

$$N = 8.969 \times 10^{22} \text{ cm}^{-3}$$

$$K_{\infty} = \nu \frac{\sigma_f}{\sigma_a}$$

$$\nu = 1.08 \times 10^4 \text{ cm}^2$$

$$\Sigma_a = N \sigma_a = 0.129155 \text{ cm}^{-1}$$

$$\nu \sigma_f = 3.32 \times 10^{-24} \text{ cm}^2$$

$$l_{\infty} = \frac{1}{\Sigma_r \nu} = \frac{1}{0.129155 \text{ cm}^{-1} \cdot 1.08 \times 10^4 \text{ cm}^2}$$

$$l_{\infty} = \frac{1}{\Sigma_r \nu}$$

$$K_{\infty} = \frac{\nu \sigma_f}{\sigma_a} = \frac{3.32 \times 10^{-24} \text{ cm}^2}{1.44 \times 10^{-24} \text{ cm}^2}$$

$$l_{\infty} = 7.169 \times 10^{-9} \text{ s}$$

$$l = l_{\infty} (0.75) = 5.3768 \times 10^{-9} \text{ s}$$

$$K_{\infty} = 2.30556$$

$$K = K_{\infty} (0.75) = 1.729167$$

$$\alpha = \frac{1.729167 - 1}{5.3768 \times 10^{-9} \text{ s}} = 1.36 \times 10^8 \text{ s}^{-1}$$

$$1.36 \times 10^8 \text{ s}^{-1}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 13. A pure U-235 fission primary device is boosted with 4.00 grams of deuterium and 6.00 grams of tritium.

- (30) a. Calculate the 100% fusion yield (in kT) to **three significant figures**. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$M_D = \frac{2g D}{mol}$$

$$M_T = \frac{3g T}{mol}$$

$$E = 1?$$

$$\frac{4g D}{M_D} = 2 mol D$$

$$\frac{6g T}{M_T} = 2 mol T$$

$$\sigma_{D,T} = 3.68 \times 10^{-12} E^{-2/3} e^{-10.44 E^{-1/3}}$$

$$= 8.05467 \times 10^{-19}$$

$$N_{reac} = \sigma_{D,T} n_D n_T = \sigma_{D,T} (2 mol \times 2 mol)$$

$$n_{D,T} = n_{D,T} = 2 mol \quad Y = N_{reac} \times Q = N_{reac} (17.6 MeV) = 1.168 \times 10^{26}$$

$$Q = 17.6 MeV$$

$$Y = 2.05632 \times 10^{26} MeV$$

$$2.613 \times 10^{25} MeV/kT$$

$$7.87 kT$$

Report your answer in this box.

- (20) b. Calculate the energy yield (in kT) resulting from the fission reactions caused by the neutrons produced by the D-T fusion reactions to **three significant figures**. Assume 100% efficiency. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

2 mol of neutrons $\Rightarrow$ 2 mol of U-235 fissioning

$$Y = N_{reactions} \times Q = (2 \times 6.02214199 \times 10^{23}) (207 MeV) = 2.4932 \times 10^{26} MeV$$

$$\times \frac{1}{2.613 \times 10^{25} MeV/kT} = 9.54 kT$$

$$9.54 kT$$

Report your answer in this box.

Wt. No.

50 14.

Cadet: _____

A 250.0-kT nuclear weapon is detonated at an altitude of 45.0 km. A target is located at an altitude of 25.0 km with a ground range of zero (i.e., target is located directly under the detonation). Assume the Planckian distribution of x-rays can be approximated as a single energy group of 0.150-MeV photons. Accounting for only geometric attenuation (ignoring atmospheric scattering and attenuation), calculate the total x-ray fluence in J/m² at the target to **three significant figures**. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

* Only Use

$F(r)$ eq

$$F(r) = \frac{Y_x}{4\pi r^2}$$

$$Y_x = 0.74$$

$$r = 45 - 25$$

$$Y_x = 0.7(250 \times 10^3) \cdot 4.186 \times 10^{12} \text{ J/kg} = 7.3255 \times 10^{14} \text{ J}$$

$$F(20) = \frac{Y_x}{4\pi(20)^2} = 1.457362 \times 10^{11} \text{ J/m}^2$$

$$1.46 \times 10^{11} \text{ J/m}^2$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 15. A 700.0-kT nuclear weapon is detonated at an altitude of 20.0 kft. Calculate the radiant exposure on the ground at a distance of 48.0 kft from ground zero to **two** significant figures. **If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.**

$$\text{Slant range} = \sqrt{H_0B^2 + D^2} = \sqrt{20^2 + 48^2} = 52 \text{ kft}$$

$$5250 \text{ ft} \approx 1 \text{ mile}$$

$$\frac{52 \text{ kft} \cdot 1000 \text{ ft/mile}}{5250 \text{ ft}} = 9.9 \text{ miles}$$

Using Figure 5-14

$$\begin{array}{l} X = 7 \times 10^2 \text{ kT} \\ Y \approx 10 \text{ miles} \end{array} \xrightarrow{\text{gets}} 3 \text{ cal/cm}^2$$

3 cal/cm²

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 50 16. A 100.0-kT nuclear weapon is detonated on the ground. Calculate the peak overpressure at your location 2500 ft from ground zero to **two** significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$D = D_1 \times W^{\frac{1}{3}}$$

$$D = 2500 \text{ ft}$$

$$D_1 = \frac{D}{W^{\frac{1}{3}}} = \frac{2500}{(100)^{\frac{1}{3}}} = 538 \text{ ft}$$

$$W = 100 \text{ KT}$$

Using Figure 5-6

$$X \approx 540 \text{ ft}$$

$$Y = 0 \text{ (ground burst)}$$

gets
→

Interpolating between 50 and 30 psi line, so 40 psi

40 psi

Report your answer in this box.

Wt. No.

Cadet: _____

- 30 17. A 1.25-MT thermonuclear weapon is detonated at 1,000 ft above the surface. Calculate the equivalent neutron dose a soldier will receive located at a slant range of 2,000 yards to two significant figures. Only determine the neutron equivalent dose. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

Using Figure S-18

$Q = 10$ (quality factor)

$X = 2000 \text{ yards} \xrightarrow{\text{gets}} \approx 1.25 \text{ Rads/KT}$
- Use left arrow

$$D = \frac{1.25 \text{ Rad}}{\text{KT}} \times \frac{1.25 \times 10^3 \text{ KT}}{1} = 15625 \text{ Rad}$$

$$H = DQ = D(10) = 15625 \text{ rem}$$

(double!)

16000 Rem

Report your answer in this box.

Wt. No.

Cadet: _____

- 50 18. The dose rate in the fallout area is measured to be 7.00 rad/hr at 4.0 hours after the nuclear detonation. Determine the dose rate at 10.0 hours after the nuclear detonation to **two** significant figures. If you need to reference a graph/table, indicate starting/ending values and specify the figure/table used.

$$R_t \approx R_1 t^{-1.2}$$

$$R_1 = \frac{R_t}{t^{-1.2}} = \frac{7.00}{4^{-1.2}} = 36.94 \frac{\text{rad}}{\text{hr}}$$

$$D \approx 5R_1 (t_a^{-0.2} - t_b^{-0.2})$$

$$D = 5(R_1 (4^{-0.2} - 10^{-0.2})) = 23.1 \text{ rad/s}$$

$$R_{10} \approx R_1 (10)^{-1.2} = 2.33 \text{ rad/hr}$$

2.3 rad/hr

Report your answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB J FINAL COURSE GRADES

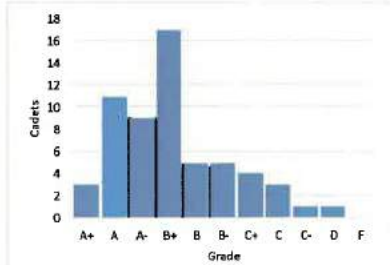
NE450, AY26-2 Final Grades

Last Name	First Name	Grad Year	Max Points	Points Earned	Percent	Letter Grade	Points Needed	FOS	CES	HWs	IPs	WPRs	TEE
		2026	1950	1859	95.3%	A+		NEN1		97%	91%	94%	97%
		2026	2000	1900	95.0%	A+		PHY1		96%	97%	96%	90%
		2026	2000	1896	94.8%	A+		PHY1	SPA0N	93%	91%	94%	99%
		2026	2000	1871	93.6%	A		SSC0		91%	91%	93%	94%
		2026	2000	1867	93.4%	A	13	CEN2		93%	100%	86%	96%
		2026	2000	1854	92.7%	A		CEN2		93%	100%	89%	92%
		2026	2000	1846	92.3%	A		CEN2		100%	100%	78%	93%
		2026	2000	1837	91.9%	A		ENL2	RSE1N	90%	94%	84%	96%
		2026	2000	1829	91.6%	A		ECN2	AST1N	93%	91%	90%	92%
		2026	2000	1826	91.3%	A		PYL2	AMF0N	92%	99%	86%	92%
		2027	2000	1825	91.3%	A		ECN2		96%	94%	86%	89%
		2026	2000	1824	91.2%	A		NEN1		92%	94%	87%	94%
		2026	2000	1813	90.7%	A		CEN2	TST1N	98%	97%	81%	89%
		2026	2000	1810	90.5%	A		BIO0		94%	94%	90%	87%
		2026	2000	1785	89.3%	A-	25	BEA0	RSE1N	92%	97%	78%	95%
		2026	2000	1778	88.9%	A-		BEA0		93%	94%	88%	82%
		2026	2000	1775	88.8%	A-		PHY1		91%	79%	87%	90%
		2026	2000	1775	88.7%	A-		NEN1		82%	97%	85%	97%
		2026	2000	1764	88.2%	A-		CHM2		96%	49%	87%	87%
		2026	2000	1755	87.8%	A-		CEN2		90%	66%	87%	92%
		2026	2000	1752	87.6%	A-		NEN1		93%	97%	82%	81%
		2026	2000	1748	87.4%	A-		CHM2A		90%	91%	80%	91%
		2026	1950	1697	87.0%	A-		PHY1		86%	85%	82%	88%
		2026	1950	1687	86.5%	B+	10	FSL2		96%	85%	82%	87%
		2026	2000	1727	86.4%	B+	13	CHM2A		89%	99%	81%	81%
		2026	2000	1722	86.1%	B+	18	ENL2H		89%	76%	87%	84%
		2027	2000	1714	85.7%	B+		SSC0		86%	100%	75%	90%
		2026	2000	1709	85.5%	B+		BIO0H		84%	91%	80%	88%
		2026	2000	1709	85.4%	B+		SSC0		91%	84%	77%	86%
		2026	2000	1705	85.3%	B+		NEN1		87%	88%	87%	74%
		2026	2000	1703	85.2%	B+		SSC0		95%	88%	72%	83%
		2026	2000	1703	85.1%	B+		GIS2		86%	76%	79%	91%
		2026	1950	1657	84.9%	B+		NEN1	RSM1N	79%	93%	79%	93%
		2026	2000	1698	84.9%	B+		SSC0		85%	66%	83%	82%
		2026	2000	1692	84.6%	B+		KIN1		90%	86%	82%	77%
		2026	2000	1689	84.5%	B+		NEN1		88%	64%	82%	86%
		2026	2000	1686	84.3%	B+		PHY1	RSU1N	91%	72%	81%	77%
		2026	2000	1684	84.2%	B+		PHY1	RSA1N	87%	94%	74%	83%
		2026	2000	1681	84.1%	B+		CHM2A		89%	100%	81%	75%
		2026	2000	1677	83.9%	B+		FLG2		94%	85%	64%	85%
		2026	2000	1660	83.0%	B	10	PII0	FLG2	94%	85%	82%	72%
		2026	2000	1636	81.8%	B		FLF2		94%	99%	79%	67%
		2026	2000	1624	81.2%	B		LLS3	TST1N	89%	85%	75%	71%
		2026	2000	1624	81.2%	B		FCR2		84%	94%	76%	79%
		2026	2000	1618	80.9%	B		NEN1		73%	67%	80%	88%
		2026	2000	1596	79.8%	B-	4	FLR2		83%	79%	78%	73%
		2026	2000	1588	79.4%	B-	12	PII0		60%	93%	77%	79%
		2026	2000	1585	79.3%	B-	15	ECN2		90%	97%	69%	69%
		2026	2000	1577	78.9%	B-		LLS3		92%	100%	67%	67%
		2026	2000	1548	77.4%	B-		HUS3T		77%	54%	78%	79%
		2026	2000	1524	76.2%	C+	6	SSC0		69%	76%	69%	78%
		2026	2000	1482	74.1%	C+		HMH3		81%	92%	61%	61%
		2026	2000	1479	74.0%	C+		BEA0		74%	69%	54%	85%
		2026	2000	1474	73.7%	C+		MGN0		85%	78%	62%	69%
		2026	2000	1443	72.2%	C	17	HNT3		72%	87%	62%	71%
		2026	2000	1439	72.0%	C		PAP3		80%	82%	72%	57%
		2026	2000	1428	71.4%	C		EPS1		89%	91%	59%	51%
		2026	1950	1313	67.3%	C-	43	CHM2A		78%	94%	48%	55%
		2026	2000	1306	65.3%	D	14	DSS2		87%	53%	62%	33%
				AVG		84.5%	B+						

Grading Scale	
A+	94.0%
A	90.5%
A-	87.0%
B+	83.5%
B	80.0%
B-	76.5%
C+	73.0%
C	69.5%
C-	66.0%
D	60.0%
F	<60%

Grade Distribution		
Graded Requirement	Marks	% of Total
TEE	500	25%
WPR 1	250	13%
WPR 2	250	13%
Problem Sets (4)	500	25%
Extension Problems (2)	300	15%
Instructor Points	100	5%
FNEK Writ (2)	100	5%
Total	2000	100%

8	NEN1
6	PHY1
6	SSC0
0	NSMN
46	NE
0	Famous



Historical Data		
AY	Average	Grade
26-1	89.26%	A-
25-2		
25-1		
24-2	87.36%	A-
24-1	86.23%	A-
23-2	87.48%	A-
23-1		
22-2		
22-1		
21-2		
21-1		
20-2		
20-1		

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UNITED STATES MILITARY ACADEMY
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SECTION III

ASSIGNMENTS



UNITED STATES MILITARY ACADEMY
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TAB K

PROBLEM SETS



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #1

Cadet _____
(First) (Last)

Section _____
(hour)

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Problem Set 1 – 125 Points

1. Instructions:

- a) For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- b) Write your name on every page.
- c) You are allowed and encouraged to collaborate with peers on problem sets. Ensure you document assistance IAW the DAW.
- d) Complete the documentation section below AFTER all work is complete. Sign using a digital signature (CAC), digital ink (stylus or finger), or ink. Do not type.

2. Documentation:

_____ I certify that I have completely documented all sources that I used to complete this assignment and that I acknowledged all assistance I received in the completion of this assignment.

_____ I certify that I did not use any sources or receive any assistance requiring documentation while completing this assignment.

AND

_____ I acknowledge that I used generative AI to complete this assignment.

_____ I acknowledge that I did not use generative AI to complete this assignment.

Cadet Signature: _____

Cadet Name (print): _____

Date: _____

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Wt. No.

Cadet: _____

- 25 1. A rogue actor wants 135 kg of uranium at 93.0% enrichment. Assuming the enrichment plant has a tails assay (waste factor) of 0.200% and feed material with a natural assay of 0.711%, calculate the following to 4 significant figures:

a. The enrichment plant feed factor $\frac{F}{P} = \frac{x_p - x_w}{x_f - x_w}$

$$\frac{F}{P} = \frac{0.93 - 0.002}{0.00711 - 0.002}$$

$$\frac{F}{P} = 181.6$$

181.6

Report your answer in this box.

- b. The amount of feed required to make the product

$$\frac{F}{P} = 181.6$$

$$F = 181.6 P$$

$$F = 181.6 (135 \text{ kg})$$

$$F = 2.45 \times 10^4 \text{ kg}$$

$2.452 \times 10^4 \text{ kg}$

Report your answer in this box.

- c. The amount of separative work required to complete the production

$$SWU = [PV(x_p) + WV(x_w) - FV(x_f)]\tau$$

$$V(x_i) = (2x_i - 1) \ln \frac{x_i}{1 - x_i}$$

$$V(x_p) = 2.225$$

$$V(x_w) = 6.188$$

$$V(x_f) = 4.869$$

Assume time = unity

Factor out P

$$SWU = \left[V(x_p) + \frac{W}{P} V(x_w) - \frac{F}{P} V(x_f) \right] P$$

$$\frac{F}{P} = 181.6$$

$$\frac{W}{P} = 180.6$$

$$SWU = [2.225 + 180.6(6.188) - 181.6(4.869)]135 \text{ kg}$$

$$SWU = 3.180 \times 10^4 \text{ kg}$$

$3.180 \times 10^4 \text{ kg}$

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 2. For the enrichment process in Problem 1, assume a gas centrifuge with a radius of 11.0 cm, a rotational speed of 50,000 rpm, and UF_6 gas heated to 325 K.

- a. Calculate the number of stages required to complete the separation

$$N = \frac{2}{\ln \alpha} \ln \left[\frac{x_p(1 - x_w)}{x_w(1 - x_p)} \right] - 1$$

$$\alpha = \exp \left[\frac{(0.352 - 0.349) \frac{\text{kg}}{\text{mol}} \left(5236 \frac{\text{rad}}{\text{s}} \right)^2 (0.11\text{m})^2}{2 \left(8.31 \frac{\text{J}}{\text{mol K}} \right) (325 \text{ K})} \right]$$

$$\alpha = \exp \left[\frac{(m_H - m_L) \omega^2 r^2}{2RT} \right]$$

$$\alpha = 1.202$$

$$\omega = 50,000 \frac{\text{rev}}{\text{min}} \left(\frac{2\pi \text{ rad}}{\text{rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ sec}} \right)$$

$$\omega = 5,236 \frac{\text{rad}}{\text{s}}$$

$$N = \frac{2}{\ln 1.202} \ln \left[\frac{0.93(1 - 0.002)}{0.002(1 - 0.93)} \right] - 1$$

$$N = 95 \text{ stages}$$

$$m_H = 0.238 + 6(0.019) = 0.352 \frac{\text{kg}}{\text{mol}}$$

$$m_L = 0.235 + 6(0.019) = 0.349 \frac{\text{kg}}{\text{mol}}$$

95 stages

Report your answer in this box.

- b. Calculate the number of stages required if gaseous diffusion is used

$$N = \frac{2}{\ln \alpha} \ln \left[\frac{x_p(1 - x_w)}{x_w(1 - x_p)} \right] - 1$$

$$\alpha = \sqrt{\frac{m_H}{m_L}} = \sqrt{\frac{0.352}{0.349}} = 1.00429$$

$$N = \frac{2}{\ln 1.00429} \ln \left[\frac{0.93(1 - 0.002)}{0.002(1 - 0.93)} \right] - 1$$

$$N = 4112 \text{ stages}$$

4112 stages

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 3. Consider a bare, fissioning sphere of ^{239}Pu . The sphere ignites at 3.0 times nominal density with a non-leakage probability of 0.475.
- a. Calculate the multiplication rate (bomb alpha) in gen/shake for this assembly to 4 significant figures.

$$\alpha = \frac{k - 1}{l}$$

$$l_{\infty} = \frac{1}{\Sigma_a v}$$

$$\Sigma_a = N\sigma_a \quad N = \frac{\rho N_A}{M}$$

$$l_{\infty} = \frac{M}{\rho N_A \sigma_a v}$$

$$l = l_{\infty} L$$

$$l = \frac{ML}{\rho N_A \sigma_a v}$$

$$l = \frac{239 \frac{g}{mol} (0.475)}{3 \left(19.6 \frac{g}{cm^3} \right) N_A (1.98 \times 10^{-24} cm^2) \left(1.08 \times 10^9 \frac{cm}{s} \right)}$$

$$l = 1.499 \times 10^{-9} s = 0.1499 \text{ shakes}$$

$$k_{\infty} = \frac{v\sigma_f}{\sigma_a} \quad k = k_{\infty} L$$

$$k = \frac{v\sigma_f}{\sigma_a} L = \frac{5.61}{1.98} 0.475 = 1.346$$

$$\alpha = \frac{k - 1}{l} = \frac{1.346 - 1}{0.1499 \text{ shakes}} = 2.307 \frac{\text{generations}}{\text{shake}}$$

$$2.307 \frac{\text{gen}}{\text{shake}}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 3. Consider a bare, fissioning sphere of ^{239}Pu . The sphere ignites at 3.0 times nominal density with a non-leakage probability of 0.475.
- b. Calculate bomb alpha for a sphere made of ^{233}U and one of ^{232}Th to 4 significant figures.

$$l = \frac{233 (0.475)}{3(19.1)N_A(2.09 \times 10^{-24})(1.08 \times 10^9)}$$

$$l = 1.421 \times 10^{-9} \text{ s} = 0.1421 \text{ shakes}$$

$$k = \frac{\nu \sigma_f}{\sigma_a} L = \frac{5.17}{2.09} 0.475 = 1.175$$

$$\alpha = \frac{k - 1}{l} = \frac{1.175 - 1}{0.1421 \text{ shakes}} = 1.232 \frac{\text{gen}}{\text{shake}}$$

$$l = \frac{232 (0.475)}{3(11.71)N_A(0.157 \times 10^{-24})(1.08 \times 10^9)}$$

$$l = 3.072 \times 10^{-8} \text{ s} = 3.072 \text{ shakes}$$

$$k = \frac{\nu \sigma_f}{\sigma_a} L = \frac{0.112}{0.175} 0.475 = 0.3389$$

$$\alpha = \frac{k - 1}{l} = \frac{0.3389 - 1}{3.072 \text{ shakes}} = -0.2152 \frac{\text{gen}}{\text{shake}}$$

^{233}U
$1.232 \frac{\text{gen}}{\text{shake}}$
Report your answer in this box.

^{232}Th
$-0.2152 \frac{\text{gen}}{\text{shake}}$
Report your answer in this box.

- c. Based on these three spheres, which will generate the most yield within the first 0.5 μs ? Justify your response.

^{239}Pu has the highest bomb alpha thus will generate the most yield in the shortest amount of time, followed by ^{233}U . A negative bomb alpha in ^{232}Th indicates criticality is not achieved, thus no yield.

Wt. No.

Cadet: _____

- 25 4. A spherical weapon assembly composed of pure ^{239}Pu has an initial density of 32.0 g/cm^3 . The fission process is initiated with one thousand neutrons, and the constant non-leakage probability of the assembly is 0.62.
- a. Calculate the number of neutrons present after the 5th neutron generation to 4 significant figures.

$$l = \frac{239 (0.62)}{(32.0) N_A (1.98 \times 10^{-24}) (1.08 \times 10^9)}$$

$$l = 3.596 \times 10^{-9} \text{ s} = 0.3596 \text{ shakes}$$

$$k = \frac{\nu \sigma_f}{\sigma_a} L = \frac{5.61}{1.98} 0.62 = 1.757$$

$$\alpha = \frac{k - 1}{l} = \frac{1.757 - 1}{0.3596 \text{ shakes}} = 2.104 \frac{\text{gen}}{\text{shake}}$$

$$n(t) = n_0 e^{\alpha t}$$

$$t = gl$$

$$n(t) = n_0 * e^{\alpha gl}$$

$$n(5) = 1,000 * e^{2.104 * 5 * 0.3596}$$

$$n(5) = 4.396 \times 10^4 \text{ neutrons}$$

$$4.396 \times 10^4$$

Report your answer in this box.

- b. Calculate the number of neutrons present after the 65th neutron generation to 4 significant figures.

$$n(t) = n_0 e^{\alpha t}$$

$$t = gl$$

$$n(65) = 1,000 * e^{\alpha gl}$$

$$n(65) = 1,000 * e^{2.104 * 65 * 0.3596}$$

$$n(65) = 2.291 \times 10^{24} \text{ neutrons}$$

$$2.291 \times 10^{24}$$

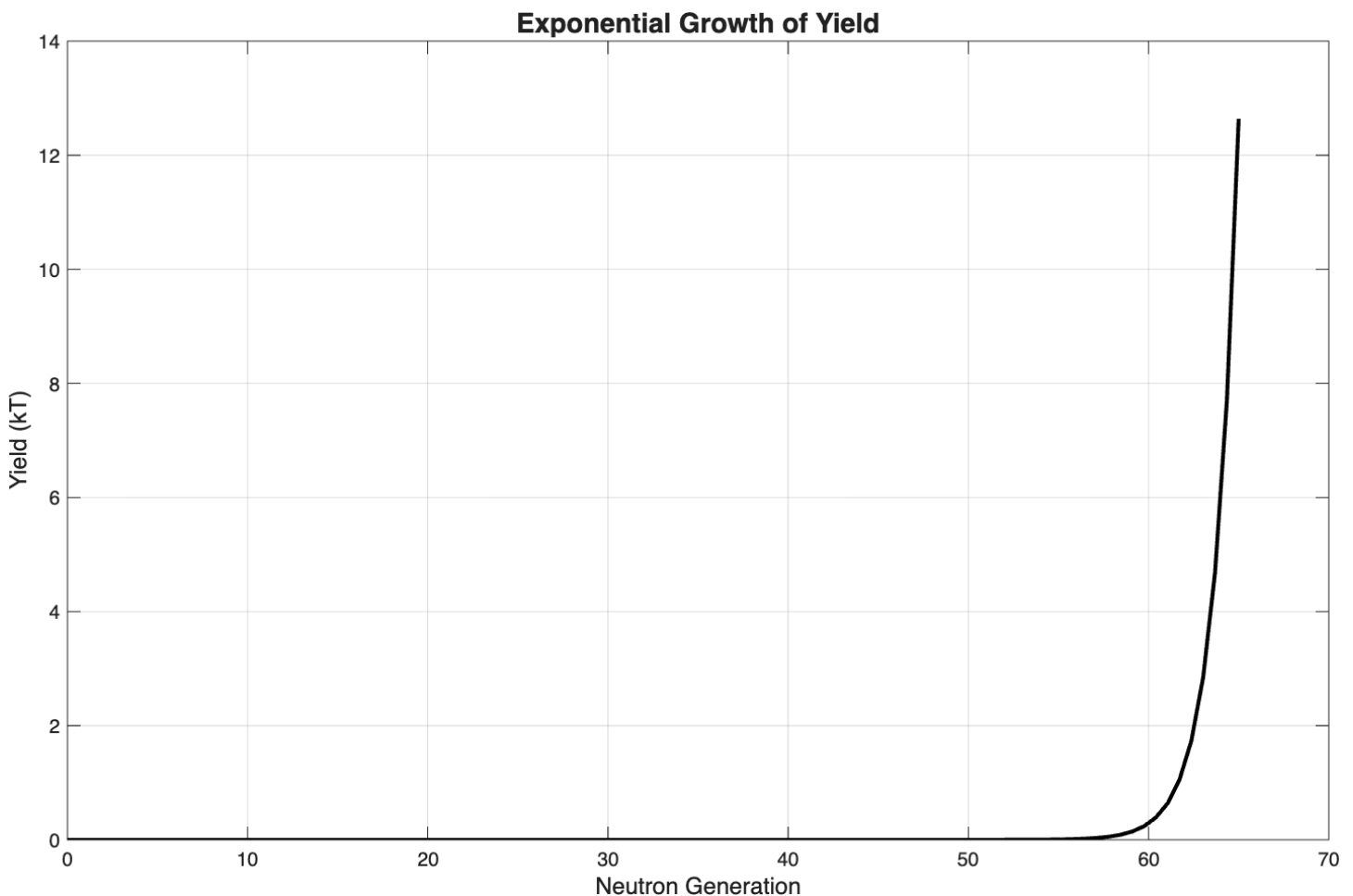
Report your answer in this box.

Wt. No.

Cadet: _____

- 25 4. A spherical weapon assembly composed of pure ^{239}Pu has an initial density of 32.0 g/cm^3 . The fission process is initiated with one thousand neutrons, and the constant non-leakage probability of the assembly is 0.62.
- c. Graph the total yield, in kT, as a function of time through 65 neutron generations.

$$Y(t) = \frac{\Sigma_f v Q n_0}{\alpha} (e^{\alpha t} - 1)$$



Cadet: _____



UNITED STATES MILITARY ACADEMY
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PROBLEM SET #2

Cadet _____
(First) (Last)

Section _____
(hour)

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Problem Set 2 – 125 Points

1. Instructions:

- a) For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- b) Write your name on every page.
- c) You are allowed and encouraged to collaborate with peers on problem sets. Ensure you document assistance IAW the DAW.
- d) Complete the documentation section below AFTER all work is complete. Sign using a digital signature (CAC), digital ink (stylus or finger), or ink. Do not type.

2. Documentation:

_____ I certify that I have completely documented all sources that I used to complete this assignment and that I acknowledged all assistance I received in the completion of this assignment.

_____ I certify that I did not use any sources or receive any assistance requiring documentation while completing this assignment.

AND

_____ I acknowledge that I used generative AI to complete this assignment.

_____ I acknowledge that I did not use generative AI to complete this assignment.

Cadet Signature: _____

Cadet Name (print): _____

Date: _____

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Wt. No.

Cadet: _____

- 25 1. A ^{239}Pu weapon assembly is at a density of $\rho_0 = 39.2 \text{ g/cm}^3$ when the fission process is initiated. Some time later, the burn-up fraction is $\text{BF} = 0.15$, and the density of atoms is $7.90 \times 10^{22} \text{ atoms/cm}^3$. Calculate the following to 3 significant figures:

- a. The macroscopic fission cross-section at initiation

$$\Sigma_f = N\sigma_f \quad \Sigma_f = \frac{39.2 \frac{\text{g}}{\text{cm}^3} N_A \frac{5.61 \times 10^{-24} \text{cm}^2}{2.98}}{239 \frac{\text{g}}{\text{mol}}}$$

$$\Sigma_f = \frac{\rho N_A \sigma_f}{M} \quad \Sigma_f = 0.1859 \text{ cm}^{-1}$$

0.186 cm⁻¹

Report your answer in this box.

- b. The macroscopic absorption cross-section at initiation

$$\Sigma_a = N\sigma_a \quad \Sigma_f = \frac{39.2 \frac{\text{g}}{\text{cm}^3} N_A \frac{1.98 \times 10^{-24} \text{cm}^2}{2.98}}{239 \frac{\text{g}}{\text{mol}}}$$

$$\Sigma_a = \frac{\rho N_A \sigma_a}{M} \quad \Sigma_f = 0.1956 \text{ cm}^{-1}$$

0.196 cm⁻¹

Report your answer in this box.

- c. The macroscopic fission cross-section when $\text{BF} = 0.15$

$$\Sigma = \Sigma_0 \frac{\rho}{\rho_0} \left[1 - \text{BF} + 2\text{BF} \frac{\sigma_{ff}}{\sigma_{fuel}} \right] \quad \Sigma_f = N\sigma_f [1 - \text{BF}]$$

$$\Sigma_f = \frac{\rho_0 N_A \sigma_f}{M} \frac{NM}{\rho_0 N_A} [1 - \text{BF}] \quad \Sigma_f = 7.90 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3} \frac{5.61 \times 10^{-24} \text{cm}^2}{2.98} [1 - 0.15]$$

$$\Sigma_f = 0.1264 \text{ cm}^{-1}$$

0.126 cm⁻¹

Report your answer in this box.

- d. The macroscopic absorption cross-section when $\text{BF} = 0.15$

$$\Sigma_a = N\sigma_a \left[1 - \text{BF} + 2\text{BF} \frac{\sigma_{ff}}{\sigma_{fuel}} \right]$$

$$\Sigma_a = 7.90 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3} \frac{1.98 \times 10^{-24} \text{cm}^2}{1.98} \left[1 - 0.15 + 2(0.15) \frac{0.0711}{1.98} \right]$$

$$\Sigma_a = 0.1346 \text{ cm}^{-1}$$

0.135 cm⁻¹

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 2. A spherical secondary fusion device of D-T fuel has a 1:1 ratio of tritium to deuterium atoms, a volume of 175 cm^3 , a temperature of $1.50 \times 10^8 \text{ K}$, and produces a yield of 200. kT. Calculate the reaction rate of the D-T fuel to 3 significant figures.

$$N_{\text{reac}} = (\overline{\sigma v}) n_{\text{tar}} n_{\text{bomb}}$$

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} E^{-\frac{2}{3}} e^{-19.94 E^{-\frac{1}{3}}}$$

$$E = 1.50 \times 10^8 \text{ K} \left(\frac{1 \text{ eV}}{1.1605 \times 10^4 \text{ K}} \right) \left(\frac{1 \text{ keV}}{1000 \text{ eV}} \right)$$

$$E = 12.9 \text{ keV}$$

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} (12.9^{-\frac{2}{3}}) e^{-19.94 (12.9^{-\frac{1}{3}})}$$

$$(\overline{\sigma v})_{DT} = 1.36 \times 10^{-16} \frac{\text{cm}^3}{\text{sec}}$$

$$n_{\text{tar}} = n_{\text{bomb}} = \frac{n}{(175 \text{ cm}^3)}$$

$$n_{\text{tar}} = n_{\text{bomb}} = 1.70 \times 10^{24} \text{ cm}^{-3}$$

$$N_{\text{reac}} = (\overline{\sigma v}) n_{\text{tar}} n_{\text{bomb}}$$

$$n = \frac{Y}{Q}$$

$$Q = 17.6 \text{ MeV}$$

$$N_{\text{reac}} = 1.35 \times 10^{-16} \frac{\text{cm}^3}{\text{sec}} (1.70 \times 10^{24} \text{ cm}^{-3})^2$$

$$N_{\text{reac}} = 3.93 \times 10^{32} \text{ cm}^{-3} \text{ s}^{-1}$$

$$Y = 200 \text{ kT} \left(\frac{2.613 \times 10^{25} \text{ MeV}}{1 \text{ kT}} \right)$$

$$Y = 5.23 \times 10^{27} \text{ MeV}$$

$$R = N_{\text{reac}} V$$

$$R = 3.93 \times 10^{32} \text{ cm}^{-3} \text{ s}^{-1} (175 \text{ cm}^3)$$

$$n = \frac{5.23 \times 10^{27} \text{ MeV}}{17.6 \text{ MeV}}$$

$$n = 2.97 \times 10^{26}$$

$$R = 6.8731 \times 10^{34} \text{ s}^{-1}$$

$$6.87 \times 10^{34} \text{ s}^{-1}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 3. The largest weapon ever tested had a yield of 50.0 MT. The designers claimed that substituting ^{238}U for the lead in the secondary would produce 100 MT. Assume the secondary fusion fuel was pure deuterium. Calculate the following to 3 significant figures:

- a. The number of ^{238}U fissions needed to produce the additional 50.0 MT of yield

$$n = \frac{Y}{Q}$$

$$n = \frac{50000 \text{ kT}}{180 \text{ MeV}} \frac{2.613 \times 10^{25} \text{ MeV}}{1 \text{ kT}}$$

$$n = 7.2583 \times 10^{27}$$

$$7.26 \times 10^{27}$$

Report your answer in this box.

- b. The mass of deuterium needed to fission that amount of ^{238}U

$$n_D = 5(7.26 \times 10^{27})$$

$$n_D = 3.629 \times 10^{28}$$

$$m_D = \frac{n_D M_D}{N_A}$$

$$m_D = \frac{3.629 \times 10^{28} \left(2 \frac{\text{g}}{\text{mol}} \right)}{6.022 \times 10^{23} \frac{\text{atoms}}{\text{mol}}}$$

$$m_D = 1.2053 \times 10^5 \text{ g}$$

$$121 \text{ kg}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 4. A D-T fusion assembly consists of 50.0 g of equal a/o deuterium-tritium fuel and has a compressed volume of 0.500 cm³ at the time of ignition. Once ignition occurs, assume the fusion assembly burns completely. Calculate the following to 3 significant figures:

- a. The 100% fusion yield, in kT, resulting from the deuterium fuel assembly

$$Y = n_D Q$$

$$n_D = \frac{m_D N_A}{M}$$

$$Y = 6.022 \times 10^{24} \left(17.6 \frac{\text{MeV}}{D} \right) \left(\frac{1 \text{ kT}}{2.613 \times 10^{25} \text{ MeV}} \right)$$

$$n_D = \frac{20.0 \text{ g} \left(6.022 \times 10^{23} \frac{\text{atoms}}{\text{mol}} \right)}{2 \frac{\text{g}}{\text{mol}}}$$

$$Y = 4.0562 \text{ kT}$$

$$n_D = 6.022 \times 10^{24} \text{ atoms}$$

4.06 kT

Report your answer in this box.

- b. The yield, in kT, that could result from a ²³⁸U tamper surrounding the fuel

$$n_{\text{fissions}} = 6.022 \times 10^{24} D$$

$$Y = n_{\text{fissions}} Q$$

$$Y = 6.022 \times 10^{24} (180 \text{ MeV})$$

$$Y = 41.4834 \text{ kT}$$

41.5 kT

Report your answer in this box.

Wt. No.

Cadet: _____

25 4. A D-T fusion assembly consists of 50.0 g of equal a/o deuterium-tritium fuel and has a compressed volume of 0.500 cm³ at the time of ignition. Once ignition occurs, assume the fusion assembly burns completely. Calculate the following to 3 significant figures:

c. The radius for sustained burn for this assembly if the plasma temperature is 4.038x10⁷ K. Based on your answer, was the assumption of complete burn correct?

$$R_{SB} = \frac{3\sigma_{sb}T^4}{n_b n_t (\overline{\sigma v}) Q_l}$$

$$E = 4.038 \times 10^7 \text{ K} \left(\frac{1 \text{ eV}}{1.1605 \times 10^4 \text{ K}} \right) \left(\frac{1 \text{ keV}}{1000 \text{ eV}} \right)$$

$$E = 3.4795 \text{ keV}$$

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} E^{-\frac{2}{3}} e^{-19.94 E^{-\frac{1}{3}}}$$

$$(\overline{\sigma v})_{DT} = 3.68 \times 10^{-12} 3.48^{-\frac{2}{3}} e^{-19.94 (3.48^{-\frac{1}{3}})}$$

$$(\overline{\sigma v})_{DT} = 3.0904 \times 10^{-18} \frac{\text{cm}^3}{\text{s}}$$

$$R_{SB} = \frac{3 \left(5.67 \times 10^{-11} \frac{\text{J}}{\text{K}^4 \text{cm}^2 \text{s}} \right) (4.038 \times 10^7 \text{ K})^4}{3.63 \times 10^{49} \text{ cm}^{-6} \left(3.09 \times 10^{-18} \frac{\text{cm}^3}{\text{s}} \right) 5.6077 \times 10^{-13} \text{ J}}$$

$$N_D = \frac{n_D}{V}$$

$$R_{SB} = 7.196 \text{ cm}$$

$$N_D = \frac{6.022 \times 10^{24}}{0.5 \text{ cm}^3}$$

$$N_D = 1.2044 \times 10^{25} \text{ cm}^{-3}$$

$$n_t n_b = \frac{N_D^2}{4}$$

$$n_t n_b = 3.6264 \times 10^{49} \text{ cm}^{-6}$$

$$Q_l = 3.5 \text{ MeV}$$

$$Q_l = 5.6077 \times 10^{-13} \text{ J}$$

7.20 cm

Report your answer in this box.

☐ The assumption was correct

☒ The assumption was incorrect

Wt. No.

Cadet: _____

- 25 5. A spherical weapon primary composed of pure ^{239}Pu is compressed to 52.0 g/cm^3 . The fission process is initiated with ten thousand neutrons, and the non-leakage probability of the assembly is 0.645. Assume the multiplication factor and non-leakage probability remain unchanged. The primary is next to a secondary with 75.0 g of pure deuterium surrounded by a tamper of ^{238}U . Calculate the following yields in kT to 3 significant figures:

- a. The primary yield, assuming 57 generations before disassembly

$$l_{\infty} = \frac{1}{\Sigma_a v}$$

$$l = \frac{ML}{\rho N_A \sigma_a v}$$

$$\Sigma_a = N \sigma_a \quad N = \frac{\rho N_A}{M}$$

$$l_{\infty} = \frac{M}{\rho N_A \sigma_a v}$$

$$l = l_{\infty} L$$

$$l = \frac{239 \frac{\text{g}}{\text{mol}} (0.645)}{\left(52.0 \frac{\text{g}}{\text{cm}^3}\right) N_A (1.98 \times 10^{-24} \text{cm}^2) \left(1.08 \times 10^9 \frac{\text{cm}}{\text{s}}\right)}$$

$$l = 2.3021 \times 10^{-9} \text{ s}$$

$$\Sigma_f = N \sigma_f = \frac{\rho N_A \sigma_f}{M} = 0.2467$$

$$\alpha = \frac{k - 1}{l} = \frac{1.8275 - 1}{2.302 \times 10^{-9} \text{ s}} = 3.5945 \times 10^8 \frac{\text{gen}}{\text{sec}}$$

$$k_{\infty} = \frac{v \sigma_f}{\sigma_a} \quad k = k_{\infty} L$$

$$t = gl$$

$$t = 1.312 \times 10^{-7} \text{ s}$$

$$k = \frac{v \sigma_f}{\sigma_a} L = \frac{5.61}{1.98} 0.645 = 1.8275$$

$$Y(t) = \frac{\Sigma_f v Q n_0}{\alpha} (e^{\alpha t} - 1)$$

$$Y(t) = \frac{0.2467 \left(1.08 \times 10^9 \frac{\text{cm}}{\text{s}}\right) (7.08 \times 10^{-24} \text{ kT}) (1 \times 10^4)}{3.5945 \times 10^8 \frac{\text{gen}}{\text{sec}}} \left(e^{3.5945 \times 10^8 \frac{\text{gen}}{\text{sec}} (1.312 \times 10^{-7} \text{ s})} - 1\right)$$

$$Y(t) = 16.0137 \text{ kT}$$

16.0 kT

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 5. A spherical weapon primary composed of pure ^{239}Pu is compressed to 52.0 g/cm^3 . The fission process is initiated with ten thousand neutrons, and the non-leakage probability of the assembly is 0.645. Assume the multiplication factor and non-leakage probability remain unchanged. The primary is next to a secondary with 75.0 g of pure deuterium surrounded by a tamper of ^{238}U . Calculate the following yields in kT to 3 significant figures:

- b. The 100% fusion yield resulting from the deuterium fuel assembly

$$Y = n_D Q$$

$$n_D = \frac{m_D N_A}{M}$$

$$Y = 2.2583 \times 10^{25} \left(4.98 \frac{\text{MeV}}{D} \right) \left(\frac{1 \text{ kT}}{2.613 \times 10^{25} \text{ MeV}} \right)$$

$$n_D = \frac{75 \text{ g} \left(6.022 \times 10^{23} \frac{\text{atoms}}{\text{mol}} \right)}{2 \frac{\text{g}}{\text{mol}}}$$

$$Y = 4.3039 \text{ kT}$$

$$n_D = 2.2583 \times 10^{25} \text{ atoms}$$

4.30 kT

Report your answer in this box.

- c. The tamper yield, assuming 100% efficiency

$$Y = n_{\text{fissions}} Q$$

$$Y = \frac{2.2583 \times 10^{25}}{5} (180 \text{ MeV})$$

$$Y = 31.1125 \text{ kT}$$

31.1 kT

Report your answer in this box.

- d. The total yield of the device

$$Y_t = Y_{\text{primary}} + Y_{\text{secondary fusion}} + Y_{\text{secondary fission}}$$

$$Y_t = 16 \text{ kT} + 4.30 \text{ kT} + 31.1 \text{ kT}$$

$$Y_t = 51.4301 \text{ kT}$$

51.4 kT

Report your answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #3

Cadet _____
(First) (Last)

Section _____
(hour)

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Problem Set 3 – 125 Points

1. Instructions:

- a) For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- b) Write your name on every page. **Only add pages to the end of your submitted PDF.**
- c) You are allowed and encouraged to collaborate with peers on problem sets. Ensure you document assistance IAW the DAW.
- d) Complete the documentation section below AFTER all work is complete. Sign using a digital signature (CAC), digital ink (stylus or finger), or ink. Do not type.

2. Documentation:

_____ I certify that I have completely documented all sources that I used to complete this assignment and that I acknowledged all assistance I received in the completion of this assignment.

_____ I certify that I did not use any sources or receive any assistance requiring documentation while completing this assignment.

AND

_____ I acknowledge that I used generative AI to complete this assignment.

_____ I acknowledge that I did not use generative AI to complete this assignment.

Cadet Signature: _____

Cadet Name (print): _____

Date: _____

- 25 1. A 25.0 kT device is detonated at an altitude of 554 km. A Cubesat surrounded by a thin aluminum shield is directly below the point of burst at an altitude of 315 km. Calculate the following to 3 significant figures:

- a. The x-ray fluence at the Cubesat

$$F(r) = \frac{Y_x}{4\pi r^2}$$

$$F(r) = \frac{0.7(25kT)}{4\pi(239 \times 10^3 m)^2}$$

$$F(r) = 2.4380 \times 10^{-11} \frac{kT}{m^2}$$

$$102 \frac{J}{m^2}$$

Report your answer in this box.

- b. The energy deposition in the shield at a depth 5.50 mm, assuming the x-ray energy spectrum can be approximated as a single group of 1.50 MeV photons

$$\rho = 2.699 \frac{g}{cm^3}$$

$$E_{dep}(x) = F_0 e^{-\mu x} \left(\frac{\mu_a}{\rho} \right)$$

$$\frac{\mu_a}{\rho} = 0.0248 \frac{cm^2}{g}$$

$$E_{dep}(x) = 102 \frac{J}{m^2} e^{-0.05 \frac{cm^2}{g} * 2.699 \frac{g}{cm^3} * 0.55 cm} \left(0.00248 \frac{m^2}{kg} \right)$$

$$\frac{\mu}{\rho} = 0.0500 \frac{cm^2}{g}$$

$$E_{dep}(x) = 0.2350 \frac{J}{kg}$$

$$0.235 \frac{J}{kg}$$

Report your answer in this box.

- c. The maximum pressure in psi at the depth due to the x-ray energy deposition

$$p(x) = E_{dep}(x) \rho$$

$$p(x) = 0.235 \frac{J}{kg} \left(0.002699 \frac{kg}{cm^3} \right)$$

$$p(x) = 6.3424 \times 10^{-4} \frac{J}{cm^3}$$

$$p(x) = 0.0920 \text{ psi}$$

$$0.0920 \text{ psi}$$

Report your answer in this box.

Wt. No.

Cadet: _____

25 2. A 45.0 kT nuclear weapon is detonated in the atmosphere. Calculate the following in feet to 3 significant figures:

a. The minimum height of burst to avoid significant local fallout

$$H = 180W^{0.4}$$

$$H = 180(45^{0.4})$$

$$H = 825.1968 \text{ ft}$$

825 ft

Report your answer in this box.

b. The radius of the fireball at breakaway, at this height of burst

$$R = 100W^{0.4}$$

$$R = 100(45^{0.4})$$

$$R = 458.4426 \text{ ft}$$

458 ft

Report your answer in this box.

c. The radius of the fireball at thermal minimum

$$R = 90W^{0.4}$$

$$R = 90(45^{0.4})$$

$$R = 412.5984 \text{ ft}$$

413 ft

Report your answer in this box.

d. The radius of a fireball at breakaway from a contact surface burst Tsar Bomba (58 MT)

$$R = 145W^{0.4}$$

$$R = 145(58000^{0.4})$$

$$R = 11,661 \text{ ft}$$

11,700 ft

Report your answer in this box.

Wt. No.

Cadet: _____

25 3. List and *briefly* describe the five phases of fireball growth in an atmospheric nuclear weapon burst.

- **Burn-Out:** The weapons case and surrounding air are completely ionized by the x-rays from fission. This is typically confined to a radius of 25 m from the point of burst.
- **Diffusion Growth:** The primary X-rays are absorbed in the surrounding air, and the fireball appears to grow through re-radiation, which heats the surrounding air.
- **Hydrodynamic Separation:** The radiation front is slowed as photons are absorbed and re-radiated, allowing the blast front to overtake it. The shock wave compresses the air and makes it opaque to the fireball, causing the fireball to appear dimmer.
- **Hydrodynamic Growth:** Air behind the shock front heats and expands due to energy being absorbed from the radiation front.
- **Breakaway:** The shock wave energy will dissipate spherically, and the air becomes less dense, allowing observation of the fireball within. The shock front and radiation front now appear as two distinct spherical waves.

Wt. No.

Cadet: _____

25 4. A 500 kT weapon is detonated at an altitude of 40,000 feet. Calculate the following to 3 significant figures:

a. The rate of thermal energy emission 1.50 seconds after the detonation

$$P_{max} = \frac{3.56W^{0.59}}{\left(\frac{\rho(h)}{\rho_0}\right)^{0.45}}$$

$$P_{max} = \frac{3.56(500)^{0.59}}{(0.24)^{0.45}}$$

$$P_{max} = 264.696 \frac{kT}{s}$$

$$t_{max} = 0.038W^{0.44} \left(\frac{\rho(h)}{\rho_0}\right)^{0.36}$$

$$t_{max} = 0.038(500)^{0.44} (0.24)^{0.36}$$

$$t_{max} = 0.35011 s$$

$$\frac{t}{t_{max}} = 4.28$$

$$\frac{P}{P_{max}} = 0.15$$

$$P = 0.15P_{max}$$

$$P = 0.15 \left(264.696 \frac{kT}{s}\right)$$

$$P = 39.7044 \frac{kT}{s}$$

$$39.7 \frac{kT}{s}$$

Report your answer in this box.

b. The total thermal energy emitted through 2.0 seconds after the detonation

$$E_{tot} = fW$$

$$\frac{E}{E_{tot}} = 0.72$$

$$f = 0.39$$

$$E = 0.75(0.39)(500)$$

$$E = 140.4 kT$$

$$140 kT$$

Report your answer in this box.

c. The radiant exposure on a target 75,000 feet from ground zero

$$Q = \frac{85.6fW\tau}{D^2}$$

$$D = \sqrt{H^2 + d^2}$$

$$D = \sqrt{40^2 + 75^2}$$

$$D = 85 \text{ kilofeet}$$

$$\tau = 0.72$$

$$f = 0.39$$

$$Q = \frac{85.6(0.39)(500)(0.72)}{85^2}$$

$$Q = 1.6634 \frac{cal}{cm^2}$$

$$1.66 \frac{cal}{cm^2}$$

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 5. A 25.0 kT weapon is detonated at an altitude of 2500. feet above the ground. A target is located on the ground at a ground distance of 7500 feet. Calculate the following:

- a. The scaled height-of-burst for a 1 kT weapon in feet (3 sig figs)

$$h_1 = \frac{h}{W^{\frac{1}{3}}}$$

$$h_1 = \frac{2500}{25^{\frac{1}{3}}}$$

$$h_1 = 854.99 \text{ ft}$$

855 ft

Report your answer in this box.

- b. The scaled ground distance from a 1 kT weapon in feet (3 sig figs)

$$d_1 = \frac{d}{W^{\frac{1}{3}}}$$

$$d_1 = \frac{7500}{25^{\frac{1}{3}}}$$

$$d_1 = 2564.96 \text{ ft}$$

2560 ft

Report your answer in this box.

- c. The peak overpressure at the target location in psi (2 sig figs)

Using 1 kT scaled distances and NERD-P fig 5-7:

4.0 psi

Report your answer in this box.

- d. The time for the shock front to arrive at the target location in sec (2 sig figs)

Using 1 kT scaled distances and NERD-P fig 5-11: $t_1 = 1.8 \text{ sec}$

$$t = t_1 W^{\frac{1}{3}}$$

$$t = 1.8s(25kT)^{\frac{1}{3}}$$

$$t = 5.26 \text{ sec}$$

5.3 sec

Report your answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #4

Cadet _____
(First) (Last)

Section _____
(hour)

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE450, Nuclear Weapons Effects
Problem Set 4 – 125 Points

1. Instructions:

- a) For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
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- d) Complete the documentation section below AFTER all work is complete. Sign using a digital signature (CAC), digital ink (stylus or finger), or ink. Do not type.

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_____ I certify that I did not use any sources or receive any assistance requiring documentation while completing this assignment.

AND

_____ I acknowledge that I used generative AI to complete this assignment.

_____ I acknowledge that I did not use generative AI to complete this assignment.

Cadet Signature: _____

Cadet Name (print): _____

Date: _____

- 25 1. A 125 kT weapon is detonated underground. Calculate the following in feet to 3 significant figures:

- a. The optimal depth-of-burst for crater radius in dry soil

$$DOB = DOB_1 W^{0.3}$$

$$DOB = 170(125)^{0.3}$$

$$DOB = 723.6389 \text{ ft}$$

724 ft

Report your answer in this box.

- b. The apparent crater radius for 85 ft DOB in dry soil

$$DOB_1 = \frac{DOB}{W^{0.3}}$$

$$DOB_1 = \frac{85}{125^{0.3}}$$

$$DOB_1 = 19.968 \text{ ft}$$

$$R_a = R_{a1} W^{0.3}$$

$$R_a = 120(125)^{0.3}$$

$$R_a = 510.804 \text{ ft}$$

511 ft

Report your answer in this box.

- c. The apparent crater depth for 725 ft DOB in wet hard rock

$$DOB_1 = \frac{DOB}{W^{0.3}}$$

$$DOB_1 = \frac{725}{125^{0.3}}$$

$$DOB_1 = 170.32 \text{ ft}$$

$$D_a = D_{a1} W^{0.3}$$

$$D_a = 80(125)^{0.3}$$

$$D_a = 340.536 \text{ ft}$$

341 ft

Report your answer in this box.

- d. The maximum distance from surface zero at which a small, well-designed structure will experience severe damage for the device in part b.

Table 6.108: 1.25 apparent crater radii

639 ft

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 2. A 500 kT thermonuclear weapon is detonated at a height of 4000 feet. 60% of the weapon's yield is derived from fission. Calculate the total initial radiation dose equivalent 8,000 feet from ground zero to 3 significant figures.

$$D = \sqrt{4000^2 + 8000^2}$$

$$D = 8944 \text{ ft}$$

$$D = 2981 \text{ yards}$$

$$D = D_n + D_{\gamma s} + D_{\gamma f}$$

$$D_n = 0.01 (500 \text{ kT})$$

$$D_n = 5 \text{ rads}$$

$$D_{\gamma s} = 0.05 (500 \text{ kT})$$

$$D_{\gamma s} = 25 \text{ rads}$$

$$D_{\gamma f} = 0.017 (8000 \text{ kT})(0.6)$$

$$D_{\gamma f} = 81.6 \text{ rads}$$

$$H = 5(10) + 25 + 81.6$$

$$H = 157 \text{ rem}$$

157 rem

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 3. Your unit is required to conduct a mission in an area affected by early fallout. You enter the fallout zone 3 hours post-detonation and measure the radiation dose rate at that time to be 2.5 rad/hr. Calculate your total absorbed dose if the mission takes six hours to complete to 2 significant figures.

$$\text{Fig 5 - 24 (9.16a): } 3 \text{ hrs} \rightarrow 0.27 \frac{DR}{UTRD}$$

$$\frac{DR}{UTRD} = 0.27$$

$$R_1 = \frac{DR}{0.27}$$

$$R_1 = \frac{2.5 \frac{\text{rad}}{\text{hr}}}{0.27}$$

$$R_1 = 9.26 \frac{\text{rad}}{\text{hr}}$$

Alternate:

$$R = R_1 t^{-1.2}$$

$$R_1 = \frac{R}{t^{-1.2}}$$

$$R_1 = \frac{2.5 \frac{\text{Rad}}{\text{hr}}}{3^{-1.2}}$$

$$R_1 = 9.34 \frac{\text{Rad}}{\text{hr}}$$

$$\text{Fig 25: } 3 \text{ hrs} \rightarrow 6.2 \frac{D_T}{UTRD}$$

$$6.2 = \frac{D_T}{9.34}$$

$$D_{T1} = 57.908 \text{ Rads}$$

$$\text{Fig 25: } 9 \text{ hrs} \rightarrow 6.9 \frac{D_T}{UTRD}$$

$$6.9 = \frac{D_T}{9.34}$$

$$D_{T2} = 64.446 \text{ Rads}$$

$$D_T = D_{T2} - D_{T1}$$

$$D_T = 64.446 - 57.908$$

$$D_T = 6.538 \text{ Rads}$$

Alternate:

$$D_T = 5R_1(t_a^{-0.2} - t_b^{-0.2})$$

$$D_T = 5(9.34)(3^{-0.2} - 9^{-0.2})$$

$$D_T = 7.39 \text{ Rads}$$

7.4 Rads

Report your answer in this box.

Wt. No.

Cadet: _____

- 25 4. Your unit is tasked to complete a patrol in a fallout area but not to exceed a total absorbed dose of 65 rad. The mission will take four hours. Calculate the soonest after detonation that you can enter the fallout area in hours to two significant figures if the measured dose rate 4 hours post detonation is 7.7 rad/hr.

$$\text{Fig 5 - 24 (9.16a): } 4 \text{ hrs} \rightarrow 0.2 \frac{DR}{UTRDR}$$

$$\frac{DR}{UTRDR} = 0.2$$

$$R_1 = \frac{DR}{0.2}$$

$$R_1 = \frac{7.7 \frac{\text{rad}}{\text{hr}}}{0.2}$$

$$R_1 = 38.5 \frac{\text{rad}}{\text{hr}}$$

$$MF = \frac{D_T}{UTRDR}$$

$$MF = \frac{65}{40.64}$$

$$MF = 1.599$$

$$\text{Fig 27: } 1.6 \text{ MF} \rightarrow 4 \text{ hrs} \rightarrow 0.7 \text{ hours}$$

Alternate:

$$R = R_1 t^{-1.2}$$

$$R_1 = \frac{R}{t^{-1.2}}$$

$$R_1 = \frac{7.7 \frac{\text{Rad}}{\text{hr}}}{4^{-1.2}}$$

$$R_1 = 40.64 \frac{\text{Rad}}{\text{hr}}$$

Alternate:

$$D_T = 5R_1(t_a^{-0.2} - t_b^{-0.2})$$

$$65 = 5(40.64)(t^{-0.2} - (t + 4)^{-0.2})$$

$$t = 0.7789 \text{ hrs}$$

0.78 hrs

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 25 5. Describe how EMP is created in a high-altitude burst and describe three practices that can be used to harden a system against EMP.

In a high altitude burst, γ -rays moving down from the point of burst are moving into a region of denser air. These γ -rays will form a circular deposition region by ionizing the air through Compton scattering. The scattering electrons will be deflected by the Earth's magnetic field. This deflection causes acceleration which causes the electrons to emit EM waves directed toward the surface of the Earth. (ENW Paragraphs 11.13-11.14)

- DISCONNECT the antenna/power source.
- Use SHORT cable runs with NO loops.
- BURY cables a minimum of 3 feet.
- SHIELD equipment with a continuous metal sheet.
- Properly GROUND all electronics.



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WEST POINT.

TAB L EXTENSION PROBLEMS



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EXTENSION PROBLEM #1



Your group will code three weapon models where yield is compared:

1. The base model where exponential growth from an initial number of neutrons, N_0 , is determined using a **constant** leakage factor, L

$$L = 0.43M^{-0.15} \ln C + 0.14 \ln M$$

Where: $L \equiv$ non-leakage probability in Pu-239

$M \equiv$ Pu-239 mass (kg)

$C \equiv$ compression factor relative to nominal density (19.6 g/cm³)

2. The mechanical disassembly model where the base model is used but now leakage factor **varies** with changing radius and burn-up is accounted for in macroscopic cross sections.
3. The final model compares the effects of adding a tamper and boosting to the weapon.



For each model you will be using NE450 Battle Drill 1 for the solution:

1. Determine multiplication factor k
2. Determine neutron lifetime l
3. Determine bomb alpha α

These factors will change as we add complexity to our model:

- Yield will cause outward expansion (Eq 11)
- Expansion will change compression (recalculate density)
- Compression changes leakage (recalculate L)
- Burnup changes macroscopic cross sections (Eq 12)
- Tamping adds non-fissile mass to Eq 11, boosting injects a large population of neutrons in a single generation. You can play with these parameters to get a feel for their effects.

Pro tips:

- UNITS, UNITS, UNITS!
- When you use discrete modeling, every generation's inputs come from the previous generation's outputs.
- Our equations are simplified models for fissioning spheres, NOT exploded spheres. You should expect your code to work up to a certain point depending on your initial conditions and then start giving you nonsensical results or errors. Once bomb alpha reaches second critical, your model has exploded. This point should be less than 100 generations.



- Your initial conditions (gen = 0) are set by this slide and your group
- Start with $n_0 = 1,000$ neutrons
- Assume we initiate at optimal compression: $u_1 = 0$ m/s
- Your group has a total of 10 units to allocate to starting fuel mass (in kg) and starting compression factor. For example:
If I want to start with 9 kg of fuel, I only have $10 - 9 = 1$ unit left to allocate to my compression factor. You are not allowed to start with a compression factor of 1 as this sphere would never go supercritical. You do not need to use whole numbers.
- You must select your mass/compression allocation during Lesson 6 and email it, with a group name if you wish, to your instructor.
- **BONUS:** The group with the highest yield before tamping and boosting (and proper code to get that yield) will get 5 bonus points!



- You are briefing in class 25 and 26 February
- The Brief: 5-10 minutes showing what code you used, how it works, results (yield graphs showing cumulative kT per generation), and what you learned along the way. Memes optional but encouraged. Slides uploaded to canvas.
- The Poster: A standalone single slide poster following the example format uploaded to canvas.
- The Code: Code file (.xlsx, .m, .py, ...) uploaded to canvas.

Outstanding Student Work

UNITED STATES MILITARY ACADEMY

EXTENSION PROBLEM 1

NE450: NUCLEAR WEAPONS EFFECTS

SECTION II

MAJ DAVE PERT

BY

[REDACTED]

WEST POINT, NEW YORK

26 FEBRUARY 2026

CP HB WG WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

 WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

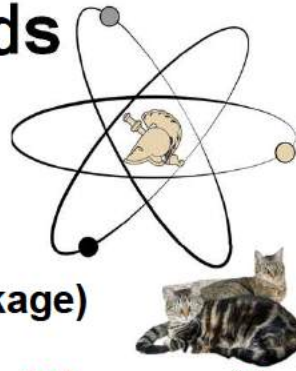
CP HB WG WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

 WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

[REDACTED]



Extension Problem 1: Modeling Weapon Yields



Introduction:

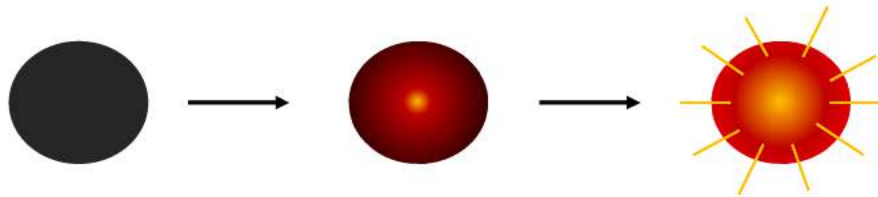
By applying different EXCEL and MATLAB models we determined theoretical yield for three different weapon systems of increasing complexion. All three use a fuel mass of 6.5kg Pu^{239} and a compression factor of 3.5.

Initial Conditions

- 1.) $N_0 = 1000$
- 2.) $M = 6.5 \text{ kg Pu} - 239$
- 3.) $C = 3.5$

Overall Assumptions

- Densities at every time step are uniform throughout sphere
- Spherical shape is maintained at every timestep
- Assume full burnup for model 3



Base Model (Constant Leakage)

The base model assumes constant leakage:

$$L = 0.43, M^{-0.15} \ln(C) + 0.14, \ln(M) \quad (1)$$

Where M is the mass of plutonium fuel and C is the compression factor.

With constant leakage, we can assume an exponential increase in yield the that utilizes:

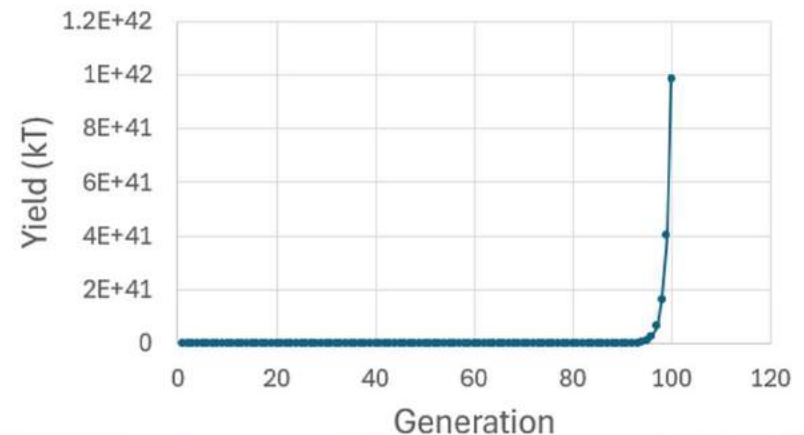
$$Y(t) = \frac{\sum_f v Q N_0 e^{\alpha t}}{\alpha} (e^{\alpha t} - 1) \quad (2)$$

Where $\sum_f$ is the macroscopic cross-section of $Pu-239$ fission, α is the bomb alpha, v is the avg. speed of neutron, Q is the energy generated per fission, and t is the time in generations.

Model Results

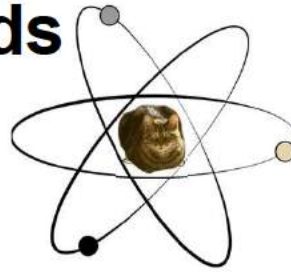
$$Y(100) = 9.859E41 \text{ kT}$$

Model 1: Yield vs. Generation





Extension Problem 1: Modeling Weapon Yields



Mechanical Disassembly Model

The second model considers the effects of mechanical disassembly on the leakage. The outward expansion of the nuclear explosion results in a change of radius. The radius of the bombs core determines the spherical volume and number density of target Pu atoms.

The equation used to derive the yield is:

$$\frac{Y}{3} = \frac{1}{2} M(u_2)^2 \quad (3)$$

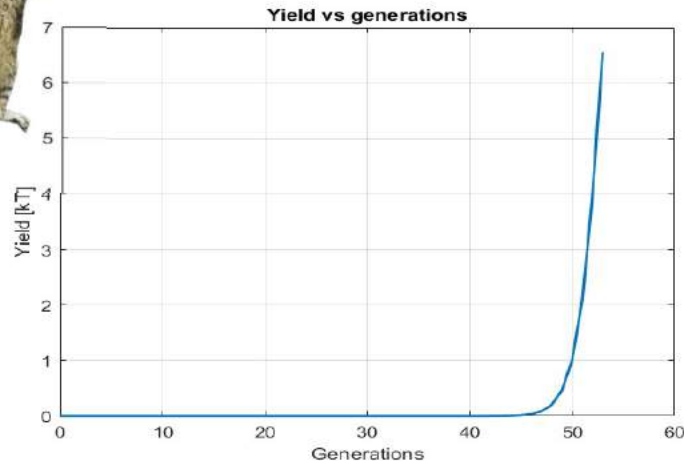
Where M is the mass of plutonium fuel and u_2 is the expansion velocity.

To account for the burn-up we use the compressed absorption equation with density being a function of time:

$$\Sigma(t) = \Sigma_0 \frac{\rho(t)}{\rho_0} \left[1 - BF + 2BF \frac{\sigma_{ff}}{\sigma_{fuel}} \right] \quad (4)$$

Model Results

Y(53) = 6.5579 kT

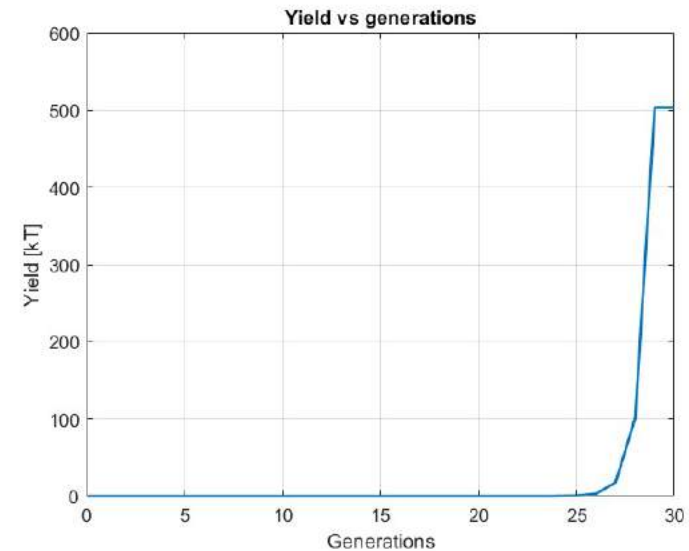


Tamper / Boosting Effects

The third model utilizes both tamper and boosting materials to increase the total yield of our fuel mass. We assume 100% burn of our tamper. We input the n_{tar} amount of neutron born from fusion at generation 30. The target amount of neutrons was calculated as 10 grams of deuterium fuel, corresponding to $2.99e24$ neutrons injected. Tamper was modeled via equation 3 the with an addition of 10 kg of tamper surrounding the primary.

Model Results

Y(29) = 503.63 kT



Potential Sources of Error:

MATLAB Code iterates over equation 3 every time, possibly constantly increasing mass leading to a much larger tamper

Table of Contents

.....	1
Problem 1:Constant Leakage factor	1
Problem 2: Varying radius and burn up (original attempted code for reference)	2
Problem 2: Variable density/compression; macroscopic XS updated each gen (ChatGPT supplied code)	3
boosting and tamper	7
Problem 3: Neutron injection at line 372 and mass in eq. 11 +10	7

```
%NE450EP1.m
%
% SYNOPSIS: Script with 3 differen functions to calculate the
% multiplication factor, neutron lifetime, and bomb alpha for 3 different
% scenarios
%
%
% CONTRIBUTION: - MATLAB 2024b 28JUN2026
%               - [REDACTED]
%               - Attribution, License (if applicable)
%
% CHANGE LOG: Helpful if you have made major revisions. List previous
% versions and major updates (especially if a shared/git document)
% -----
```

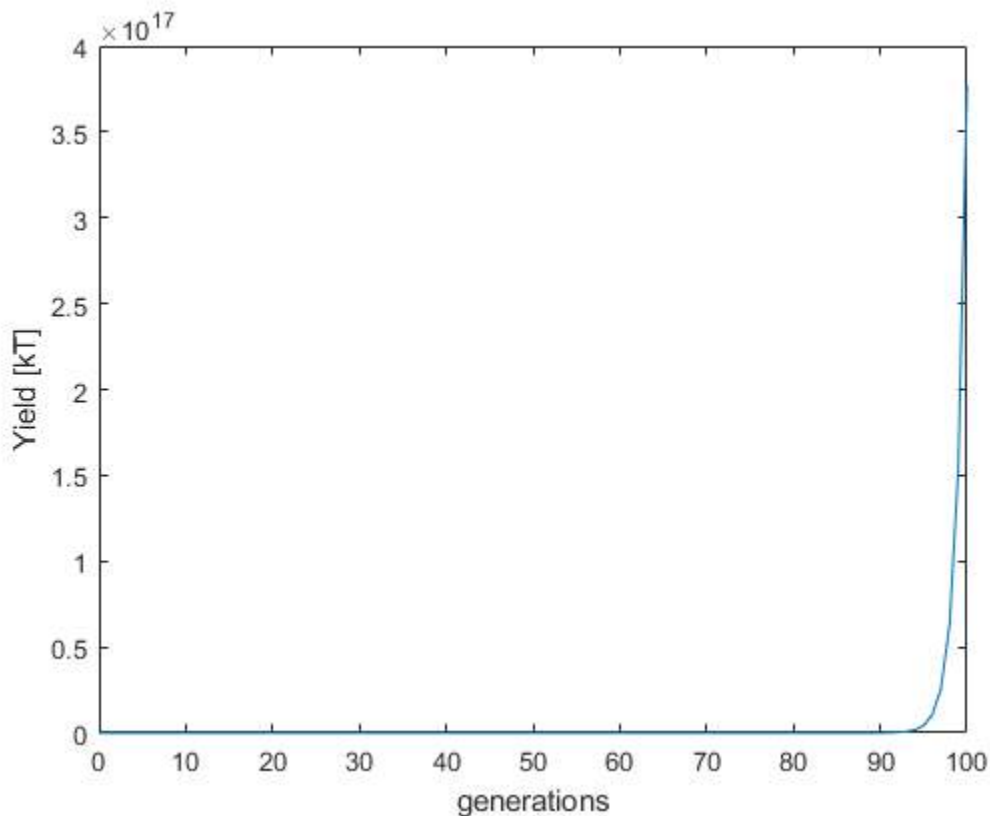
Problem 1:Constant Leakage factor

```
%known:mass and compression factor
clear
clc
close all
M = 6.5; %[kg]
C = 3.5; %[compression]
ro = 19.6; %[dens]
leak_prob = ((0.43*M^(-0.15))*log(C))+(0.14*log(M));
k_inf = (5.61/1.98);
k = k_inf*leak_prob;
N = (ro * (6.022*10^23))/239;
sigma_a = N * 1.98;
l_inf = 1/(sigma_a * (1.08*10^9));
l_val = l_inf * leak_prob;
bomb_alpha = (k-1)/l_val;
%below is calc for yield of infinite model
n_0 = 1000;
gen = 0:100;
time = l_val.*gen;
yield = zeros(size(time));
sigma_f = (5.61/2.98)*N;
yieldcalc = ((sigma_f * (1.08*10^8)*185*n_0)/bomb_alpha)/(2.613*10^25);
for qq = 1:length(time)
    yield(qq) = yieldcalc * ((exp(bomb_alpha.*time(qq)))-1);
```

```

end
%ChatGPT, OpenAI LLM, Assistance Given to the author, coding assistance.
%When I attempted to iterate across the vector for time, I recieved an
%error. I uploaded the lines 31-39 into ChatGPT to see if it could help me
%correct the code. It told me to covert line 34 to a matrix of zeros to be
%filled in the size of the time vector. it then told me to modify the time
%variable in line 38 using the qq portion of the for loop. A transcript of
%our conversation can be located here:
% https://chatgpt.com/share/699d1135-8710-8012-a9e0-53c1150dc88e
plot(gen,yield)
xlabel("generations")
ylabel("Yield [kT]")

```



Problem 2: Varying radius and burn up (original attempted code for reference)

```

clc
clear
close all

% BF = num fissions taken place /n_0
% BFinitial = 1
%function [mod_sigma] = SigmaModder(Sigma_og)
%     BF = 5;%placeholder
%     fisfrag = 3; % placeholder

```

```

%      ro = 19.6;
%      newro = 3*ro;
%      fuelsig = 5.61;
%      mod_sigma = Sigma_og * (newro/ro)*(1-BF+2*BF(fisfrag/fuelsig));
%
%end

% while alpha>1, while loop runs through code and resets all variables
% within the code to include k,l, alpha
M = 6.5; %[kg]
C = 3.5; %[compression]
ro = 19.6; %[dens]
leak_prob = ((0.43*M^(-0.15))*log(C))+(0.14*log(M));
k_inf = (5.61/1.98);
k = k_inf*leak_prob;
N = (ro * (6.022*10^23))/239;
sigma_a = N * 1.98;
l_inf = 1/(sigma_a * (1.08*10^9));
l_val = l_inf * leak_prob;
bomb_alpha = (k-1)/l_val;
while bomb_alpha > 0
    %must recalc certain values then replug them into certain equations
    %involving both density and compression factor. Compression factor is
    %dens over nominal dens, and is a control of radius. radius is calced
    %via eq 12 and is dependent upon
    exp_vel = sqrt(((2/3)*yield)/M);

    leak_prob = ((0.43*M^(-0.15))*log(C))+(0.14*log(M));

    mod_sigma = Sigma_ * (newro/ro)*(1-BF+2*BF*(fisfrag/fuelsig));
end

plot(gen,yield)
xlabel("generations")
ylabel("Yield")

```

Unrecognized function or variable 'yield'.

```

Error in NE450EP1 (line 87)
    exp_vel = sqrt(((2/3)*yield)/M);
                        ^^^^^

```

Problem 2: Variable density/compression; macroscopic XS updated each gen (ChatGPT supplied code)

```

clear; clc; close all

% -----
% GIVEN / CONSTANTS
% -----

```

```

M    = 6.5;           % [kg]
ro0  = 19.6;          % nominal density [g/cm^3]
C_init = 3.5;         % initial compression factor (as in Problem 1)

NA = 6.022e23;        % [1/mol]
A   = 239;            % [g/mol]

k_inf = (5.61/1.98);

v_abs = 1.08e9;        % [cm/s] used in lifetime
v_fis = 1.08e8;        % [cm/s] used in yield rate (per your original code)

% Microscopic cross sections (barns) -> convert each step to macroscopic
sigma_a_b = 1.98;      % [b]
sigma_f_b = (5.61/2.98); % [b] (as in your original sigma_f form)
BARN = 1e-24;          % [cm^2/b]

% Energy per fission: 185 MeV -> J
E_f = 185e6 * 1.602176634e-19; % [J/fission]

% Yield conversion to kT per your assignment constant
KT_DEN = 2.613e25;        % [J per kT] (your convention)

% Initial neutron population
n = 1000;

% -----
% INITIAL GEOMETRY from initial compressed density
% -----
rho_init_gcm3 = C_init * ro0; % [g/cm^3]
rho_init_kgm3 = rho_init_gcm3*1000; % [kg/m^3]
V0 = M / rho_init_kgm3; % [m^3]
r = (3*V0/(4*pi))^(1/3); % [m]

% -----
% STORAGE
% -----
maxGen = 100; % your generation cap
gen_hist = zeros(1,maxGen);
time_hist = zeros(1,maxGen);
yieldKT_hist = zeros(1,maxGen);
alpha_hist = zeros(1,maxGen);
C_hist = zeros(1,maxGen);
k_hist = zeros(1,maxGen);

% -----
% LOOP STATE
% -----
gen = 0;
t = 0;
Y_J = 0;
bomb_alpha = +Inf;

while bomb_alpha > 0 && gen < maxGen

```

```

% --- geometry -> density ---
V_m3 = (4/3)*pi*r^3;           % [m^3]
rho_kgm3 = M / V_m3;           % [kg/m^3]
rho_gcm3 = rho_kgm3 / 1000;    % [g/cm^3]

% --- compression factor ---
C = rho_gcm3 / ro0;
if C <= 0
    warning('C <= 0 encountered; stopping.');
```

break

```
end

% --- leakage and k ---
leak_prob = (0.43*M^(-0.15))*log(C) + (0.14*log(M));
leak_prob = max(min(leak_prob,1), 1e-12); % keep usable

k = k_inf * leak_prob;

% --- number density (atoms/cm^3) ---
N = (rho_gcm3 * NA) / A;       % [atoms/cm^3]

% --- macroscopic cross sections (1/cm) from microscopic barns ---
Sigma_a = N * sigma_a_b * BARN; % [1/cm]
Sigma_f = N * sigma_f_b * BARN; % [1/cm]

% --- lifetimes ---
l_inf = 1/(Sigma_a * v_abs);    % [s]
l_val = max(l_inf * leak_prob, 1e-20);
dt = l_val;                    % generation time step

% --- alpha ---
bomb_alpha = (k - 1)/l_val;     % [1/s]

% --- yield increment using MACROSCOPIC Sigma_f each generation ---
% Convert volume to cm^3 because Sigma_f is 1/cm and v is cm/s:
V_cm3 = V_m3 * 1e6;            % [cm^3]

% Fission rate ~ (Sigma_f * v_fis) * n * V
dY_J = (Sigma_f * v_fis) * n * V_cm3 * E_f * dt; % [J]
Y_J = Y_J + dY_J;

% --- neutron update ---
n = n * exp(bomb_alpha * dt);

% --- expansion velocity from yield in Joules ---
exp_vel = sqrt(((2/3) * Y_J) / M); % [m/s]
r = r + exp_vel * dt;           % [m]

% --- time advance ---
t = t + dt;

% --- store ---
idx = gen + 1;

```

```

    gen_hist(idx)      = gen;
    time_hist(idx)     = t;
    yieldKT_hist(idx)  = Y_J / KT_DEN * 4.186e12;
    alpha_hist(idx)    = bomb_alpha;
    C_hist(idx)        = C;
    k_hist(idx)        = k;

    gen = gen + 1;
end

% trim
nPts = gen;
gen_hist      = gen_hist(1:nPts);
time_hist     = time_hist(1:nPts);
yieldKT_hist  = yieldKT_hist(1:nPts);
alpha_hist    = alpha_hist(1:nPts);

% -----
% PLOTS
% -----
figure;
plot(gen_hist, yieldKT_hist, 'LineWidth', 1.5); grid on
xlabel('Generations')
ylabel('Yield [kJ]')
title('Yield vs generations')

% Pert, David MAJ, Instructor P&NE Department, Additional Instruction. MAJ
% Pert answered questions pertaining to the correctness of our first model
% as well as giving us some initial instructions about how to go about
% problem two, mainly being the fact that a while loop would need to be
% utilized to iterate until bomb alpha was negative, as well as the hint
% that our model would kinetically disassemble around 56 generations. He
% also advised us on the usage of eq. 11 expansion velocity on radius and
% how it was intertwined with the density calculations.

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% which most factors were dependent upon one another (density, macroscopic
% cross sections, neutron lifetime, leakage probability). These were all
% intertwined factors and I could not properly figure out a way to go about
% the while loop to integrate them all. I gave ChatGPT my initial code for
% both problems 1 and 2 (initial) and explained the radial expansion as well
% as the
% compression factor and how it was density dependent, which itself was
% dependent on radius ( $D=M/V$ ). I showed it what the original code looked
% like with no changing variables other than time and yield and asked it if
% it could help me write a code that integrated all these factors to give
% me a graph for yield. after some alterations the code above is what it
% gave. A transcript of the conversation can be found here:
% https://chatgpt.com/share/699d1135-8710-8012-a9e0-53c1150dc88e

```

boosting and tamper

```
%tamper, adjust mass in equation 11
% boosting, adding an initial population of neutrons at a point in a
% generatoin
```

```
%alpha goes negative around gen 56
```

Problem 3: Neutron injection at line 372 and mass in eq. 11 +10

10 grams of deuterium added at line 372 ie. neutron population increased by 2.99×10^{24} 10 kg of tamper added in eq 11 on line 334

```
clear; clc; close all

% -----
% GIVEN / CONSTANTS
% -----
M    = 6.5;           % [kg]
ro0  = 19.6;          % nominal density [g/cm^3]
C_init = 3.5;         % initial compression factor (as in Problem 1)

NA = 6.022e23;        % [1/mol]
A  = 239;             % [g/mol]

k_inf = (5.61/1.98);

v_abs = 1.08e9;        % [cm/s] used in lifetime
v_fis = 1.08e8;        % [cm/s] used in yield rate (per your original code)

% Microscopic cross sections (barns) -> convert each step to macroscopic
sigma_a_b = 1.98;      % [b]
sigma_f_b = (5.61/2.98); % [b] (as in your original sigma_f form)
BARN = 1e-24;          % [cm^2/b]

% Energy per fission: 185 MeV -> J
E_f = 185e6 * 1.602176634e-19; % [J/fission]

% Yield conversion to kT per your assignment constant
KT_DEN = 2.613e25;        % [J per kT] (your convention)

% Initial neutron population
n = 1000;

% -----
% INITIAL GEOMETRY from initial compressed density
% -----
rho_init_gcm3 = C_init * ro0; % [g/cm^3]
rho_init_kgm3 = rho_init_gcm3*1000; % [kg/m^3]
V0 = M / rho_init_kgm3; % [m^3]
```

```

r = (3*V0/(4*pi))^(1/3);           % [m]

% -----
% STORAGE
% -----
maxGen = 100;                       % your generation cap
gen_hist = zeros(1,maxGen);
time_hist = zeros(1,maxGen);
yieldKT_hist = zeros(1,maxGen);
alpha_hist = zeros(1,maxGen);
C_hist = zeros(1,maxGen);
k_hist = zeros(1,maxGen);

% -----
% LOOP STATE
% -----
gen = 0;
t = 0;
Y_J = 0;
bomb_alpha = +Inf;

while bomb_alpha > 0 && gen < maxGen

    % --- geometry -> density ---
    V_m3 = (4/3)*pi*r^3;           % [m^3]
    rho_kgm3 = (M) / V_m3;         % [kg/m^3]
    if gen == 10
        M = M+10;
    end
    rho_gcm3 = rho_kgm3 / 1000;    % [g/cm^3]

    % --- compression factor ---
    C = rho_gcm3 / ro0;
    if C <= 0
        warning('C <= 0 encountered; stopping.');
```

break

```

    end

    % --- leakage and k ---
    leak_prob = (0.43*M^(-0.15))*log(C) + (0.14*log(M));
    leak_prob = max(min(leak_prob,1), 1e-12); % keep usable

    k = k_inf * leak_prob;

    % --- number density (atoms/cm^3) ---
    N = (rho_gcm3 * NA) / A;       % [atoms/cm^3]

    % --- macroscopic cross sections (1/cm) from microscopic barns ---
    Sigma_a = N * sigma_a_b * BARN; % [1/cm]
    Sigma_f = N * sigma_f_b * BARN; % [1/cm]

    % --- lifetimes ---
    l_inf = 1/(Sigma_a * v_abs);    % [s]
    l_val = max(l_inf * leak_prob, 1e-20);

```

```

dt = l_val; % generation time step

% --- alpha ---
bomb_alpha = (k - 1)/l_val; % [1/s]

% --- yield increment using MACROSCOPIC Sigma_f each generation ---
% Convert volume to cm^3 because Sigma_f is 1/cm and v is cm/s:
V_cm3 = V_m3 * 1e6; % [cm^3]

% Fission rate ~ (Sigma_f * v_fis) * n * V
dY_J = (Sigma_f * v_fis) * n * V_cm3 * E_f * dt; % [J]
Y_J = Y_J + dY_J;

% --- neutron update ---
n = n * exp(bomb_alpha * dt);
if gen == 10
    n = n + 2.99e24;
end
% --- expansion velocity from yield in Joules ---
exp_vel = sqrt((2/3) * Y_J / M); % [m/s]
r = r + exp_vel * dt; % [m]

% --- time advance ---
t = t + dt;

% --- store ---
idx = gen + 1;
gen_hist(idx) = gen;
time_hist(idx) = t;
yieldKT_hist(idx) = Y_J / KT_DEN * 4.186e12;
alpha_hist(idx) = bomb_alpha;
C_hist(idx) = C;
k_hist(idx) = k;

gen = gen + 1;
end

% trim
nPts = gen;
gen_hist = gen_hist(1:nPts);
time_hist = time_hist(1:nPts);
yieldKT_hist = yieldKT_hist(1:nPts);
alpha_hist = alpha_hist(1:nPts);

% -----
% PLOTS
% -----
figure;
plot(gen_hist, yieldKT_hist, 'LineWidth', 1.5); grid on
xlabel('Generations')
ylabel('Yield [kJ]')
title('Yield vs generations')

```

Published with MATLAB® R2024b

Works Cited:

ChatGPT, OpenAI LLM, Assistance Given to the author, coding assistance.

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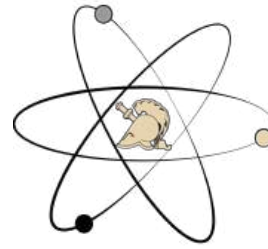
compression factor and how it was density dependent, which itself was dependent on radius ($D=M/V$). I showed it what the original code looked like with no changing variables other than time and yield and asked it if it could help me write a code that integrated all these factors to give me a graph for yield. after some alterations the code above is what it gave. A transcript of the conversation can be found here:

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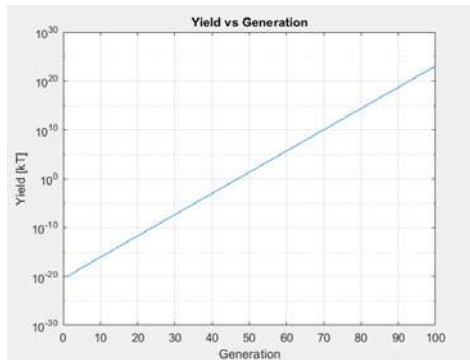
Poor Student Work



NE450 Extension Problem 1



Model 1



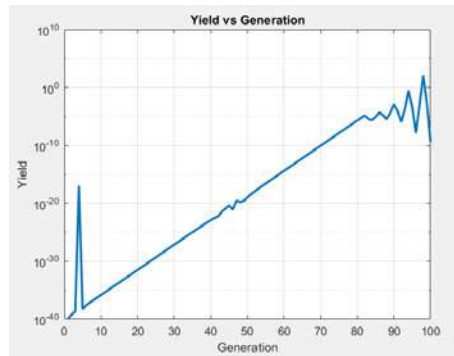
Constant leakage factor

No Iteration

$$\alpha = \frac{k - 1}{l}$$

$$Y(t) = \frac{\Sigma_f v Q n_0}{\alpha} (e^{\alpha t} - 1)$$

Model 2



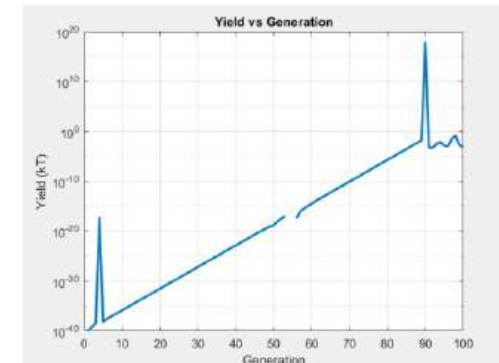
Variable Leakage factor and burnup
Iteration process introduced

Iterating through every neutron due to variable factors

$$\left(\frac{Y}{3}\right) = \frac{1}{2} M (u_2)^2$$

$$\Sigma = \Sigma_0 \frac{\rho}{\rho_0} \left[1 - BF + 2BF \frac{\sigma_{ff}}{\sigma_{fuel}} \right]$$

Model 3



Adding 1 kg tamper and boosting at generation 95 to the weapon

Mass increased from 3.5 kg to 4.5 kg

Tamper introduced into iteration created in model 2

At generation 95 the iteration was broken and an more neutrons were introduced into the simulation

This models all the 14.1 MeV neutrons that would be created from injecting deuterium into the weapon

NE495

Simulated GPF of the A2A3 AMPV

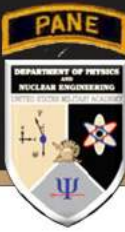


- Group Members:



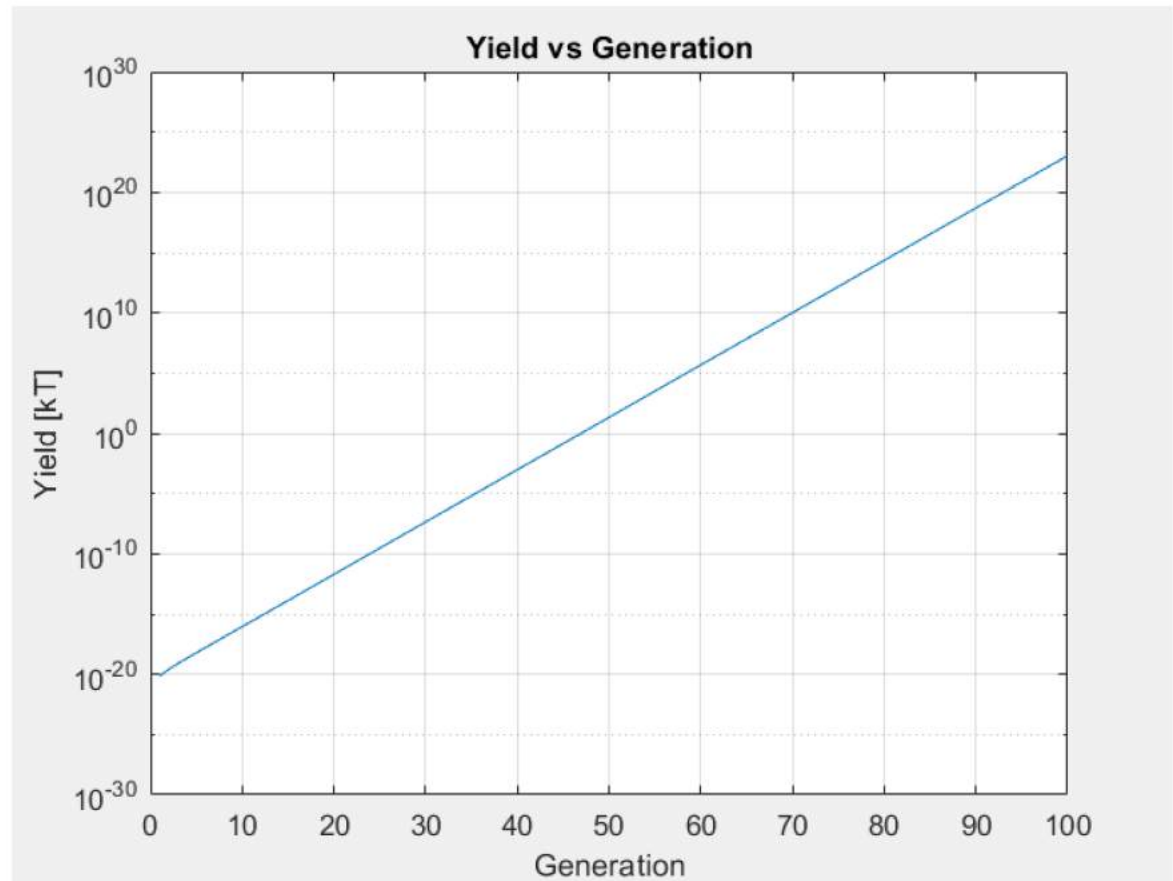


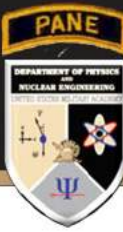
- Model 1
- Model 2
- Model 3
- Lessons learned



- The yield keeps going up due to the constant leakage factor
- There is nothing to make the assembly go subcritical in this model

$$Y(t) = \frac{\Sigma_f v Q n_0}{\alpha} (e^{\alpha t} - 1)$$

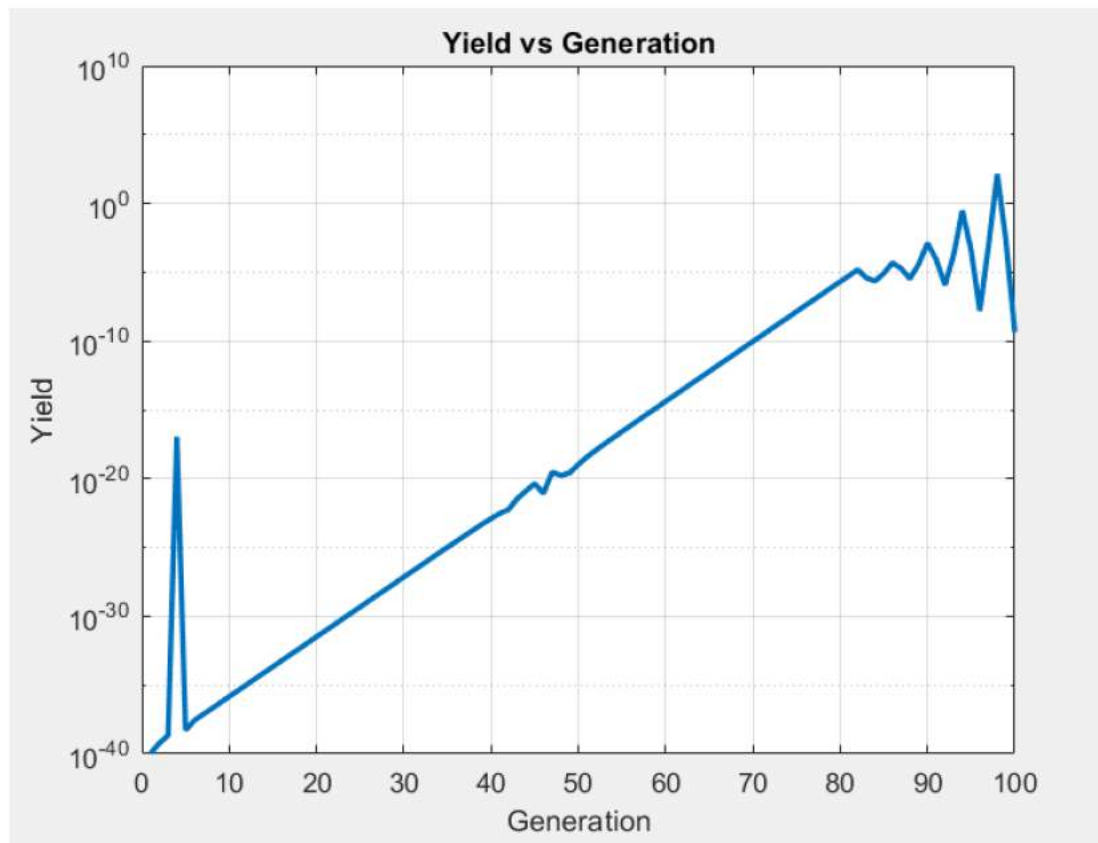


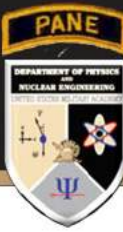


```
1
2   clear
3   clc
4   k_inf = 5.61/1.98;           %criticality inf
5   l_inf = 1/(0.097785*1080000000); %neutron life cycle inf
6   M = 3.5;                     %mass factor
7   C = 6.5;                     %compression factor
8   L = 0.43*M^-0.15 * log(C) + 0.14 * log(M); %Leakage
9   k = k_inf*L;                 %criticality
10  l = l_inf*L;                  %neutron life cycle
11  ba = (k-1)/l;                 %bomb alpha
12
13  M_f = 0.09296;                %macroscopic fission
14  Q = 185;                      %energy per fission
15  v = 1.08E9;                   %neutron speed
16  n_0 = 1000;                   %initial neutrons
17  g = (1:100);                  %generations
18  Y_1 = (((M_f*v*Q*n_0)/ba) .* (exp(g)-1))/2.613E25; %yield
19
20  figure;
21  semilogy(g, Y_1);
22  grid on;
23  xlabel('Generation');
24  ylabel('Yield [kT]');
25  title('Yield vs Generation');
26
27
```



- Variable Leakage factor and burnup
- Iteration process introduced
- Iterating through every neutron due to variable factors
- 138 kT yield (neglecting burnup)





```

4 % -----
5 % Constants (MATLAB-safe only)
6 % -----
7 rho0 = 19.6e3;           % nominal density (kg/m^3)
8 M = 3.5;                % mass (kg)
9 NA = 6.02214199e23;
10 v_n = 2.98;
11 sigma_a = 1.98 / 1e28;   % m^2
12 sigma_f = 5.61 / (v_n * 1e28); % m^2
13 v = 1.08e9;             % m/s
14 Q = 185;                % J
15 n_o = 1000;
16 gen = 1:100;
17
18 % -----
19 % Initial Conditions
20 % -----
21 u = 0;                  % expansion velocity (m/s)
22 C0 = 6.5;               % compression factor
23 R = 3.5 / 100;          % convert cm → m
24 N = rho0 * C0 * NA / M;
25 k = 1;
26
27 % Preallocate yield array
28 Y = zeros(size(gen));
29
30 % -----
31 % Plot Setup
32 % -----
33

```

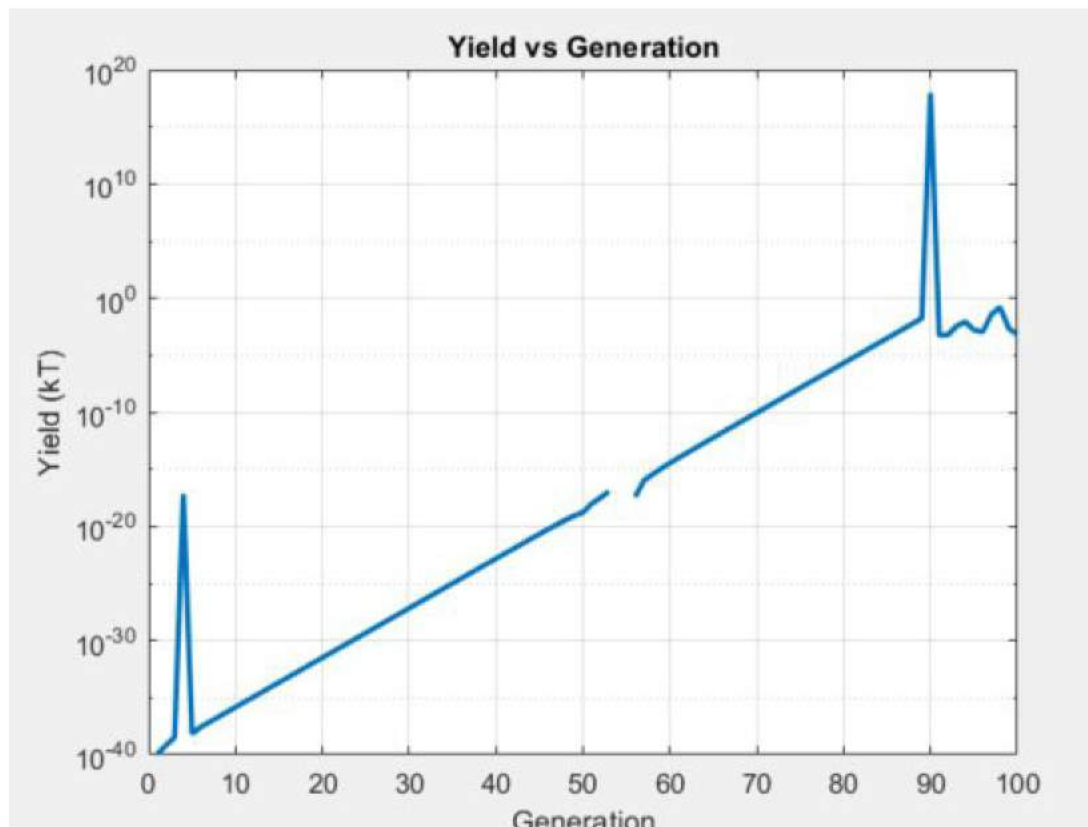
```

34 % -----
35 % Iteration Loop
36 % -----
37 for i = 1:length(gen)
38
39     % Macroscopic cross sections (1/m)
40     Sigma_a = N * sigma_a;
41     Sigma_f = N * sigma_f;
42
43     % alpha must be defined BEFORE use
44     if i == 1
45         alpha = 0; % placeholder to avoid undefined variable
46     end
47
48     % Compression factor update
49     C = C0 * (R / (R + u * i))^3;
50
51     % Lifetime
52     l_inf = 1 / (Sigma_a * v);
53
54     % Leakage factor
55     L = 0.43 * M^(-0.15) * log(C) + 0.14 * log(M);
56
57     % Effective lifetime
58     l = l_inf * L;
59
60     % Multiplication factor
61     k_inf = v_n * Sigma_f / Sigma_a;
62     k = k_inf * l;
63
64     % Growth rate
65     alpha = (k - 1) / l;
66
67     % Yield (placeholder form)
68     Y(i) = Sigma_f * v * Q * (n_o / max(alpha, 1e-12)) * (exp(i) - 1);
69
70     % Plot point
71     plot(i, Y(i), 'bo');
72     drawnow;
73
74     % Update number density
75     N = rho0 * C * NA / M;
76
77     % Update expansion velocity (MATLAB uses sqrt, not SQRT)
78     u = sqrt(abs(2 * Y(i) / (3 * M)));
79
80     % Update radius (must use scalar increment, not vector)
81     R = R + u * alpha * 1e-9; % placeholder timestep
82 end
83
84 % -----
85 % Final Plot
86 % -----
87 figure;
88 semilogy(gen, Y/(2.613E25 * 6.241457E19), 'LineWidth', 2);
89 grid on;
90 xlabel('Generation');
91 ylabel('Yield');
92 title('Yield vs Generation');
93

```



- Tamper added to the outside of the weapon
- Boosting added to generation 95
- Mass increased from 3.5 kg to 4.5 kg
- Tamper introduced into iteration created in model 2
- At generation 95 the iteration was broken and an more neutrons were introduced into the simulation
- This models all the 14.1 MeV neutrons that would be created from injecting deuterium into the weapon





```

15 % -----
16 % Constants (MATLAB-safe only)
17 % -----
18 rho0 = 19.6e3;           % nominal density (kg/m^3)
19 M = 4.5;                 % mass (kg)
20 NA = 6.02214199e23;
21 v_n = 2.98;
22 sigma_a = 1.98 / 1e28;    % m^2
23 sigma_f = 5.61 / (v_n * 1e28); % m^2
24 v = 1.08e9;              % m/s
25 Q = 185;                 % J
26 n_o = 1000;
27 gen = 1:100;
28 n_boost = 1*10^24
29
30 % -----
31 % Initial Conditions
32 % -----
33 u = 0;                   % expansion velocity (m/s)
34 C0 = 6.5;                % compression factor
35 R = 3.5 / 100;           % convert cm → m
36 N = rho0 * C0 * NA / M;
37 k = 1;
38
39 % Preallocate yield array
40 Y = zeros(size(gen));
41
42 % -----
43 % Plot Setup
44 % -----

```

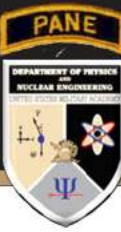
```

46 % -----
47 % Iteration Loop
48 % -----
49
50 injectstep = 90;
51 injectamount = n_boost;
52
53 for i = 1:length(gen)
54
55     % Macroscopic cross sections (1/m)
56     Sigma_a = N * sigma_a;
57     Sigma_f = N * sigma_f;
58
59     % alpha must be defined BEFORE use
60     if i == 1
61         alpha = 0; % placeholder to avoid undefined variable
62     end
63
64     % Compression factor update
65     C = C0 * (R / (R + u * i))^3;
66
67     % Lifetime
68     l_inf = 1 / (Sigma_a * v);
69
70     % Leakage factor
71     L = 0.43 * M^(-0.15) * log(C) + 0.14 * log(N);
72
73     % Effective lifetime
74     l = l_inf * L;
75
76     % Multiplication factor
77     k_inf = v_n * Sigma_f / Sigma_a;
78     k = k_inf * l;
79
80     % Growth rate
81     alpha = (k - 1) / l;
82
83     % Yield (placeholder form)
84     Y(i) = Sigma_f * v * Q * (n_o / max(alpha, 1e-12)) * (exp(i) - 1);
85
86     % Plot point
87     plot(i, Y(i), 'bo');
88     drawnow;
89
90     % Update number density
91     N = rho0 * C * NA / M;
92
93     % Update expansion velocity (MATLAB uses sqrt, not SQRT)
94     u = sqrt(abs(2 * Y(i) / (3 * M)));
95
96     % Update radius (must use scalar increment, not vector)
97     R = R + u * alpha * 1e-9; % placeholder timestep
98
99     if i == injectstep
100         Y(i) = Sigma_f * v * Q * ((n_o + injectamount) / max(alpha, 1e-12)) * (exp(i) - 1);
101     end
102 end
103
104 % -----
105 % Final Plot
106 % -----
107 figure;
108 semilogy(gen, Y/(2.613E25 * 6.241457E19), 'LineWidth', 2);
109 grid on;
110 xlabel('Generation');
111 ylabel('Yield (KT)');
112 title('Yield vs Generation');

```




- Accounting for all variables is detrimental to making a useful model.
- Assumptions severely weaken the model and make it irrelevant for stockpile maintenance.
- Burnup will play a larger factor than predicted in the model (Model 2).



Questions?



UNITED STATES MILITARY ACADEMY
WEST POINT.

EXTENSION PROBLEM #2



West Point has been targeted for a “tactical” 10 kT nuclear strike. Fortunately, the delivery vehicle was poorly constructed, and the weapon detonated on impact at the West Point Golf Course Clubhouse.

1. Using NUKEMAP, you need to plot the effect rings for (make separate figures for each effect):
 - 1st, 2nd, and 3rd degree burn 50% probability
 - 1, 5, and 20 PSI blast damage
 - 40, 400, and 4000 rem of initial radiation dose
2. You need to validate whether NUKEMAP can be trusted or not using the equations learned in NE450. Any calculation method is acceptable (by hand, code...). Also, discuss what you think is the greatest assumption we are making in this scenario and why.
3. Using NUKEMAP, you need to plot the fallout contours for:
 - $R_1 = 1, 10, 100, \text{ and } 1000 \text{ rads/hr}$
 - Using: 7 mph winds at 260 degrees
4. It is T+2 hrs. The BTO has scheduled an AFT at T+4 hrs to select cadets for early graduation to retaliate against the strike. You are located at Buffalo Soldier field. Assuming the fallout contours you plotted are homogenous, calculate the total dose you receive while maneuvering to Target Hill Athletic Field. The team with the least total dose will receive 10 bonus points.

*Unrealistic travel times will be disqualified (it is about 1.75 miles slant range, you cannot sprint for 1.75 miles. Even if you could, you cannot sprint straight through Michie Stadium and the forest on the other side.)



- You are briefing in class 4 and 5 May
- The Brief: 5-10 minutes showing your effect rings, validation with recommendation to use or not use NUKEMAP, fallout contours, chosen maneuver route/methods with total dose, and what you learned along the way. Memes optional but encouraged. Slides uploaded to canvas.
- The Poster: A standalone single slide poster following the example format uploaded to canvas.
- The Products: Maps/calculations uploaded to canvas.

Outstanding Student Work

UNITED STATES MILITARY ACADEMY

EXTENSION PROBLEM 2

NE450: NUCLEAR WEAPONS EFFECTS

SECTION I1

MAJ DAVE PERT

BY

[REDACTED]

WEST POINT, NEW YORK

4 MAY 2026

CP HB WG WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

 WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

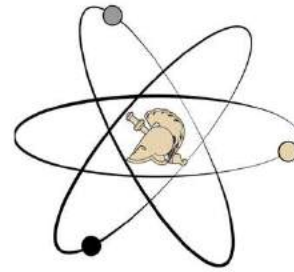
CP HB WG WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

 WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

[REDACTED]



Extension Problem 2



Problem Statement: A 10 Kt Groundburst nuke has been detonated on the golf course Clubhouse. The BTO still needs his AFT cards! What are we going to do?

Assumptions: Groundburst fission weapon, Whole class moving to Target hill, 15-min/mile



Thermal Radiation

NukeMap vs Hand Calculation
Percent Differences for distance of thermal rings

1st deg: 24% (1.61 mi vs 2.136 mi)
2nd deg: 14.9% (1.15 vs 1.35 mi)
3rd deg: 1.6% (.94 mi vs .955 mi)

Governing Eq:

$$Q \left(\frac{\text{cal}}{\text{cm}^2} \right) = \frac{3.07 f W \tau}{D^2} (\text{miles})$$

Method:

Using exposure rates for 1st 2nd and 3rd degree burns, back calculate distance for radiant exposure rings and compare



Overpressure Radius

NukeMap vs Hand Calculation
Percent Differences for distance of overpressure rings

1 psi: 2% (1.58 mi vs 1.55 mi)
5 psi: 0% (0.61 mi vs 0.61 mi)
20 psi: 0% (0.29 mi vs 0.29 mi)

Governing Eq:

$$d = d_1 * W^{\frac{1}{3}} S_d$$

Method:

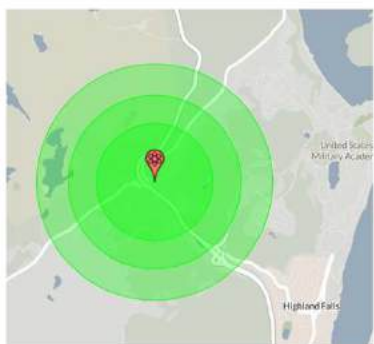
Using figures 5-6 and 5-7 of the NERD-P, determine reference distance for 1kT detonation. Scale reference distance to 10kT.



Governing Eq:
$$D \approx 5R_1(t_a^{-0.2} - t_b^{-0.2})$$

Method:

- major roads
- shorter distance
- avoid 1000 rad/hr
- hangout at BS field for ~85 minutes before leaving



Radiation Radius

NukeMap vs Hand Calculation
Percent Differences for distance of radiation rings

40 rem: 9.2% (1.1 mi vs 1.2 mi)
400 rem: 9% (.8 mi vs .88 mi)
4000 rem: 4.7% (.54 mi vs .49 mi)

Governing Eq:

$$D = D_n + D_{ys} + D_{yff}, H = DQ$$

Method:

Split does into neutron, gamma fission, and secondary gammas, and back calculate distance using rad/KT graphs



"Cadets will be taking the AFT with acute radiation syndrome and they will love it!, We couldn't be prouder of you!"-Big Mike
#thisismysquad
#leadership



Route and Fallout Contours

Fallout	Time	Distance	Dose
1 rad/hr	7.5 min	0.5 mil	0.12 rads
10 rad/hr	7.5 min	0.5 mil	0.27 rads
100 rad/hr	19.5 min	1.3 mil	7.02 rads
Totals:	34.5 min	2.3 miles	7.41 rads

~~Exposure~~, initial rad, ~~overpressure~~

initial rad

$$D = 40, 400, 4000 \text{ rem}$$

$$D_n \approx 33\% \text{ of dose} = 13.2 \text{ rem}, 132 \text{ rem}, 1320 \text{ rem}$$

$$D_{yf} \approx 52\% \text{ of dose} = 20.8 \text{ rem}, 208 \text{ rem}, 2080 \text{ rem}$$

$$D_{ys} \approx 15\% \text{ dose} = 6 \text{ rem}, 60 \text{ rem}, 600 \text{ rem}$$

$$D_n = H = DQ \Rightarrow \text{divide by } Q \text{ to get } D, Q=10 \Rightarrow D = 1.32 \text{ rad}, 13.2 \text{ rad}, 132 \text{ rad}$$

$$\text{rads/KT} = D/10 \text{ KT} \Rightarrow .132, 1.32, 13.2$$

fig 5-17 in! .132 rads/KT out! Slant range (dist from ground zero)

1.32

13.2

$$40 \text{ rem} = 2250 \text{ yds}$$

$$400 = 1750 \text{ yds}$$

$$4000 = 1100 \text{ yds}$$

$$D_{yf} = \text{rad/KT} = 2.08, 20.8, 208$$

Fig 5-21 in! rads/KT

out! slant range

$$40 \text{ rem} : 1900 \text{ yds}$$

$$400 \text{ rem} : 1400 \text{ yds}$$

$$4000 \text{ rem} : 850 \text{ yds}$$

Avg slant range

$$40 \text{ rem} = \frac{2250 + 1900 + 2250}{3} = 2133 \text{ yds}$$

$$400 \text{ rem} = \frac{1750 + 1400 + 1500}{3} = 1550 \text{ yds}$$

$$4000 \text{ rem} = \frac{1100 + 850 + 650}{3} = 867 \text{ yds}$$

$$D_{ys} = \text{rad/KT} = .6, 6, 60$$

fig 5-14 in! rads/KT

out! slant range

$$40 \text{ rem} = 2250$$

$$400 \text{ rem} = 1500 \text{ yds}$$

$$4000 \text{ rem} = 650 \text{ yds}$$

Actual dist

$$4000 - 0.54 \text{ miles } 0.79$$

$$400 - 0.8 \text{ mi } - 0.88 \text{ mi}$$

$$40 - 1.1 \text{ guess } - 1.2 \text{ mi}$$

percent diff

$$40 - 9.2\%, 400 - 9\% \text{ diff}, 4000 - 4.7\%$$

E xposure

1st deg: 2 cal/cm^2

$0.94 \text{ mi} \cdot 1.61 = 8500 \text{ ft} = 2.5 \text{ kft}$

2nd deg: 5 cal/cm^2

$1.15 \text{ mi} = 6072 \text{ ft} = 6 \text{ kft}$

3rd deg: 10 cal/cm^2

$0.94 \text{ mi} = 4963 = 5 \text{ kft}$

$$Q \frac{\text{cal}}{\text{cm}^2} = \frac{3.07 f W T}{D^2}$$

$$D = \sqrt{\frac{3.07 f W T}{Q}} \text{ (miles)}$$

fig 5-16

$f = 0.35 \quad W = 10 \quad T = 0.85 \text{ (approximation)}$

1st deg: $D = \sqrt{\frac{3.07 (0.35) (10) (0.85)}{2}} = 2.136 \text{ mi}$

2nd deg: $D = \sqrt{\frac{3.07 (0.35) (10) (0.85)}{5}} = 1.35 \text{ mi}$

3rd deg: $D = \sqrt{\frac{3.07 (0.35) (10) (0.85)}{10}} = 0.95 \text{ mi}$

Percent diff:

1st deg $\frac{2.136 - 1.61}{2.136} = 24\%$

2nd deg $\frac{1.35 - 1.15}{1.35} = 14.9\%$

3rd deg $\frac{0.95 - 0.94}{0.95} = 1.6\%$

Extension problem 2 - Peak overpressure calculations/validation

Given:

$$W = 10 \text{ kt}$$

$$S_p = S_d = S_r = 1 \text{ (ground burst)}$$

Pressure rings at: 1, 5, 20 psi

Governing Equations:

$$d = d_1 \times W^{\frac{1}{3}} S_d$$

$$P = P_1 S_p$$

Solve: Use figures S-6 and S-7 to see where pressure ring intersect HOB = 0:

1 psi

Figure S-7:

input: 1 psi, 0 ft HOB

$$\rightarrow d_1 \approx 3800 \text{ ft}$$

Calculate distance for 1 psi:

$$d = d_1 \times W^{\frac{1}{3}} S_d$$

$$d = (3800)(10)^{\frac{1}{3}}(1) = 8187 \text{ ft}$$

5 psi

Figure S-7

input: 5 psi, 0 ft HOB

$$\rightarrow d_1 \approx 1500 \text{ ft}$$

Calculate distance for 5 psi:

$$d = (1500)(10)^{\frac{1}{3}}(1) = 3232 \text{ ft}$$

20 psi

Figure S-6

input: 20 psi, 0 ft HOB

$$\rightarrow d_1 \approx 710 \text{ ft}$$

Calculate distance for 20 psi:

$$d = (710)(10)^{\frac{1}{3}}(1) = 1530 \text{ ft}$$

Extension Problem 2 Route Planning Calculations

		Route Data		Route Dosage	
		Leg time	Accumulative time	real time (hr)	Dose
	leg 1	0.075	0.075	3.375	0.01766
	leg 2	0.05	0.125	3.425	0.115135
	leg 3	0.325	0.45	3.75	7.023046
	leg 4	0.075	0.525	3.825	0.151725
	leg 5	0.05	0.575	3.875	0.09918
Givens					
R_1A	1			total dose	7.406745
R_1B	10				
R_1C	100				
Start time T+3.3hr					

Dose from each Contour

1 rad	0.11683981
10 rad	0.26685943
100 rad	7.02304579

Works Cited

ChatGPT, *OpenAI*, Generative AI Model, Assistance given to the Authors, Answering questions and image generation. ChatGPT was asked what the radiant exposure would be for different degrees of burns as well as what effects would result from the level of radiation exposure encountered on the run down to target hill field. ChatGPT was also used to modify images of the BTO in order to create a humorous picture for the presentation. A link to the photo generation conversation can be located at the following link, a pdf of the photo generation conversation was not available to be generated :

<https://chatgpt.com/share/69f7fcaa-cf90-83ea-b46a-daac4677ca31> The PDF of the questions asked to ChatGPT as well as the responses can be found in the document below.

Google.com, *Alphabet*, Assistance given to the author, unit conversions. Google was used to convert yards to miles, miles to feet, and rem to rad to help solve the hand calculations.

Alex Wellerstein, *Nukemap*, <https://nuclearsecrecy.com/nukemap/> Nukemap was used to plot all effects in this project for validation and route selection purposes.

7.4 rad dose severity

A dose of 7.4 rad is considered a low acute radiation dose. It is well below the threshold for acute radiation syndrome (ARS) in most people.

General Severity

- 7.4 rad = 0.074 Gy (for gamma/X ray radiation)
- Usually produces no immediate symptoms
- Slightly above normal background/occupational exposure levels, but not life threatening

Typical Effects at This Dose

Possible effects may include:

- Temporary, very minor blood cell changes (often not noticeable)
- No expected nausea, vomiting, or radiation sickness
- No immediate incapacitation

Context

For comparison:

- ~25 rad: earliest detectable blood changes
- ~100 rad: mild radiation sickness possible
- ~300–500 rad: severe ARS and potentially fatal without treatment
- 800 rad: often fatal even with treatment

So 7.4 rad would generally be categorized as a mild/low exposure, though still something radiation safety personnel would document and monitor in a controlled environment.



what is the cal/cm² for 50% 1st 2nd and 3rd degree burns



Approximate thermal radiation exposures for 50% probability of skin burns are:

Burn Severity	Thermal Dose (cal/cm ²)	Notes
1st-degree burn	~2 cal/cm ²	Redness/sunburn-type injury
2nd degree burn	~5 cal/cm ²	Blistering burns
3rd-degree burn	~10 cal/cm ²	Full-thickness skin destruction

These are common rule-of-thumb values used in nuclear weapons effects and thermal injury modeling (from sources like [The Effects of Nuclear Weapons](#) by [Samuel Glasstone](#) and [Philip J. Dolan](#)).

A more detailed estimate sometimes used is:

- 1st degree: 2–3 cal/cm²
- 2nd degree: 5–6 cal/cm²
- 3rd degree: 8–10+ cal/cm²

These values depend on:

- Exposure duration
- Skin pigmentation
- Clothing
- Angle of exposure
- Whether the pulse is from a nuclear thermal flash or conventional fire source



Poor Student Work

UNITED STATES MILITARY ACADEMY

EXTENSION PROBLEM 2

NE450: NUCLEAR WEAPONS EFFECTS

SECTION J1

MAJOR DAVID PERT

BY



WEST POINT, NEW YORK

4 MAY 2026

NB SM SM EV WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

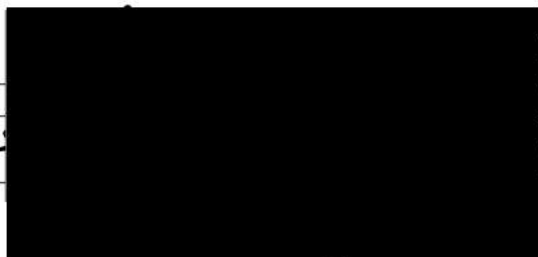
____ WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

____ WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

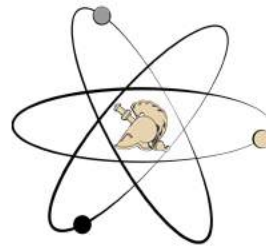
NB SM SM EV WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: _____	DATE: 4 MAY 2026 _____
SIGNATURE: _____	DATE: 4 MAY 2026 _____
SIGNATURE: _____	DATE: 4 MAY 2026 _____
SIGNATURE: _____	DATE: 4 MAY 2026 _____





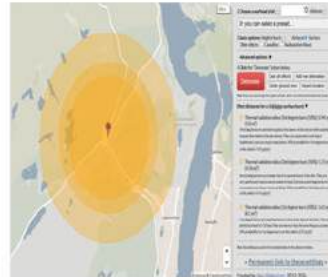
Extension Problem 2



Problem: Determine the range of effects (burn damage, blast damage, initial radiation) of a 10 KT nuclear blast on the West Point area through the software NukeMap and compare the effects through empirical calculations and determine the possibility of conducting an AFT T+4 hours after strike.

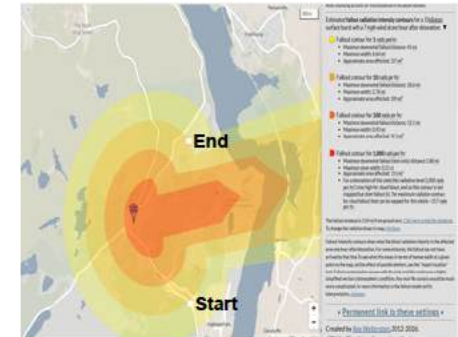
Hypothesis: Because the data used by NukeMap and the NERD-P are similarly sourced, the data should align. The nuclear effects will not substantially impact the West Point area and will permit the completion of an AFT.

Burn

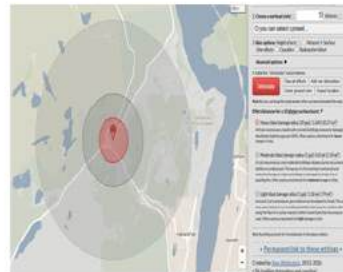


Effect	Nukemap [cal/cm ²]	Theoretical [cal/cm ²]
3rd deg.	7.19	5.63
2nd deg.	4.76	3.55
1st deg.	2.37	1.71

Fallout and AFT Completion



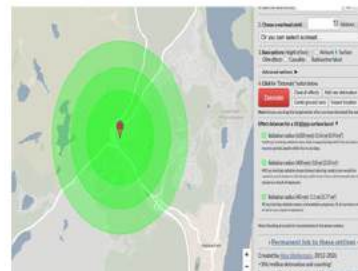
Blast



Overpressure	Nukemap [ft]	Theoretical [ft]
20 psi	1540	1530
5 psi	3221	3232
1 psi	8342	8079

Movement	Dose [Rad]
Leg 1 (5 min)	0.081
Leg 2 (5 min)	0.775
Leg 3 (20 min)	27.763
Leg 4 (5 min)	0.625
Leg 5 (5 min)	0.060
Total (40 min)	29.305

Initial Radiation



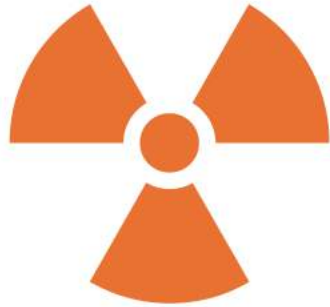
Distance [ft]	H (Nukemap)	H (Theoretical)
2851.2	4000 rem	11400 rem
4224	400 rem	950 rem
5808	40 rem	148 rem

Conclusion: While imperfect, the data provided by NukeMap is very similar to the data calculated using the NERD-P. Because of this, NukeMap is reliable software and is recommended. Overall, the nuclear effects are not existentially damaging and would allow for the completion of an AFT with minimal dose.

Extension Problem #2

The background of the slide is a faded image of a nuclear explosion, showing a large, bright, billowing cloud of smoke and fire rising from a body of water. Overlaid on this background is a large, solid black T-shaped graphic. The vertical stem of the T is centered below the text, and the horizontal bar extends to the left and right, positioned just below the text line.

Problem

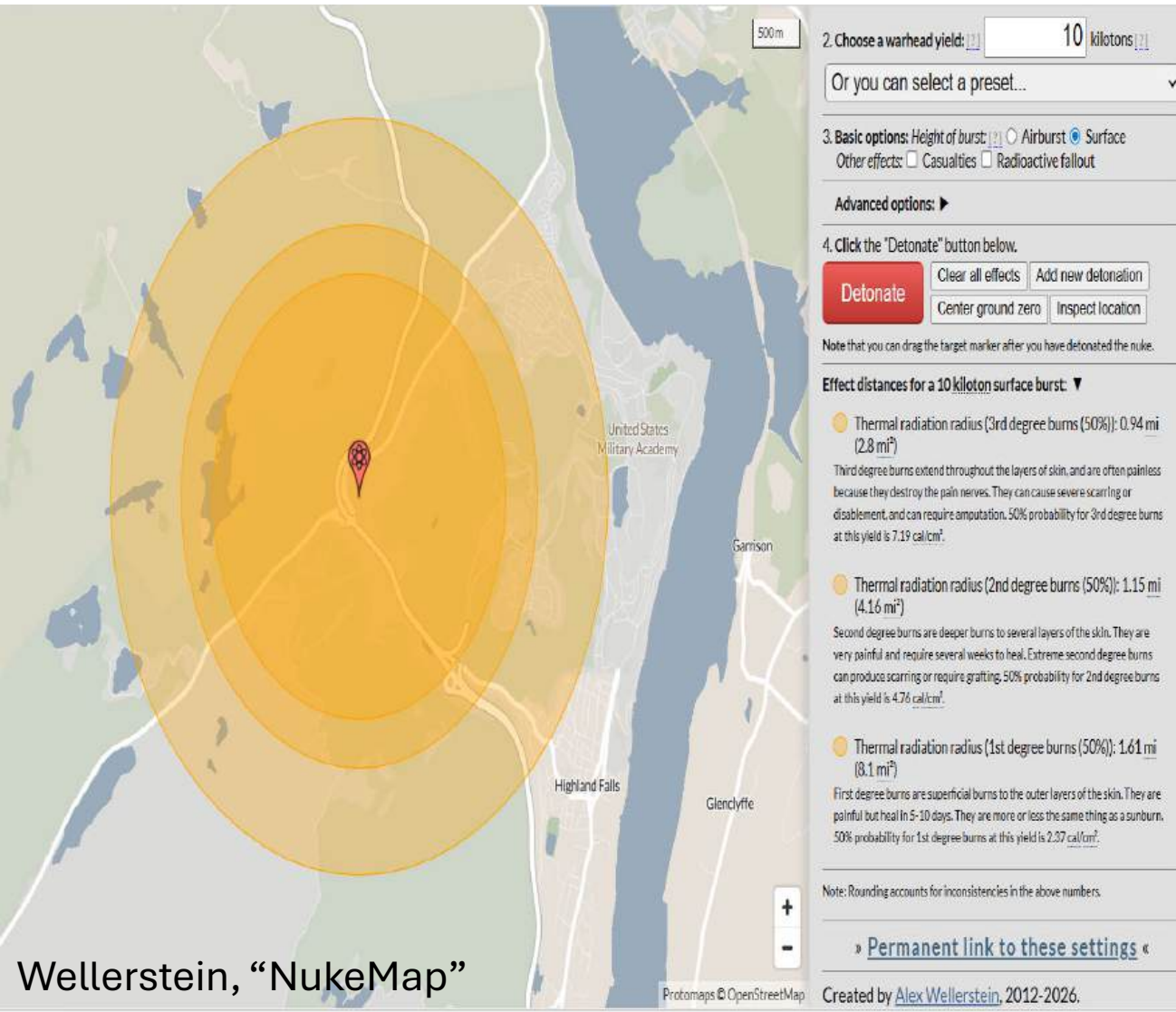


1. Is NukeMap a reliable software for calculating nuclear weapon effects?



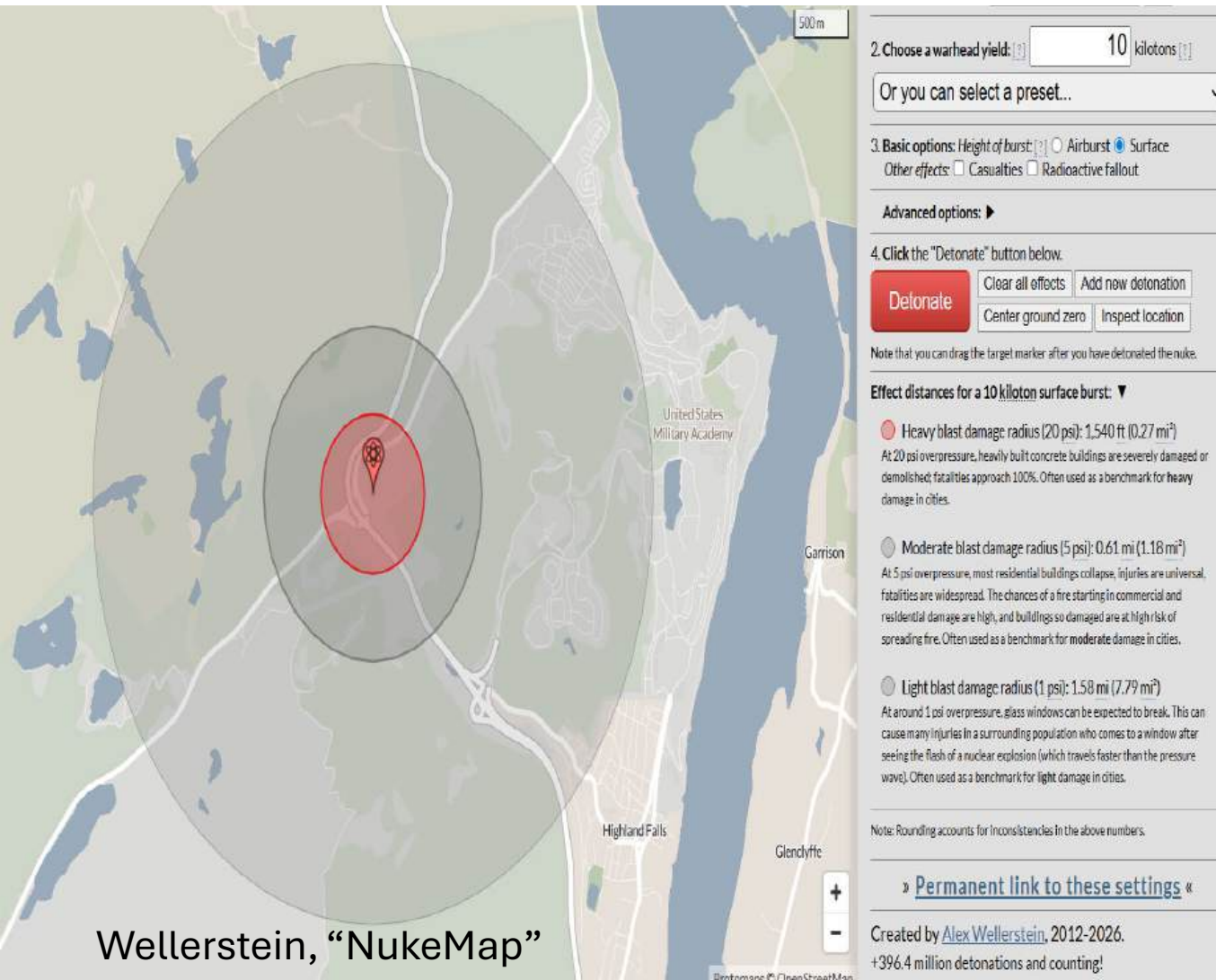
2. Can we take the AFT?

1st, 2nd, & 3rd Degree Burns



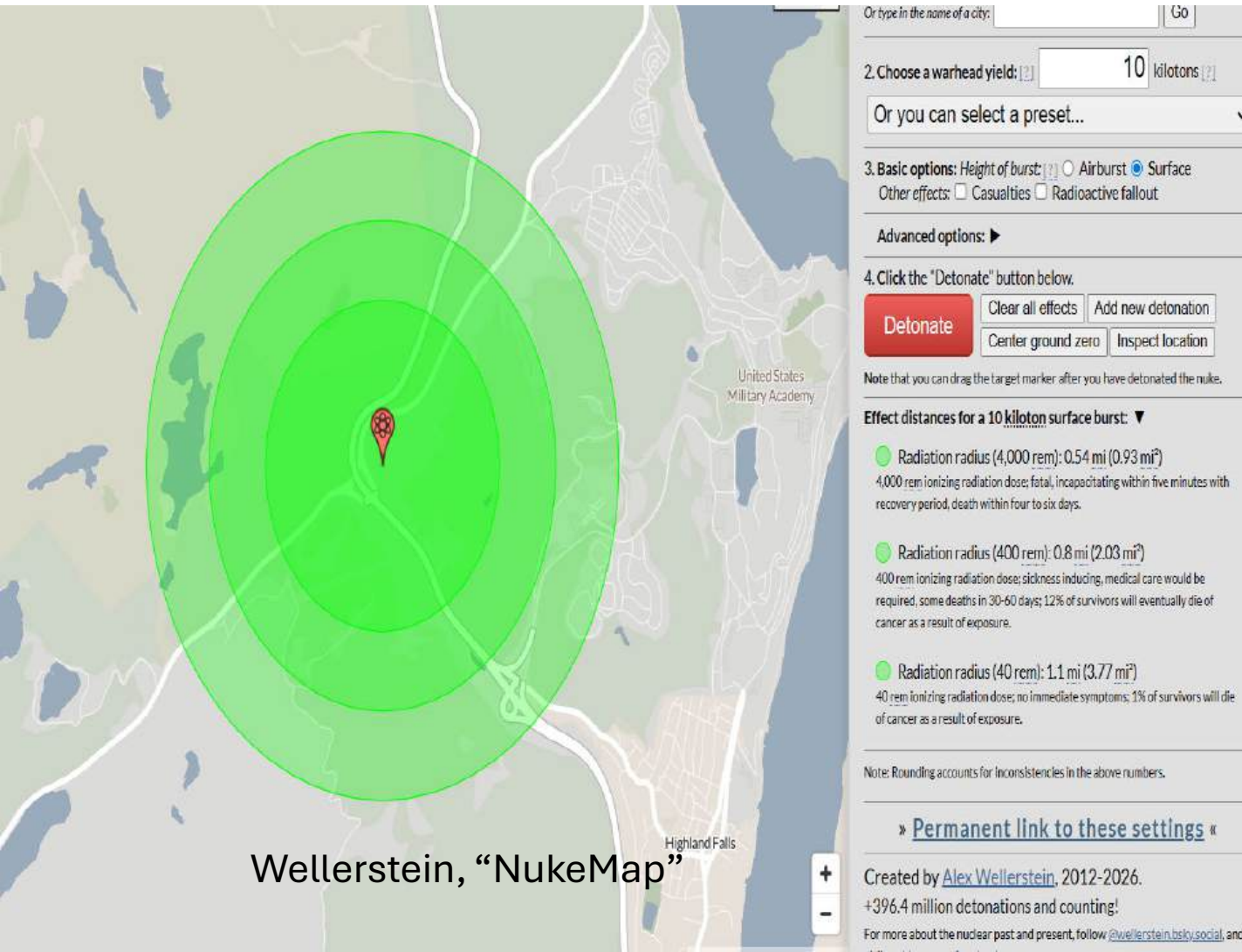
Effect	Nukemap [cal/cm ²]	Theoretical [cal/cm ²]
3rd deg.	7.19	5.63
2nd deg.	4.76	3.55
1st deg.	2.37	1.71

1, 5, & 20 PSI Blast Damage



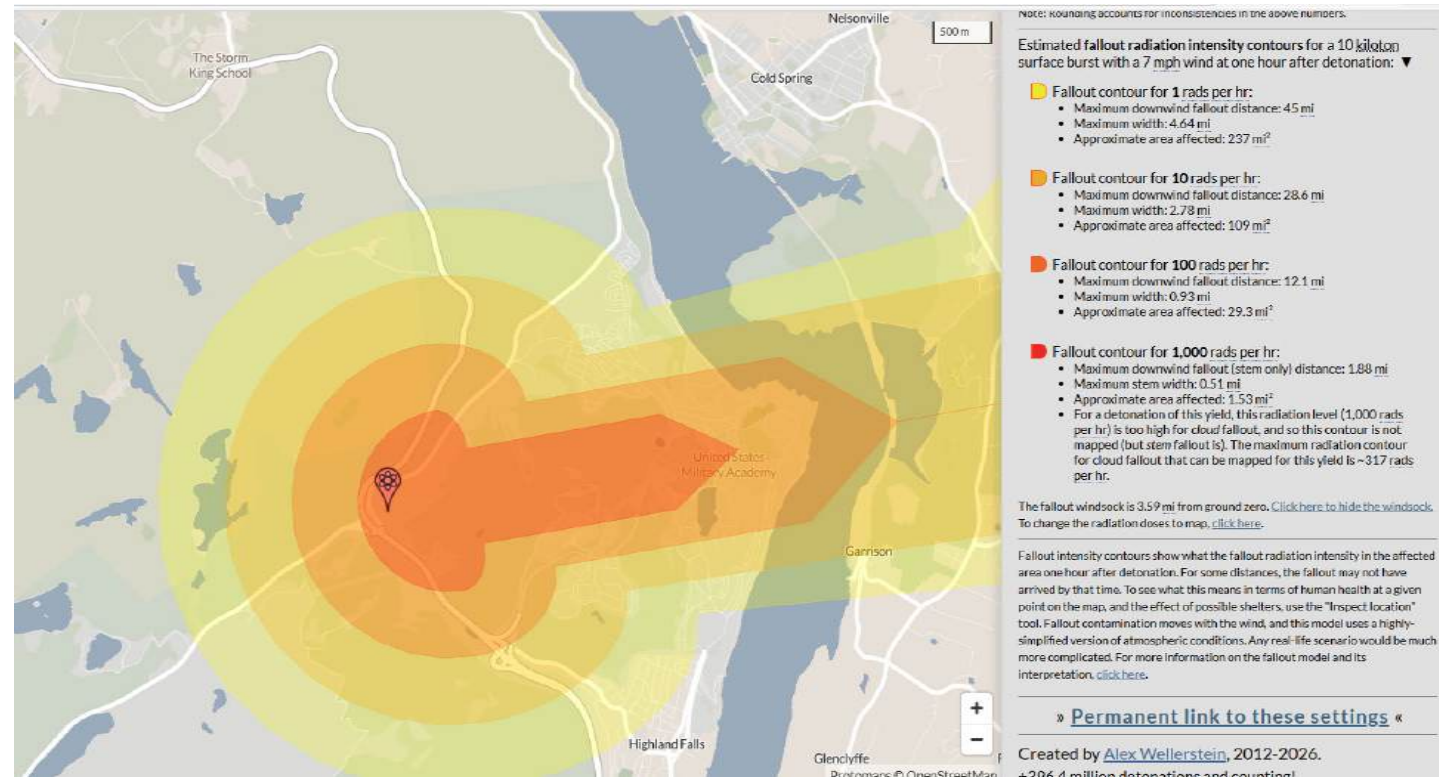
Overpressure	Nukemap [ft]	Theoretical [ft]
20 psi	1540	1530
5 psi	3221	3232
1 psi	8342	8079

40, 400 & 4000 Rem Initial Radiation



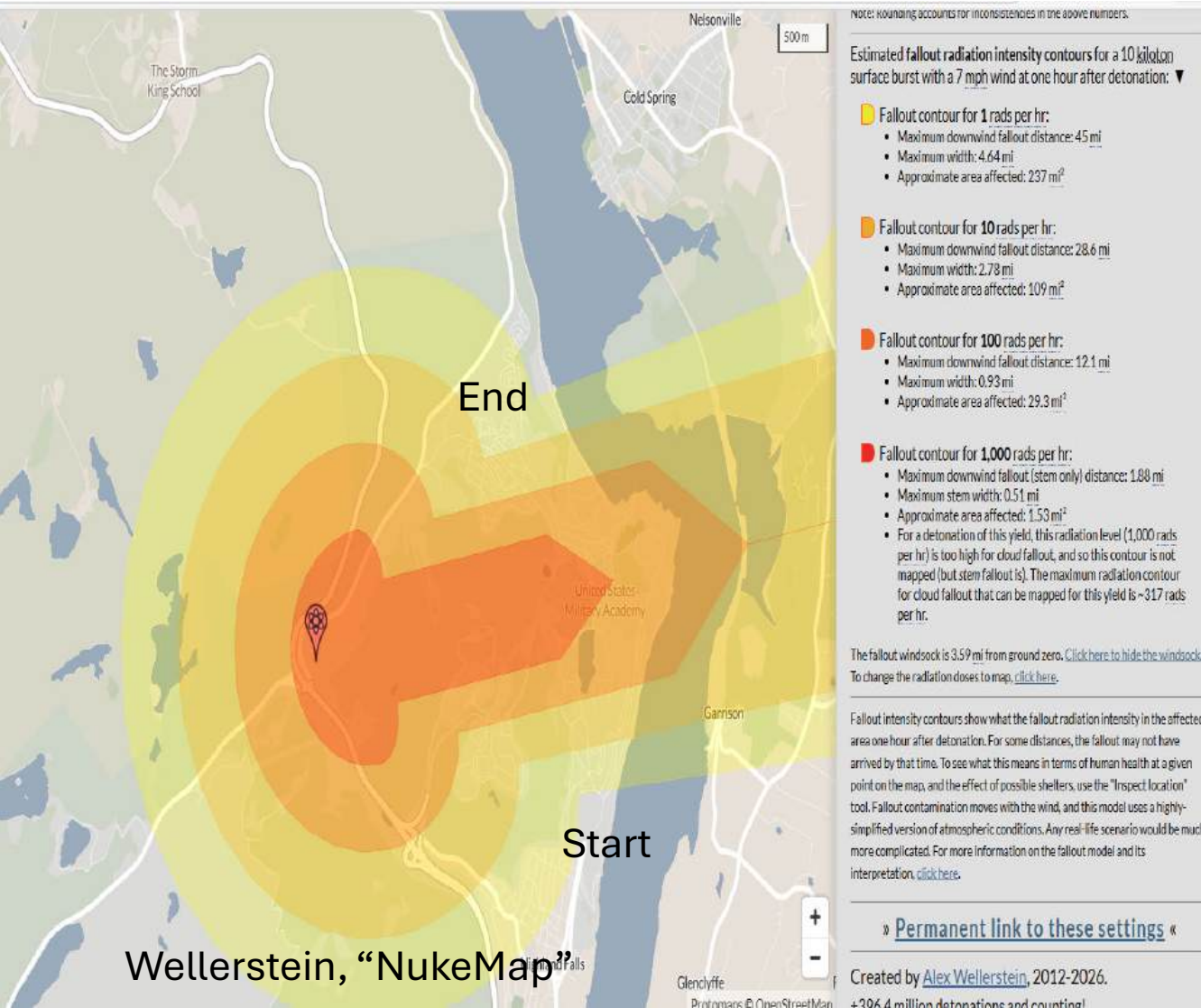
Distance [ft]	H (Nukemap)	H (Theoretical)
2851.2	4000 rem	11400 rem
4224	400 rem	950 rem
5808	40 rem	148 rem

Radioactive Fallout



Wellerstein, "NukeMap"

BTO AFT



Movement	Dose [Rad]
Leg 1 (5 min)	0.081
Leg 2 (5 min)	0.775
Leg 3 (20 min)	27.763
Leg 4 (5 min)	0.625
Leg 5 (5 min)	0.060
Total (40 min)	29.305

Works Cited

- Wellerstein, Alex. “NukeMap”, 2026, [NUKEMAP by Alex Wellerstein](#)
- Imgflip. (2026). *Meme generator*. Imgflip.
<https://imgflip.com/memegenerator>